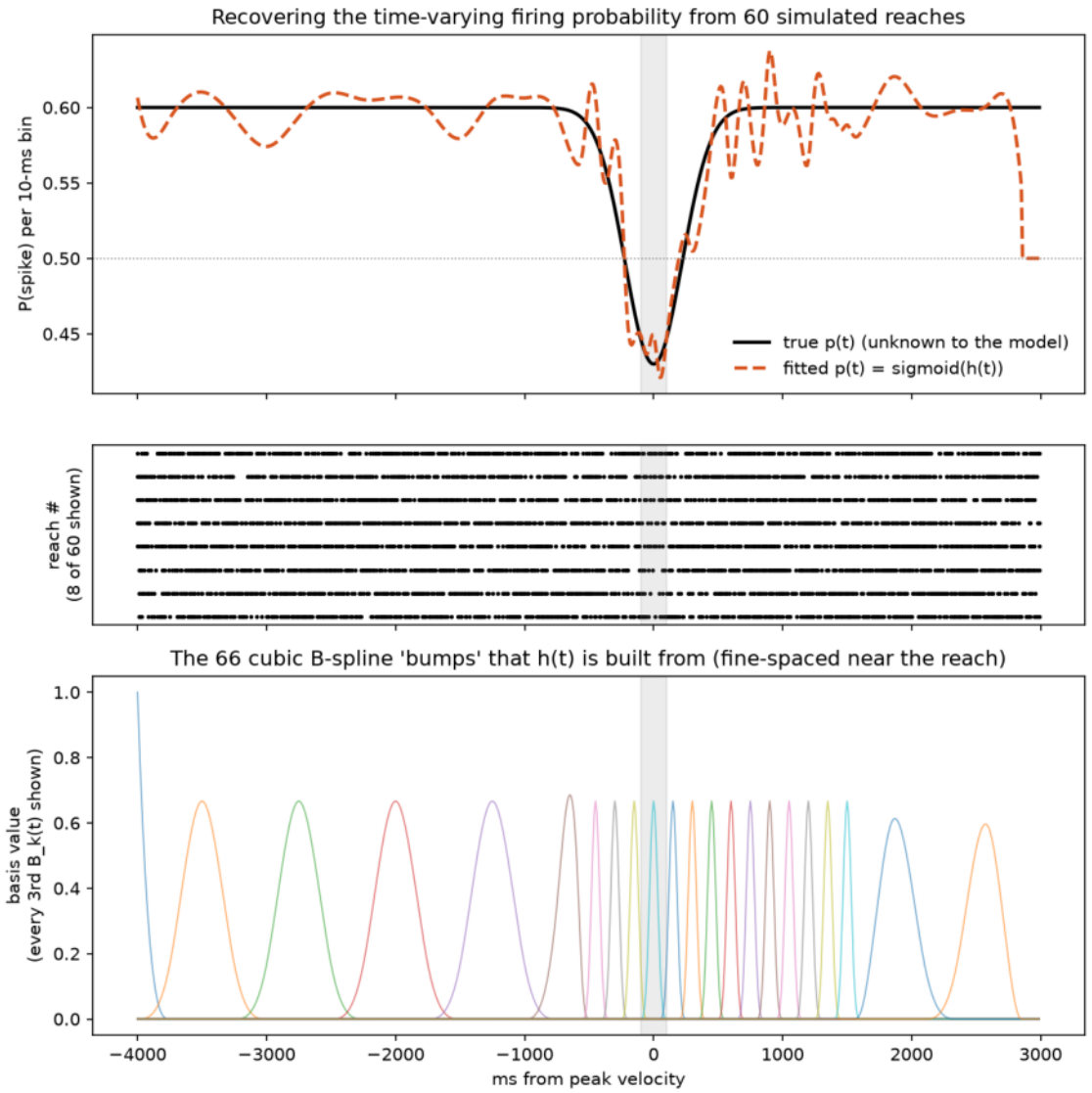
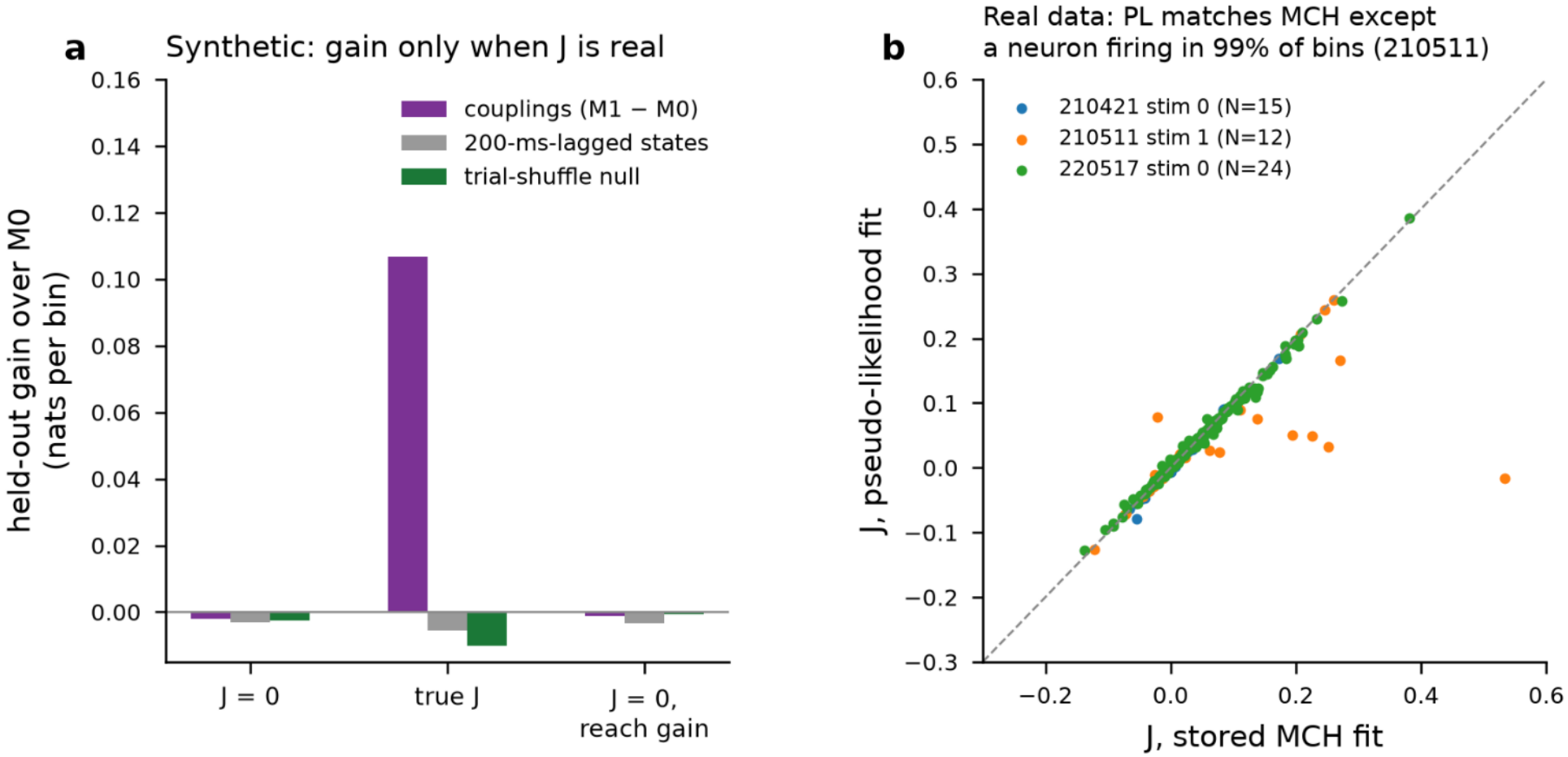
Evidence package: time-varying-field Ising model of Purkinje-cell populations

63 figures in INDEX.md order. Each page: figure + short caption. Folder X_ = withdrawn/superseded (do not cite).

1_model_and_validation / 01_tvfield_walkthrough [established (methods)]

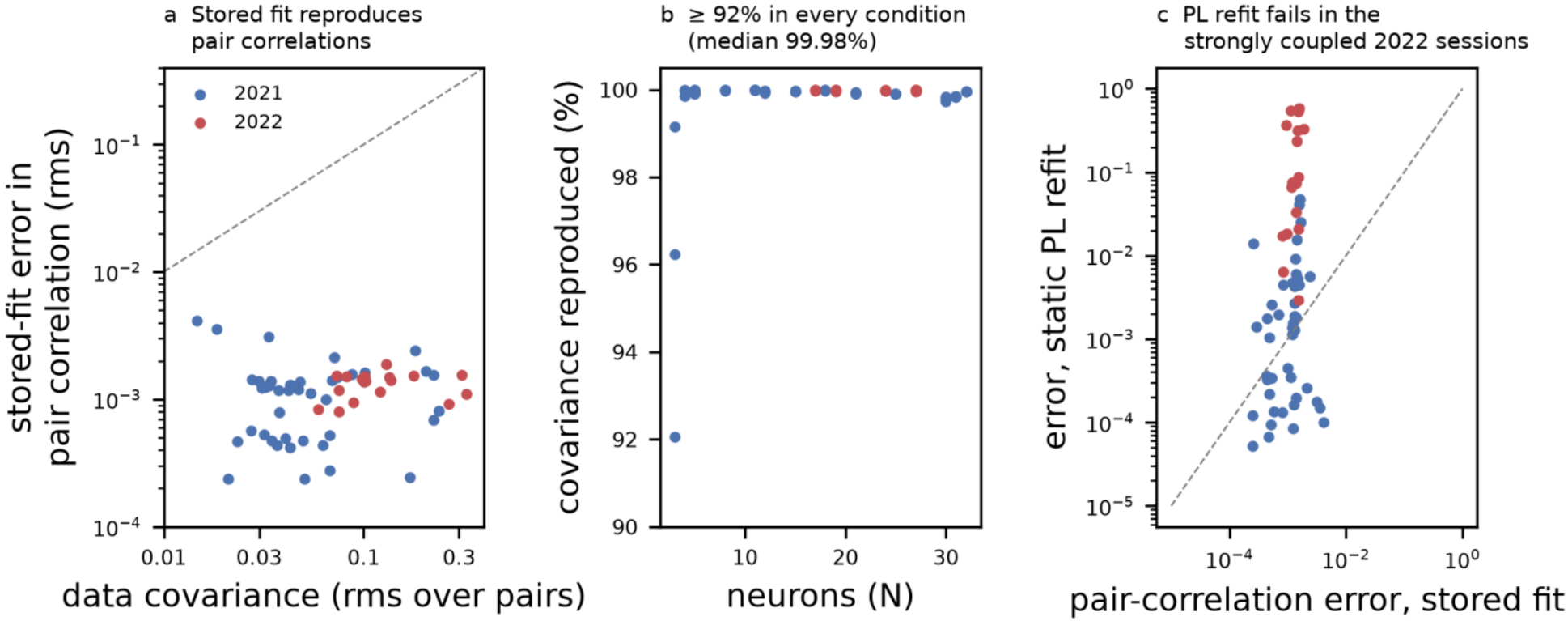


Shows: Top: P(spike) per 10-ms bin vs ms from peak velocity (true p(t) black, fitted sigmoid(h(t)) dashed). Middle: spike raster of 8 of 60 simulated reaches. Bottom: value of every 3rd cubic B-spline basis function vs time. Pattern: The fitted field follows the planted reach-locked dip in firing probability; the basis is dense (50-ms knots) near the reach and sparse elsewhere. Why: Shows how h(t) is built and that it can recover a time-varying rate from 60 reaches. Numbers: The figure title says 66 cubic B-splines; the code (tvfield_model.bspline_basis) builds 67. The title is off by one; 67 is correct.



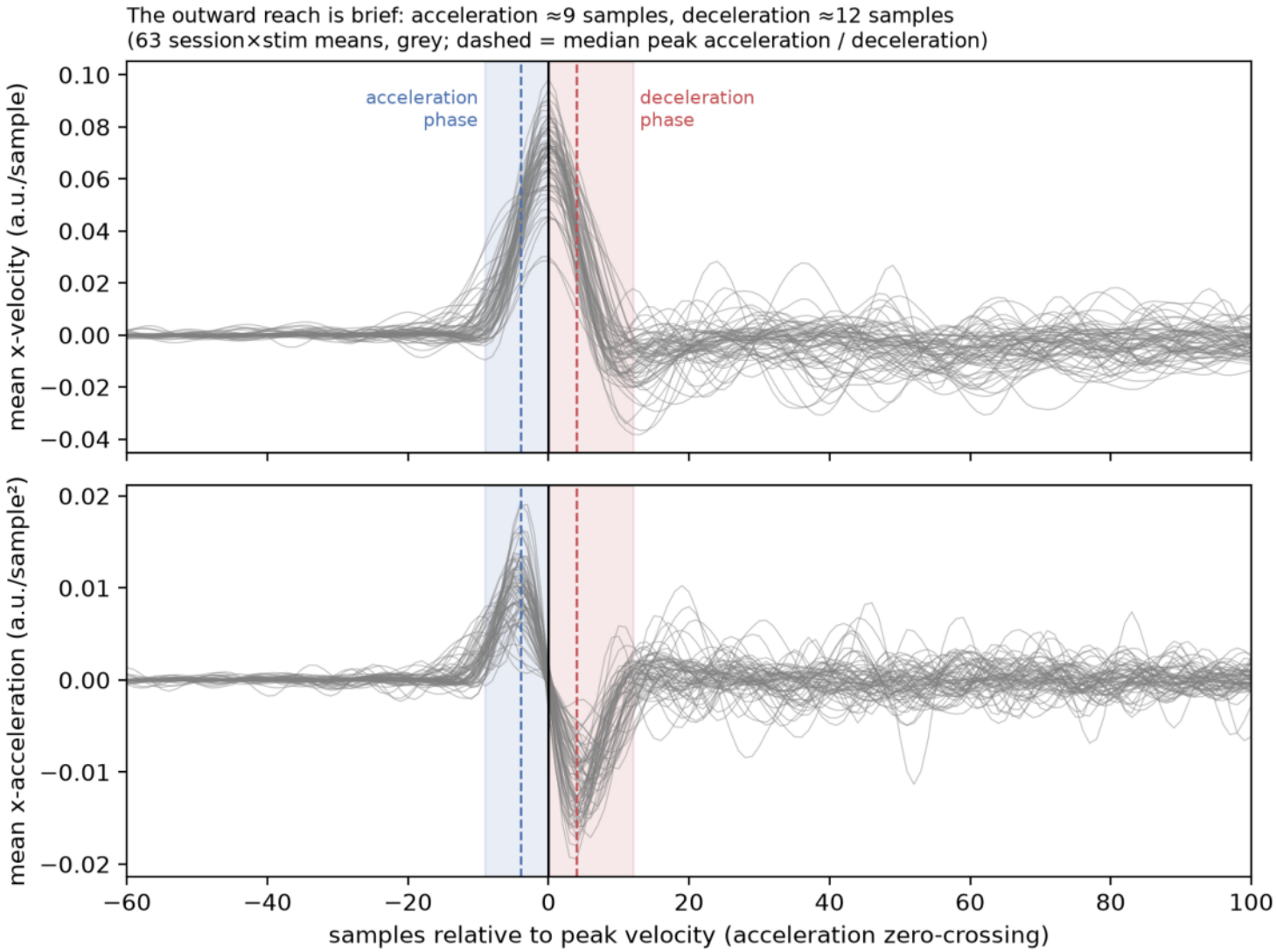
Shows: (a) Held-out gain over the no-coupling model M0 (nats/bin) for three synthetic cases (J = 0; true J; J = 0 with a shared per-reach rate gain); bars = couplings (M1 - M0), 200-ms-lagged states, trial-shuffle null. (b) Real data: J from the pseudo-likelihood fit vs the stored MCH fit for 3 conditions. Pattern: Gain appears only when J is real; the lagged control and shuffle null are at or below 0. PL and MCH couplings lie on the identity line except one neuron firing in 99% of bins (210511). Why: The coupling gain is not produced by shared rate fluctuations, and the PL estimator agrees with the established MCH solver. Numbers: dPL(M1-M0): true J +0.107 nats/bin; J = 0 -0.002; J = 0 + per-reach gain -0.001. J recovery r = 0.978 (tvfield_validation.csv). PL vs MCH r_J 0.995 for 210421 (tvfield_validation_vs_MCH.csv).

1_model_and_validation / 03_pipeline_verification [established (methods)]



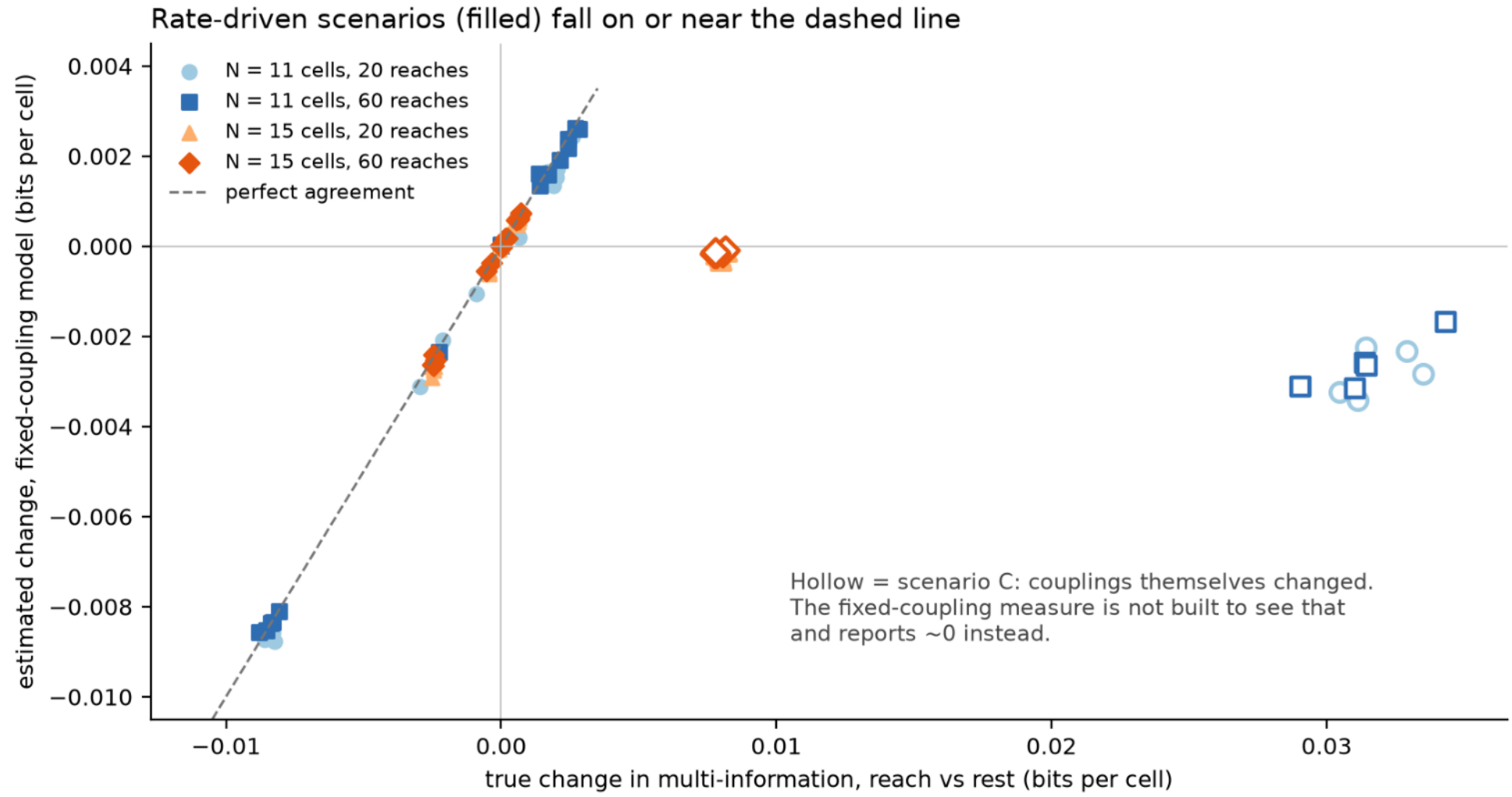
Shows: (a) Stored-fit error in pair correlation (rms) vs data covariance (rms over pairs), log-log, coloured by cohort. (b) Covariance reproduced (%) vs N. (c) Error of a static PL refit vs stored-fit error. Pattern: Stored fits reproduce pair correlations with errors far below the data covariance; >= 92% of covariance reproduced in every condition (median 99.98%); a static PL refit fails in the strongly coupled 2022 sessions. Why: Supports A1 (the stored stationary fits are correct) and documents where PL alone is unreliable. Numbers: Covariance reproduced >= 92% in every condition, median 99.98% (from figure).

1_model_and_validation / 04_phase_landmarks_qc [established (methods)]



Shows: Mean x-velocity (top) and x-acceleration (bottom) for 63 session x stim conditions vs samples (10 ms) from peak velocity. Pattern: Acceleration ~9 samples, deceleration ~12 samples: the outward reach is brief. Why: Defines the kinematic landmarks used everywhere (B1). Numbers: Acceleration median 90 ms, deceleration median 120 ms (RESULTS_LOG B1).

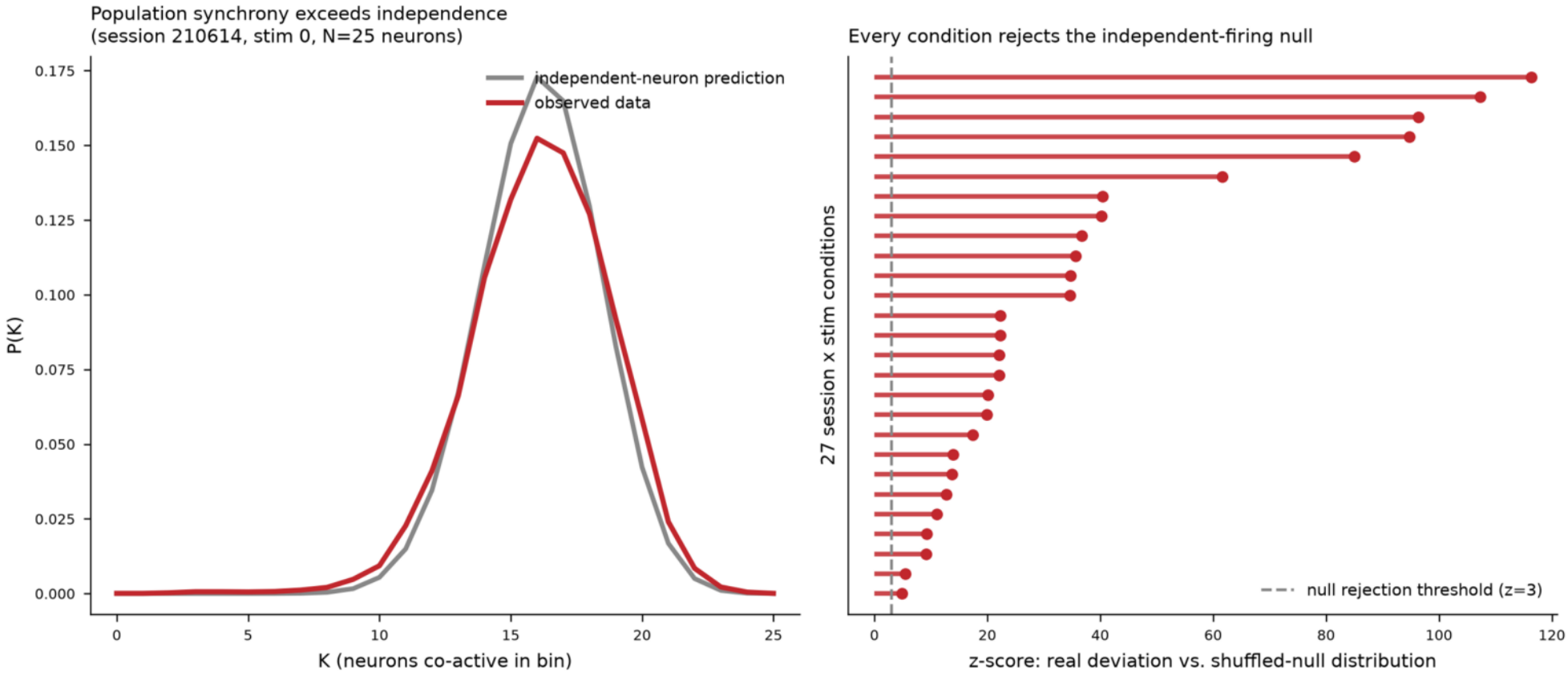
1_model_and_validation / 05_synthetic_validation_key_measure [established (methods)]



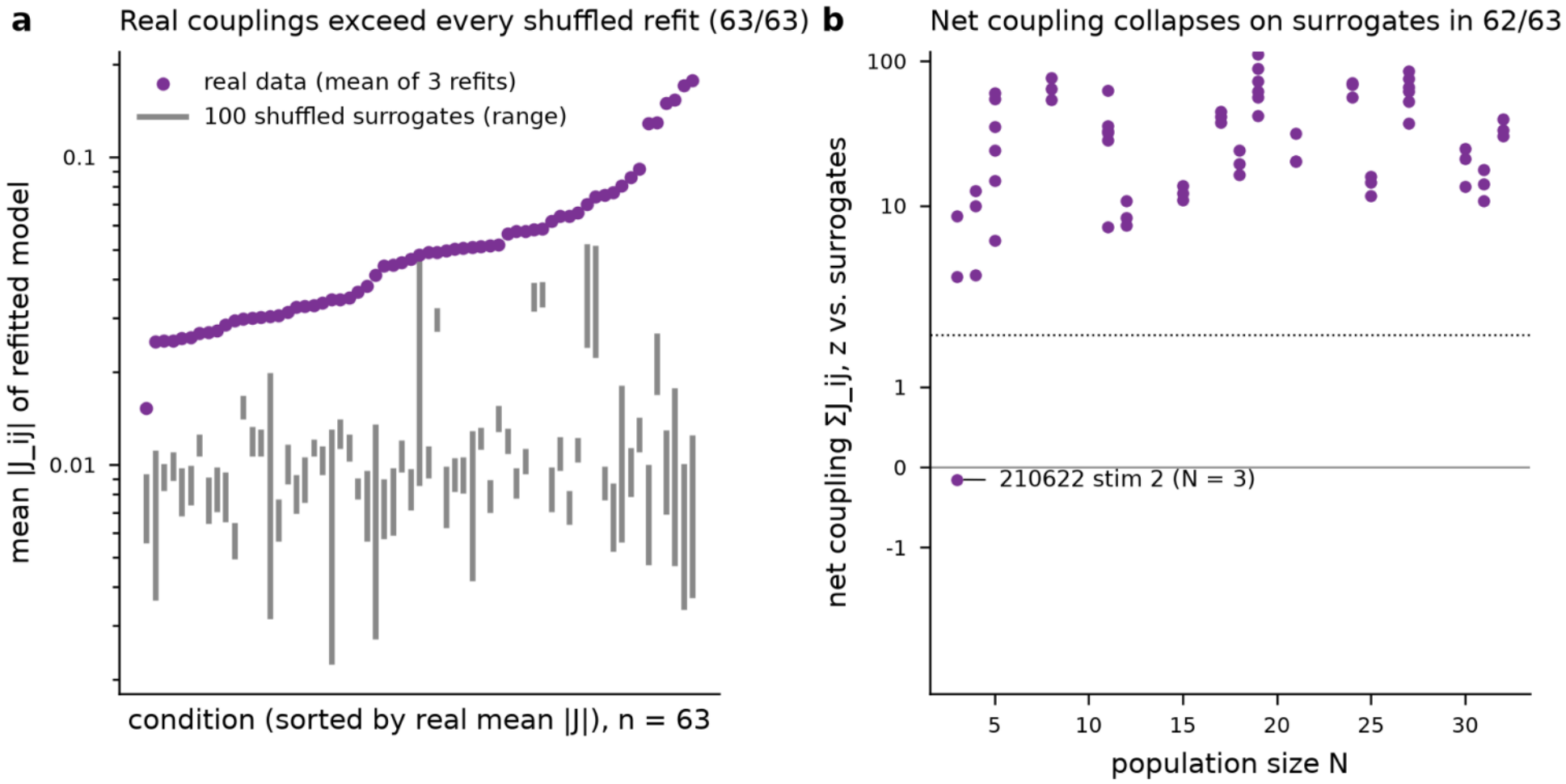
Method validation on SIMULATED data (E09 H0). Source: 09_claude_science_handoff/assets/tables/h0_results.csv.

Shows: SYNTHETIC (E09 H0). Estimated vs true change in multi-information, reach vs rest (bits per cell), fixed-coupling model; colour/marker = N cells x reaches; hollow = scenario C (couplings truly changed). Pattern: Rate-driven scenarios lie on the identity line; scenario C is reported as ~0, because the fixed-coupling measure cannot see a coupling change by design. Why: The key E10 measure is accurate when rates drive the change, and its blind spot is known. Numbers: Sign correct / within 50% of truth: A 20/20 and 20/20, B 20/20 and 20/20 (= the HANDOFF's 40/40); E 18/20 and 17/20; D has a true change of exactly 0 (sign undefined); C 0/20 by design. Redraw of panel a of 5/03.

2_coupling_beyond_rates_at_rest / 01_PK_beyond_independence [established]

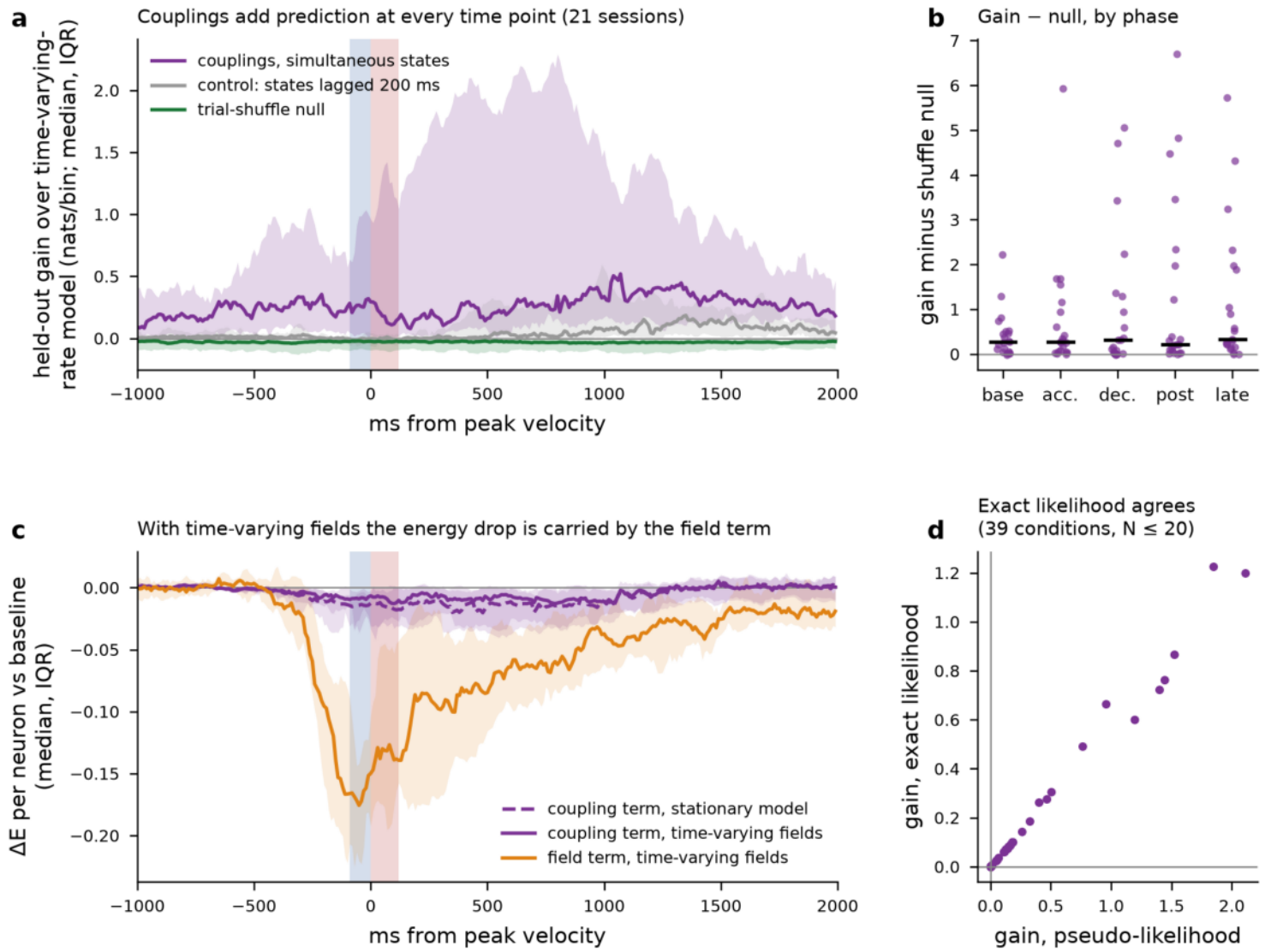


Shows: Left: P(K), probability that K neurons are co-active in a bin, observed vs independent-neuron prediction (210614 stim 0, N = 25). Right: z-score of the real deviation vs the shuffled-null distribution for each of 27 session x stim conditions; dashed line z = 3. Pattern: Observed P(K) is wider than the independent prediction (more synchronous and more co-silent bins); every condition exceeds z = 3. Why: Model-free evidence of collective structure beyond stationary independent firing (A2). Numbers: 27/27 conditions, z = 4.9 to 116 (A2). The brief's '21/26 K values outside the chance band in the example' is not shown in this figure; that number is reproduced in 04_P_K_difference_210614 below.



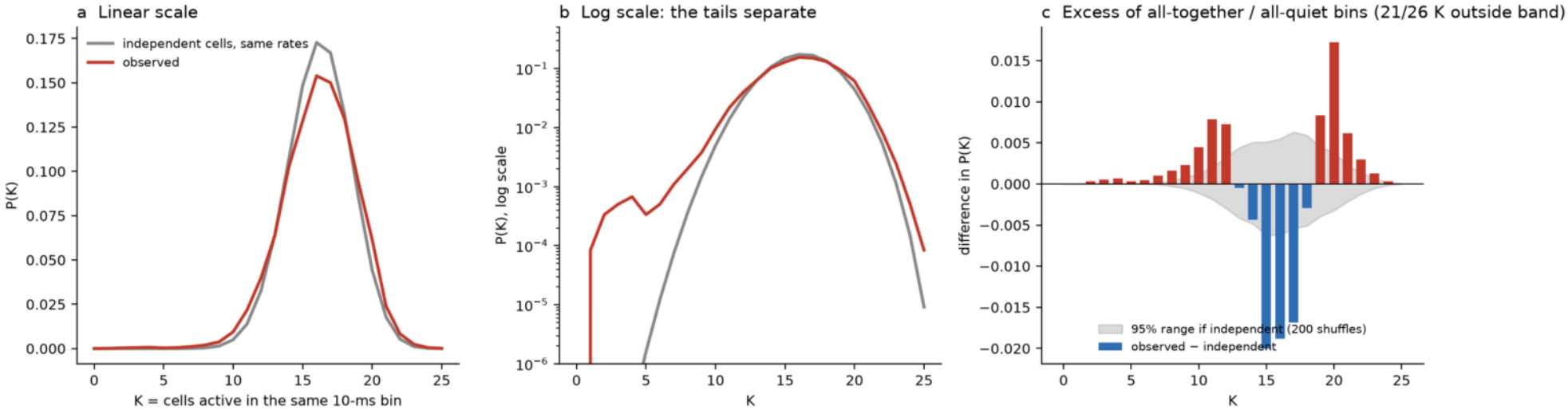
Shows: (a) Mean |J_ij| of the refitted model per condition (sorted), real data (dots) vs range over 100 rate-preserving shuffled surrogates (grey bars), log y. (b) Net coupling sum J_ij as z vs surrogates, vs population size N. Pattern: Real |J| sits above every surrogate in all 63 conditions; net coupling collapses on surrogates in 62/63 (exception 210622 stim 2, N = 3). Why: Fitted couplings are not a fitting artefact (A3). Numbers: 63/63 conditions; median 4.6x, min z = 3.6 (A3).

2_coupling_beyond_rates_at_rest / 03_tvfield_vs_independent [established]



Shows: (a) Held-out gain over the time-varying-rate independent model (nats/bin; median, IQR over 21 sessions) vs ms from peak velocity: couplings with simultaneous states, 200-ms-lagged control, trial-shuffle null. (b) Gain minus shuffle null by phase (baseline, acc., dec., post, late). (c) Energy change per neuron vs baseline: coupling term (stationary and time-varying models) and field term (time-varying). (d) Gain from exact likelihood vs pseudo-likelihood (39 conditions, N <= 20). Pattern: Couplings add prediction at every time point including the pre-reach baseline, with no rise during the reach; the lagged control recovers a small fraction; with time-varying fields the energy dip is in the field term; PL and exact gains agree. Why: Main figure for the coupling question (A7, A8, B2 note): coupling exists beyond shared rates, is present at rest and does not detectably change during the reach. Numbers: Gain over shuffle null 21/21 sessions, median 0.259 nats/bin, p_FDR 8e-6 (tvfield_phase_summary.csv); 63/63 conditions above the null 95th percentile (A7). Baseline 20/21. Lagged control 0.051 vs 0.259 (about 1/5; RESULTS_LOG: 0.05 vs 0.18, about 1/4). Exact LL agrees 38/39. rho = 0.999 PL vs exact: not found in a saved table (unverified; do not cite).

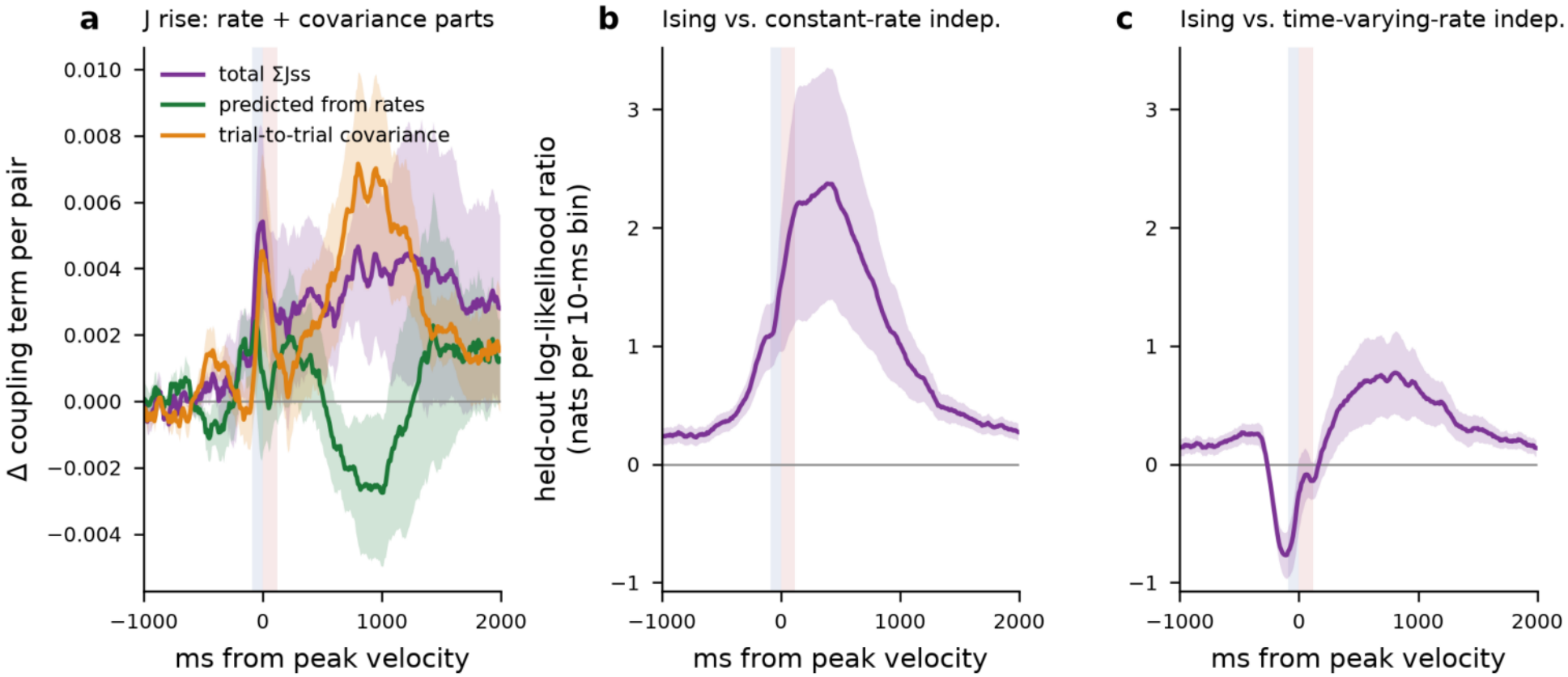
2_coupling_beyond_rates_at_rest / 04_P_K_difference_210614 [established]



Session 210614 baseline block, N = 25 cells, 11,900 bins (exclusions applied). Independent = Poisson-binomial with each cell's rate. Shuffle band: each cell's bins permuted independently. Note: this shuffle also removes reach-locked rate modulation.

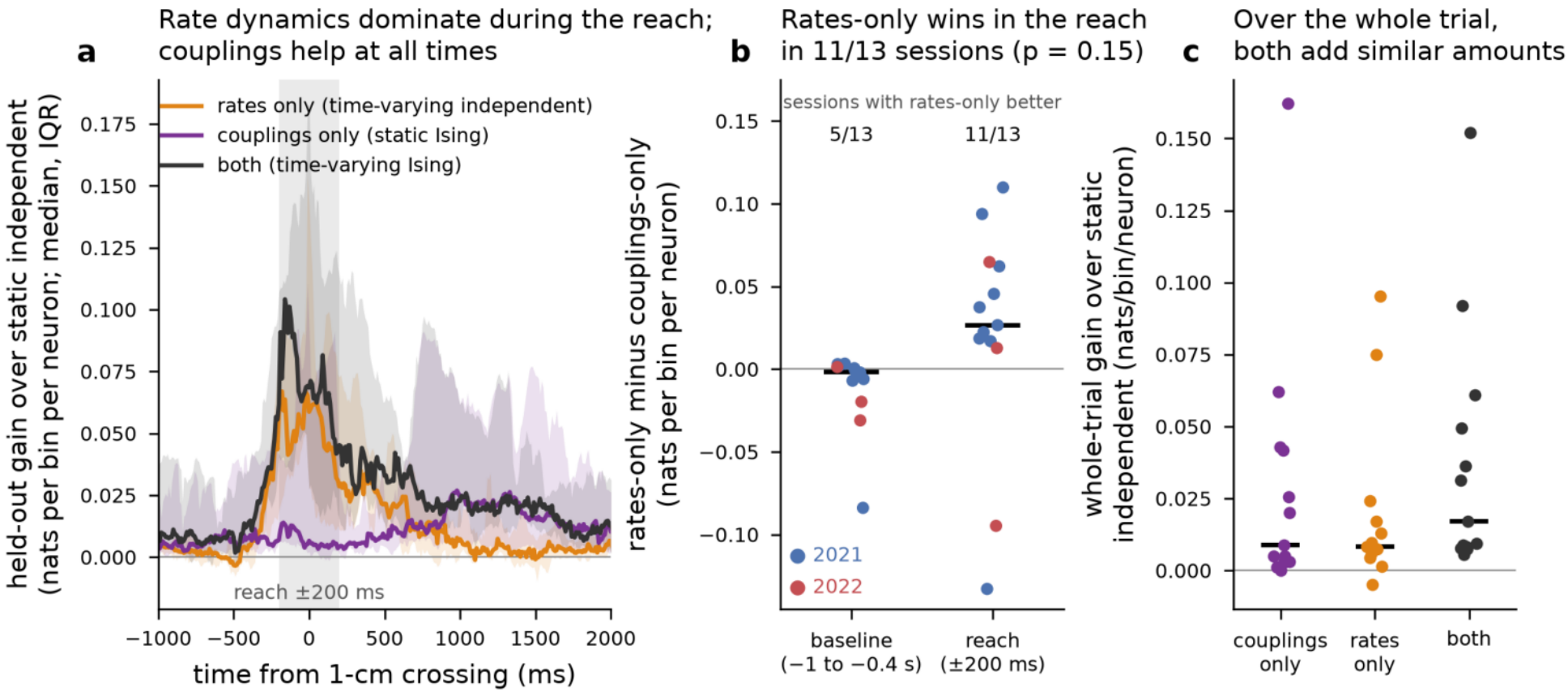
Shows: Session 210614 baseline, N = 25. (a) P(K) observed vs independent cells with the same rates (Poisson-binomial), linear. (b) Same, log scale. (c) Observed minus independent, with the 95% range of 200 within-cell shuffles. Pattern: The data have more all-quiet and more many-together bins than independent cells; the middle K values are depleted. Why: Data-only (no model fitted) picture of collective structure; complements 01_PK_beyond_independence. Numbers: 21/26 K values outside the shuffle band (verified). The shuffle also removes reach-locked rate modulation, so this panel alone does not separate coupling from shared rate changes; 03_tvfield_vs_independent does.

3_rates_vs_coupling_over_reach / 01_coupling_source_checks [established / robust]



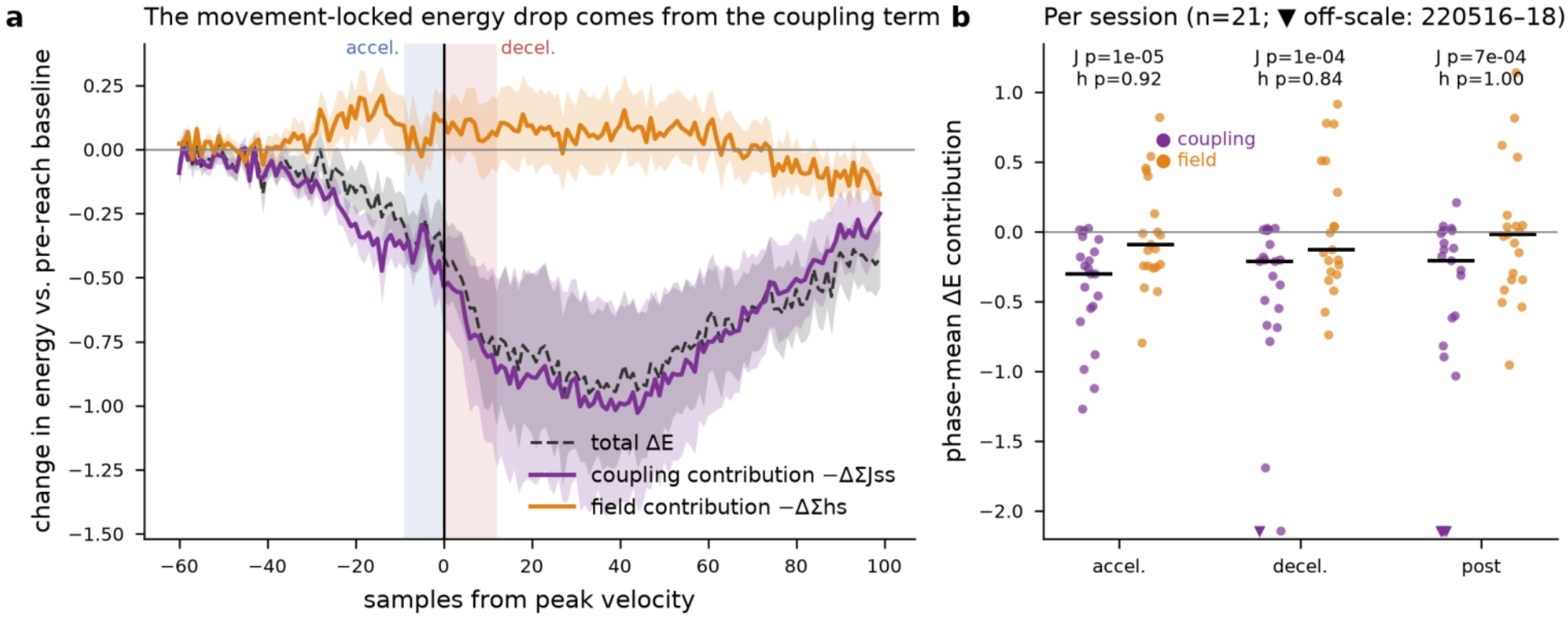
Shows: (a) Change in coupling term per pair vs ms from peak velocity, split into the part predicted from rates and the trial-to-trial covariance part. (b) Held-out LLR (nats per 10-ms bin), Ising vs constant-rate independent. (c) Ising vs time-varying-rate independent. Pattern: Against constant rates Ising gains strongly during the reach; against time-varying rates the advantage dips below zero around acceleration and recovers afterwards. Why: The movement-time Ising advantage of the stationary model comes from reach-locked rate modulation (B10, B11). Numbers: Ising worse during acceleration vs time-varying rates (2/21, p_FDR = 0.004); covariance part median 36% of the J rise (B10, B11).

3_rates_vs_coupling_over_reach / 02_rates_vs_couplings_heldout [established / robust]



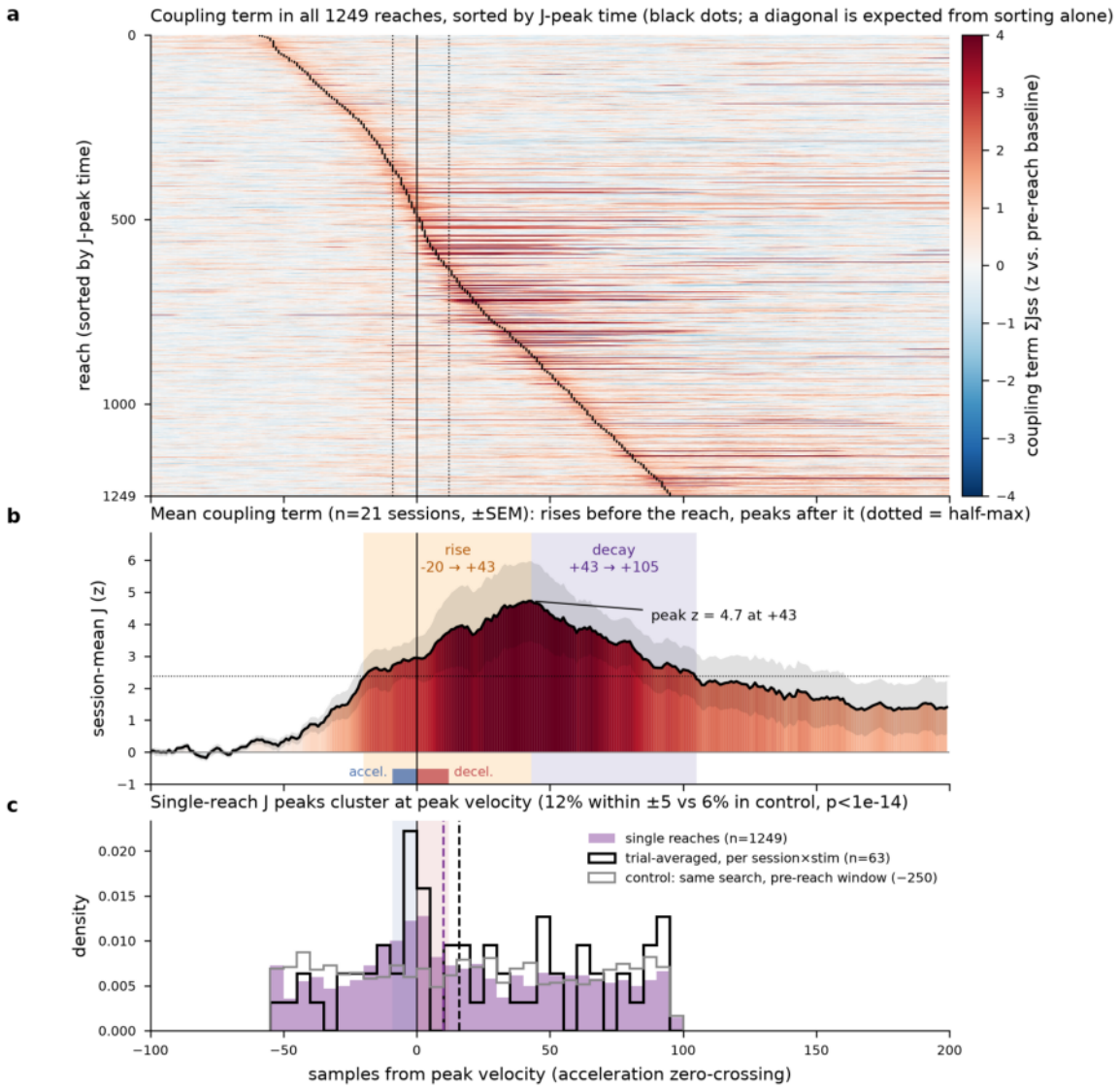
Shows: (a) Held-out gain over static independent model (nats/bin/neuron; median, IQR) vs time from 1-cm crossing for rates-only, couplings-only and both. (b) Rates-only minus couplings-only per session, baseline vs reach (+-200 ms), coloured by cohort. (c) Whole-trial gain per session for each model. Pattern: Rates dominate during the reach; couplings help at all times; the combined model is highest. Why: Shows that rate dynamics, not coupling changes, carry the reach. Numbers: Rates-only wins in the reach in 11/13 sessions; the figure states p = 0.15 (the brief gives no p for 11/13). 'Time-varying Ising beats time-varying independent, 12/13, p = 0.0005' is stated in HANDOFF_CLAUDE_CODE.md section 7 and matches tvfield_phase_summary.csv (exact LL, N <= 20: 13 sessions, p_FDR = 0.0005); it is not drawn in this figure. Only 13 sessions are in this analysis.

3_rates_vs_coupling_over_reach / 03_energy_decomposition_by_phase_STATIONARY [established / robust]



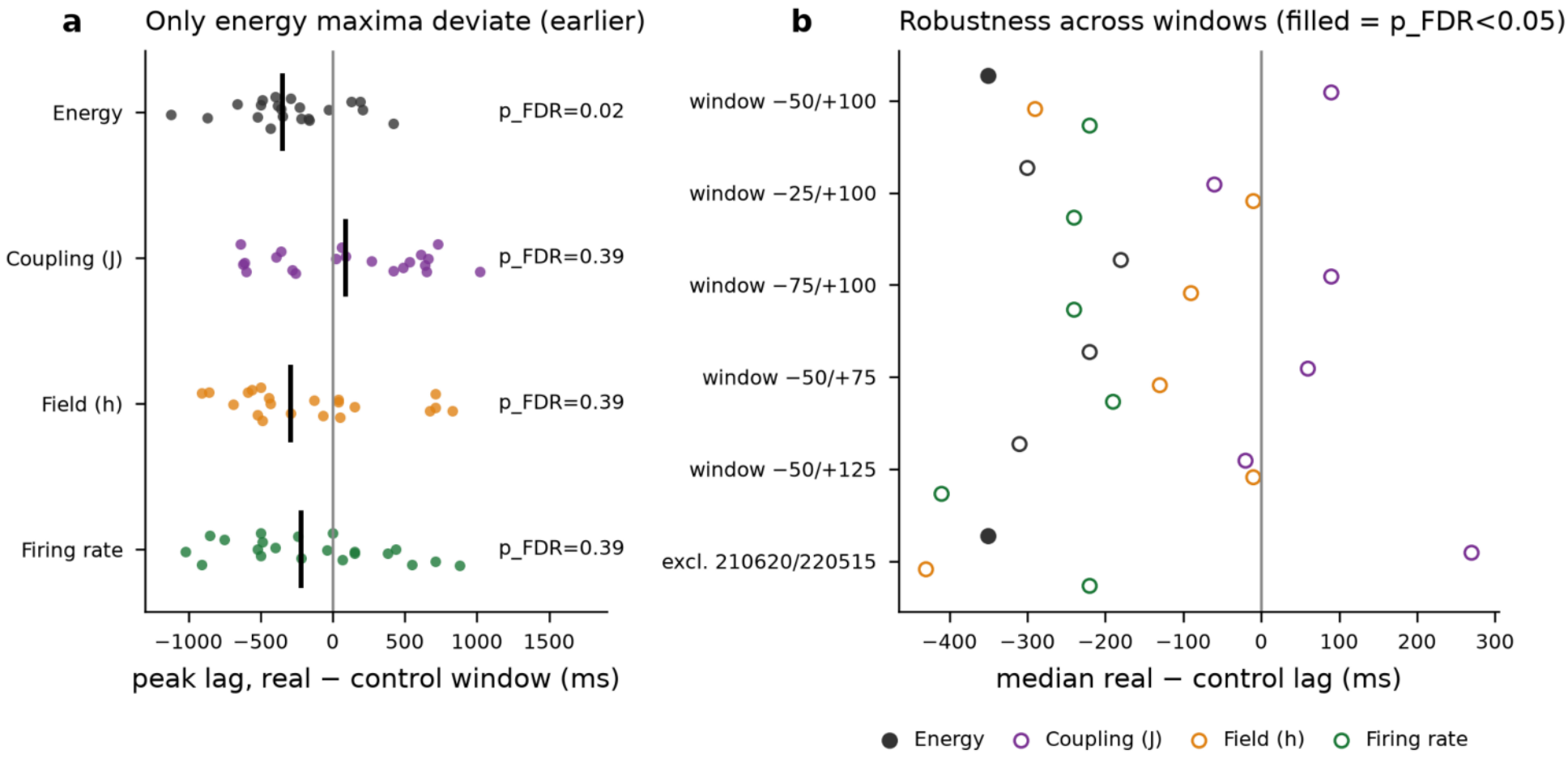
Shows: (a) Change in energy vs pre-reach baseline vs samples (10 ms) from peak velocity: total, coupling contribution, field contribution, STATIONARY model. (b) Per-session phase-mean contributions (accel., decel., post). Pattern: In the stationary model the energy drop is in the coupling term. Why: Shown ONLY with 03_tvfield_vs_independent panel c: with time-varying fields the drop moves to the field term. The coupling-carried drop is a property of the stationary model (B2, model-dependent). Numbers: Stationary coupling: accel. p = 1e-5, decel. 1e-4, post 7e-4; field p >= 0.84. Time-varying model: field term 20/21, p_FDR = 2.3e-5 at acceleration (energy_decomposition_tvfield_summary.csv); in the time-varying model the coupling term changes about 20x less than the field term (about -0.01 vs -0.17 per neuron; RESULTS_LOG). This is NOT a comparison with the stationary model's coupling dip.

3_rates_vs_coupling_over_reach / 04_J_timing_rise_decay_STATIONARY [established / robust]



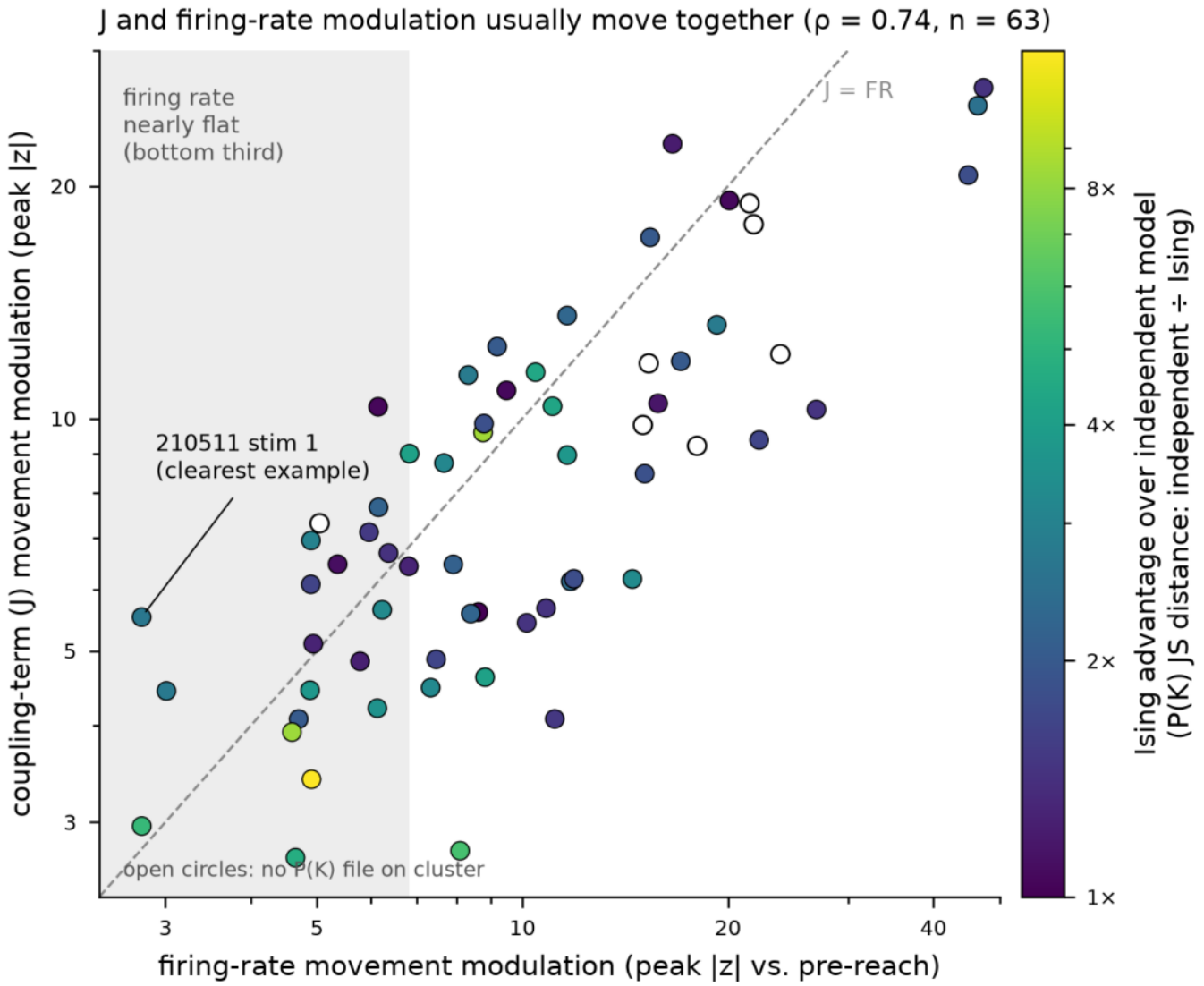
Shows: (a) Coupling term z per reach (1249 reaches) sorted by J-peak time vs samples from peak velocity. (b) Session-mean J (z, +-SEM). (c) Density of single-reach J-peak times vs control window. Pattern: Stationary-model J is a broad elevation spanning the reach (peak z = 4.7 at +430 ms), not a timed peak; single-reach maxima cluster at peak velocity. Why: Context for B3/B7; stationary model, so the rise is largely rate-driven (B10, B11). Numbers: 12.4% of J maxima within +-50 ms vs 6.4% in control, p < 1e-14 (B7).

3_rates_vs_coupling_over_reach / 05_peak_timing_vs_control [established / robust]



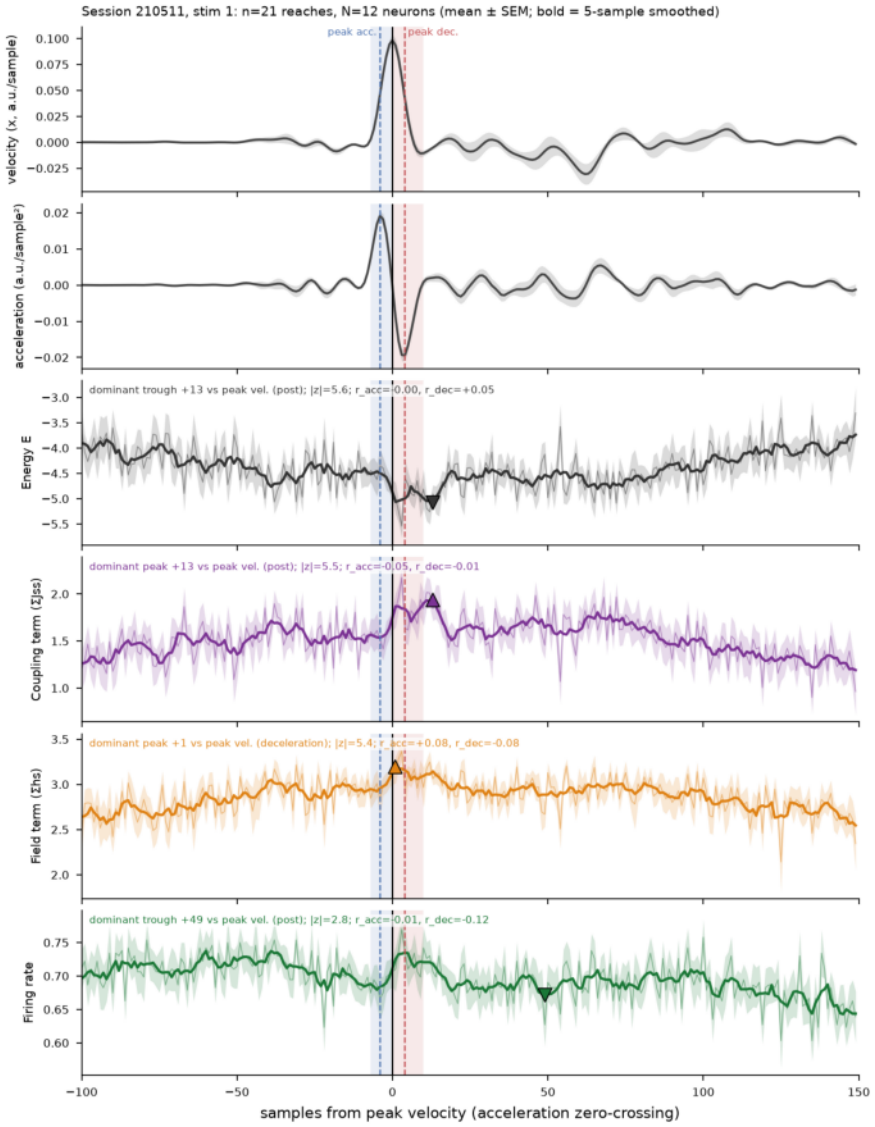
Shows: (a) Per-session peak lag, real minus control window (ms), for energy, J, h, firing rate. (b) Median lag across 6 window/exclusion variants. Pattern: Only energy maxima deviate (earlier), and only in 2 of 6 variants. Why: Timing of J, h and rate maxima matches chance (B12); supersedes earlier timing claims (see X_). Numbers: Energy p_FDR = 0.02; J, h, FR p_FDR = 0.39.

3_rates_vs_coupling_over_reach / 06_flat_FR_vs_J_modulation [established / robust]



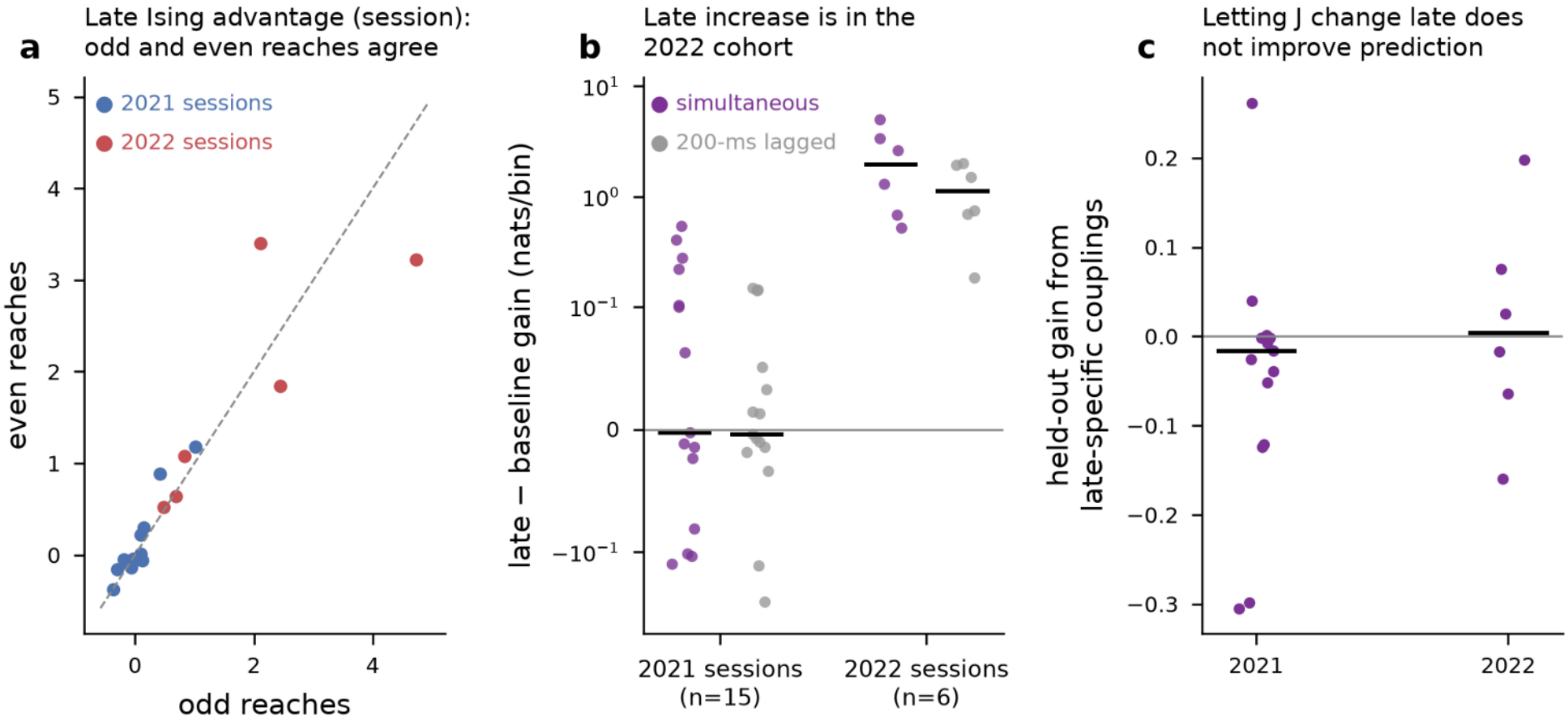
Shows: Coupling-term movement modulation (peak |z|) vs firing-rate movement modulation (peak |z|), log-log, 63 conditions, colour = Ising advantage in P(K). Pattern: J and rate modulation move together; J modulation without rate modulation is rare. Why: B9: coupling changes almost always come with rate changes. Numbers: rho = 0.74, n = 63; one clear example 210511 stim 1.

3_rates_vs_coupling_over_reach / 07_example_flatFR_210511_stim1 [established / robust]



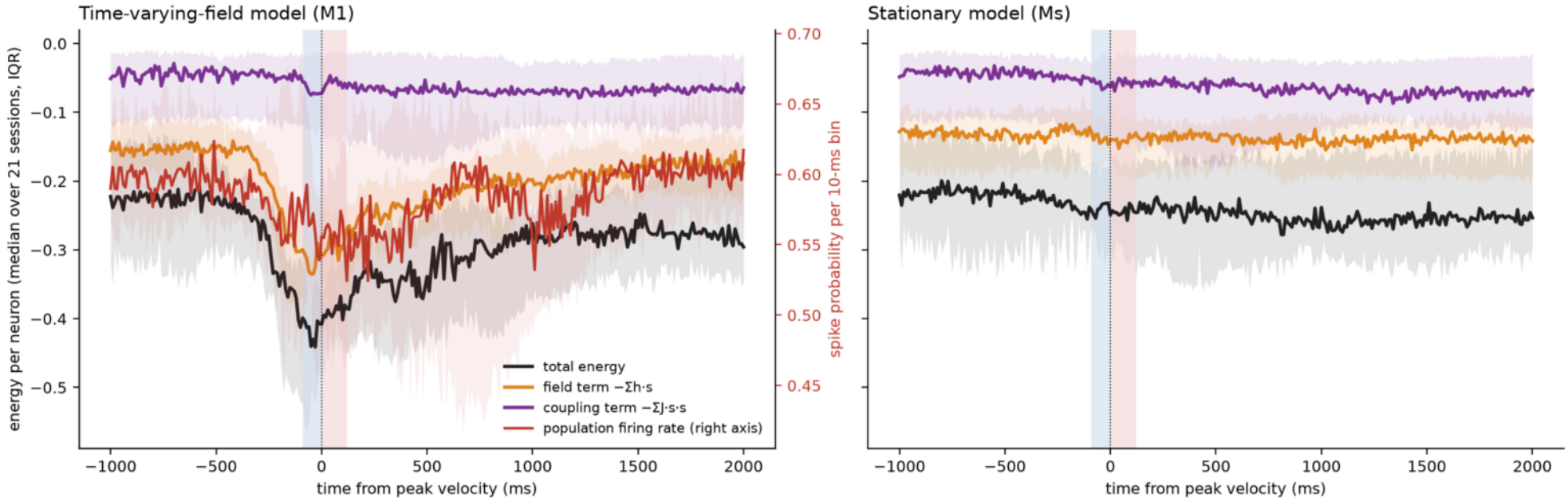
Shows: Example session 210511 stim 1: kinematics, firing rate and Ising terms vs time from peak velocity (stationary model). Pattern: The clearest case of J modulation with a nearly flat mean rate. Why: Single example for B9; note 210511 contains a neuron firing in 99% of bins (see 1/02 panel b). Numbers: Example only.

3_rates_vs_coupling_over_reach / 08_late_epoch_consistency_SUGGESTIVE [established / robust]



Shows: (a) Late Ising advantage, odd vs even reaches per session. (b) Late minus baseline gain (nats/bin), simultaneous vs 200-ms-lagged, by cohort. (c) Held-out gain from late-specific couplings. Pattern: Late (+0.5 to +1.5 s) increase only in the 2022 cohort; lagged control also rises; epoch-specific J does not help. Why: SUGGESTIVE ONLY (log section C). Shown so the late epoch is not over-read. Numbers: 13/21, p_FDR = 0.023, carried by 2022 (6/6); 2021 7/15.

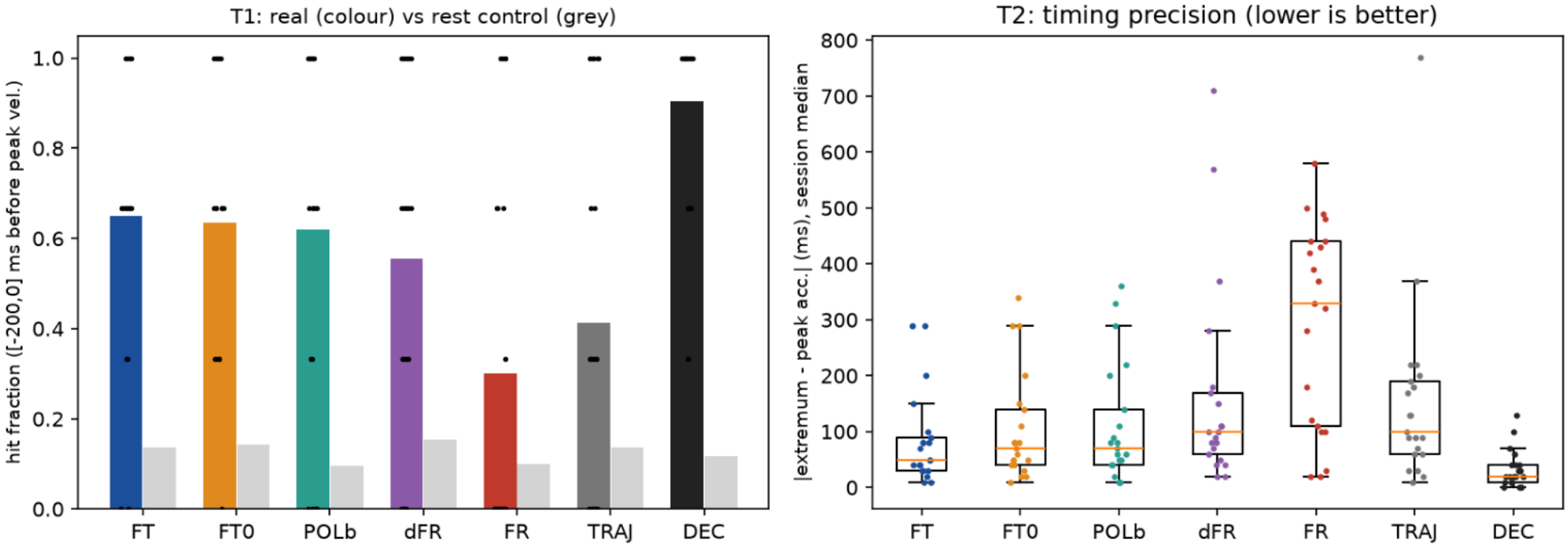
3_rates_vs_coupling_over_reach / 09_energy_absolute_with_firing_rate [established / robust]



Absolute energy (not baseline-subtracted); lower = more probable pattern. Pseudo-likelihood fits, exclusions applied; blocks averaged within session. Blue/red = acceleration/deceleration.

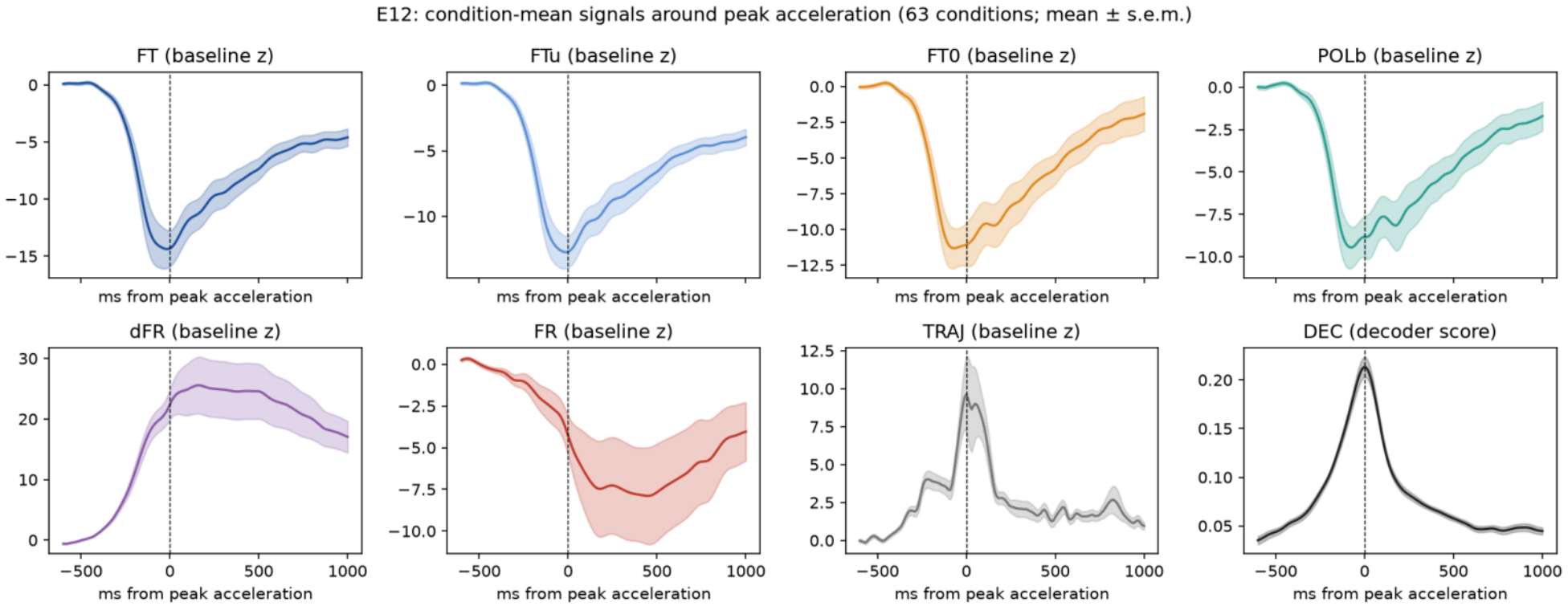
Shows: Absolute energy per neuron (not baseline-subtracted) vs ms from peak velocity, median and IQR over 21 sessions: total, field term and coupling term, for the time-varying-field model (left, with population firing rate on the right axis) and the stationary model (right). Blue/red = acceleration/deceleration. Pattern: With time-varying fields, energy drops sharply just before peak velocity and the drop is in the field term; the coupling term barely moves. The stationary model shows almost no reach-locked structure. Why: Shows the raw levels behind the change-from-baseline plots: the reach is carried by the fields (rates), with a nearly constant coupling contribution. Numbers: M1 field term -0.15 at baseline to -0.335 (minimum at -50 ms); total -0.23 to -0.44 (-30 ms); coupling term about -0.04 to -0.08 with no comparable dip. Population spike probability per 10-ms bin 0.60 at baseline, minimum 0.52 at +120 ms. Exploratory (pseudo-likelihood fits).

4_polarization_at_peak_acceleration / 01_E12_T1_T2_by_signal [robust (E12 preregistered; not unique to the Ising model)]



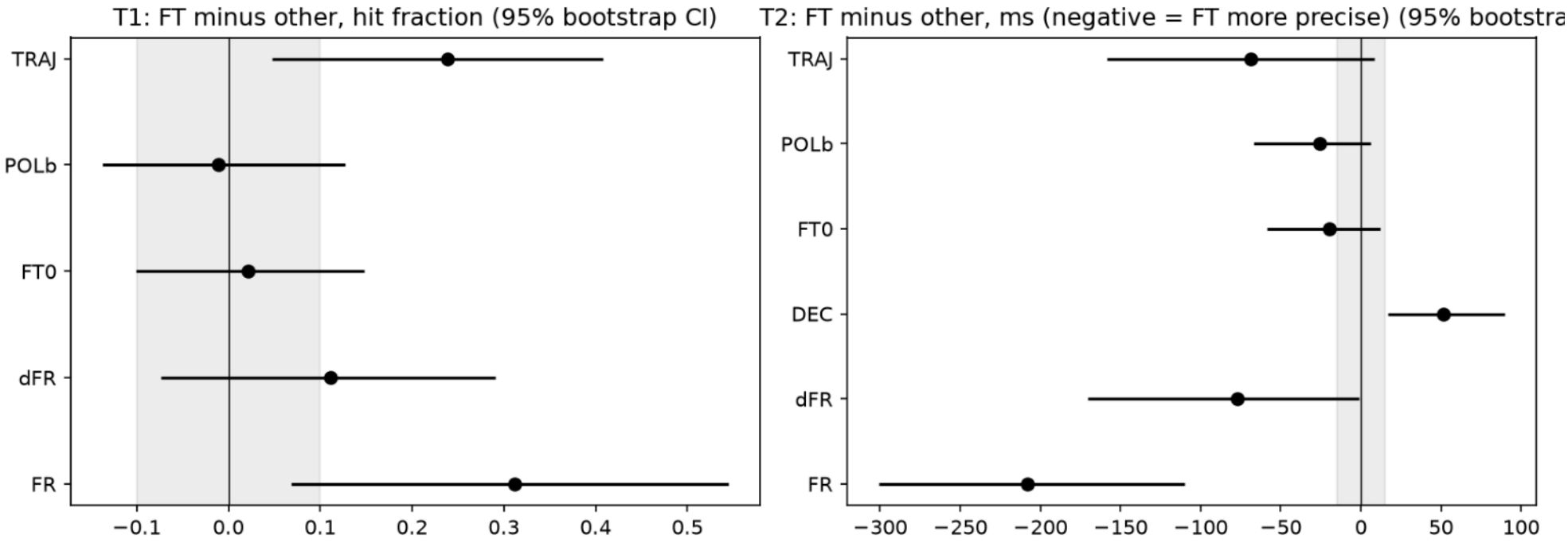
Shows: Left (T1): fraction of conditions whose trough falls in [-200, 0] ms before peak velocity, real (colour) vs rest control (grey), per signal (FT field term, FT0 uncoupled field term, POLb polarization index, dFR, FR mean rate, TRAJ PCA speed, DEC supervised decoder). Right (T2): |extremum - peak acceleration| (ms), session medians. Pattern: FT, FT0 and POLb hit far above rest controls and are close to peak acceleration; FR is late and imprecise; DEC is best but supervised. Why: Preregistered E12 test of timing (B13). Numbers: Hit rate FT 0.65 vs 0.14 rest; FR 0.30. Median error FT 50 ms, FT0 70, POLb 70, FR 330; FT vs FR -208 ms, p_BH = 0.024; hit-rate difference p_BH = 0.078 (not FDR significant).

4_polarization_at_peak_acceleration / 02_E12_signals_aligned [robust (E12 preregistered; not unique to the Ising model)]



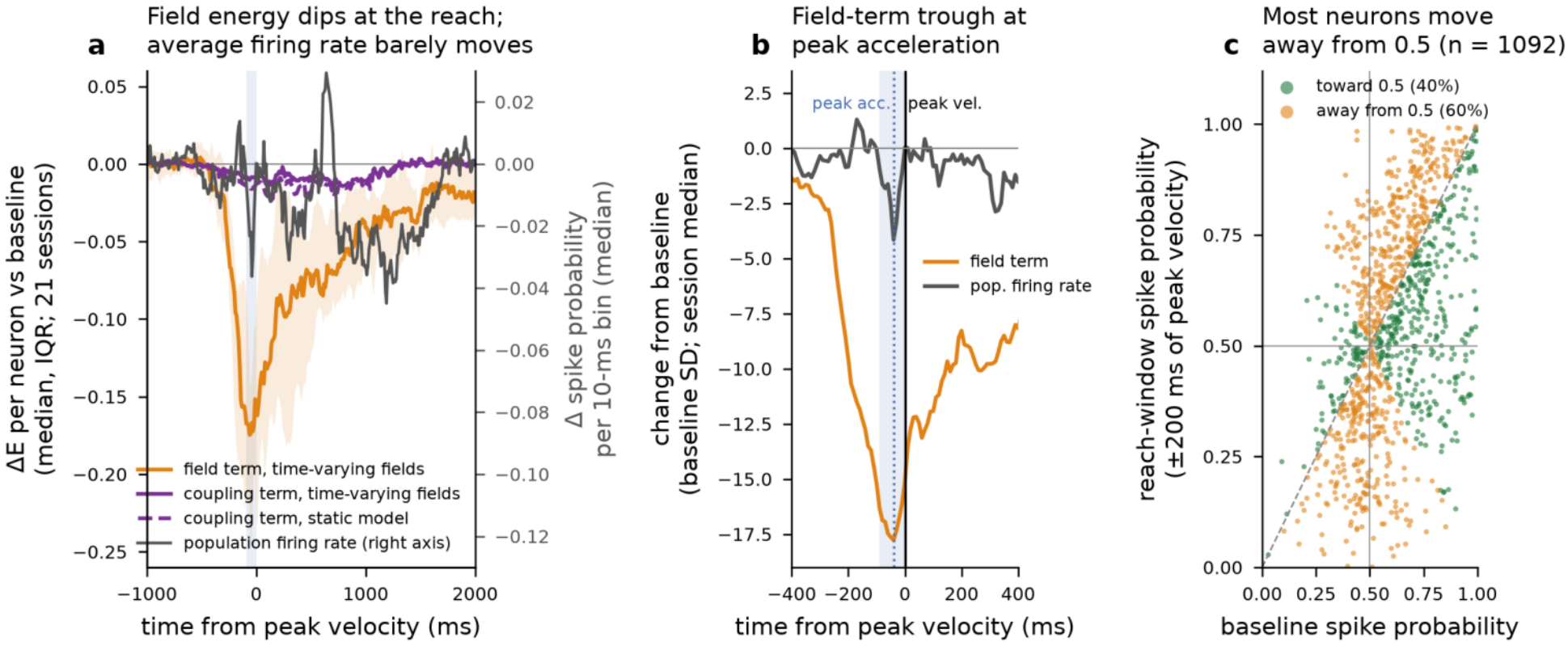
Shows: Condition-mean signals (baseline z, or decoder score) vs ms from peak acceleration, 63 conditions, mean +- s.e.m. Pattern: FT, FTu, FT0 and POLb all trough near peak acceleration; FR falls later and broadly; dFR rises. Why: Shows the shapes behind T1/T2. Numbers: Do not read the trough DEPTH as evidence (withdrawn; basis inflation, E12 R6-R7).

4_polarization_at_peak_acceleration / 03_E12_primary_contrasts_forest [robust (E12 preregistered; not unique to the Ising model)]



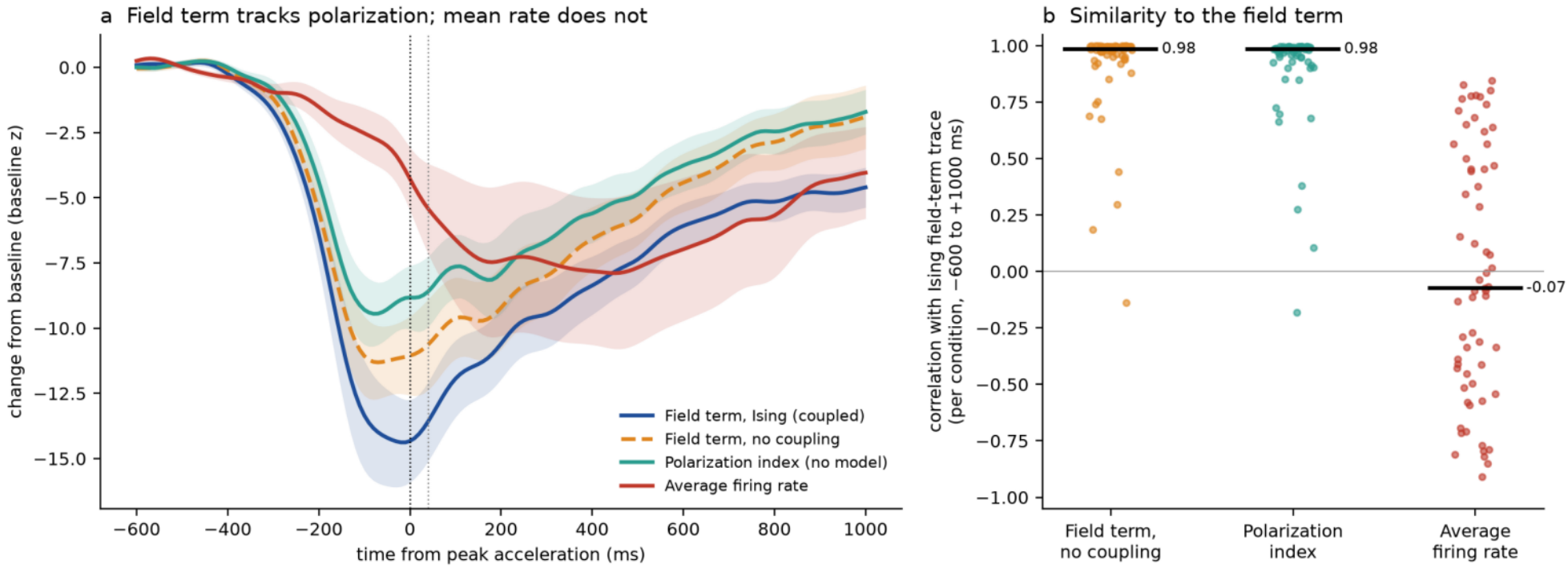
Shows: FT minus each other signal with 95% bootstrap CIs: T1 hit fraction (left) and T2 ms (right); grey = equivalence margins. Pattern: FT beats FR clearly on T2; FT vs FT0 and FT vs POLb straddle zero. Why: The dip is not unique to the coupled model: no detectable advantage over FT0 or the model-free index (equivalence not shown). Numbers: FT vs FT0: T1 +0.02 (CI -0.10 to +0.15), T2 -20 ms (CI -59 to +12). FT vs POLb: T2 -26 ms (CI -67 to +6). Trace correlations (r = 0.98 field term vs polarization index; r = -0.07 vs mean rate) are reproduced in 05_field_term_vs_polarization_vs_rate below (not part of E12 RESULTS.md).

4_polarization_at_peak_acceleration / 04_field_vs_firing_rate_polarization_CAVEAT [robust (E12 preregistered; not unique to the Ising model)]



Shows: (a) Energy change per neuron (field and coupling terms) and population firing-rate change vs time from peak velocity. (b) Field term and population rate in baseline SD vs time from peak velocity. (c) Reach-window spike probability vs baseline spike probability, 1092 neurons, coloured toward/away from 0.5. Pattern: Field term troughs at peak acceleration; 60% of neurons move away from p = 0.5 during the reach. Why: Panel c illustrates polarization. CAVEAT: panel a's title ('average firing rate barely moves') and panel b's baseline-SD depth are WITHDRAWN claims (log D: the mean rate does dip, and the depth is basis-inflated). Cite panel c only. Numbers: 60% of 1092 neurons away from 0.5. E10 H1: change in p(1-p) negative in 19/21 sessions (median -0.027, p = 7e-6; e10_tests.csv H1_dv).

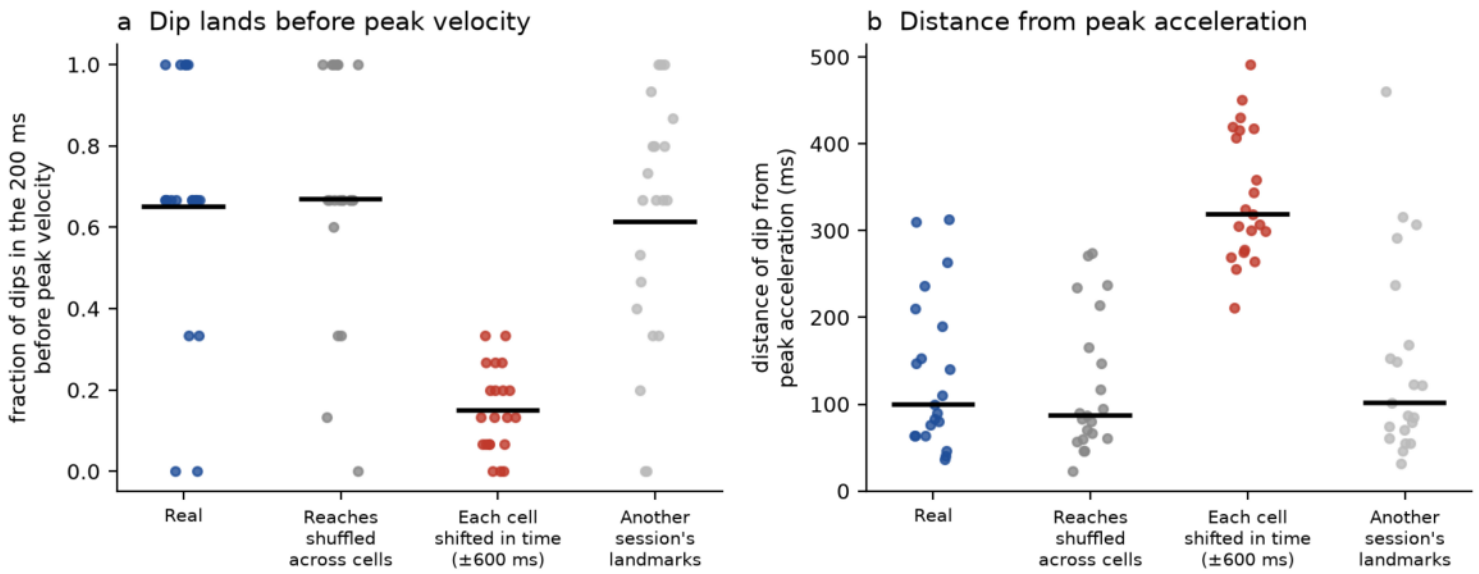
4_polarization_at_peak_acceleration / 05_field_term_vs_polarization_vs_rate [robust (E12 preregistered; not unique to the Ising model)]



63 conditions, exclusions applied; baseline-z traces aligned to each condition's peak acceleration (mean ± s.e.m.). EXPLORATORY (E12 pipeline).

Shows: (a) Baseline-z traces vs ms from peak acceleration, 63 conditions, mean +- s.e.m.: Ising field term, field term without coupling, polarization index (no model), average firing rate. (b) Per-condition correlation of each trace with the Ising field term (-600 to +1000 ms). Pattern: Field term, uncoupled field term and polarization index are almost the same curve; the mean rate has a different shape. Why: The field-term dip is a polarization signal, not a mean-rate signal, and it does not need the coupled model. Numbers: Median r with the Ising field term: uncoupled field term 0.98, polarization index 0.98, mean rate -0.07. EXPLORATORY.

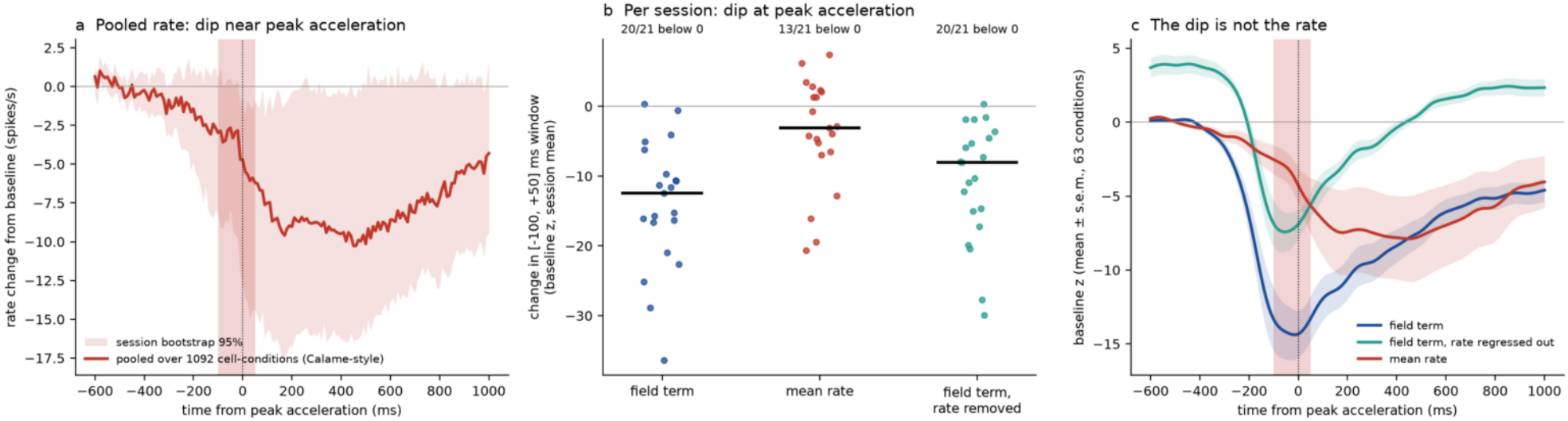
4_polarization_at_peak_acceleration / 06_fielddip_shuffle_controls [robust (E12 preregistered; not unique to the Ising model)]



EXPLORATORY. Field-term dip timing (E12 search) for real data and 5 shuffles of each type per block; one dot per session. Landmark shuffle is weak because peak acceleration sits at nearly the same bin in every session.

Shows: (a) Fraction of field-term dips (E12 search window) in the 200 ms before peak velocity and (b) their distance from peak acceleration, for real data and three shuffles (5 each per block); one dot per session. Pattern: Shifting each cell in time destroys the timing; shuffling reaches across cells (each cell keeps its average time course) does not; using another session's landmarks barely changes anything. Why: The dip timing comes from each cell's reach-locked time course, not from reach-by-reach co-variation between cells. The landmark shuffle is a weak control because peak acceleration sits at nearly the same bin in every session. Numbers: Hit rate real 0.65, trial shuffle 0.67 (p = 0.60), per-cell time shift 0.15 (p = 1e-4), landmark shuffle 0.61 (p = 0.40). Median distance 100 / 87 / 319 / 101 ms. Burster/pauser table: the rate extremum matching the cell type is not detectably closer to the field-term trough than the other extremum (280 vs 420 ms, p = 0.14; 24 burster and 39 pauser conditions). EXPLORATORY.

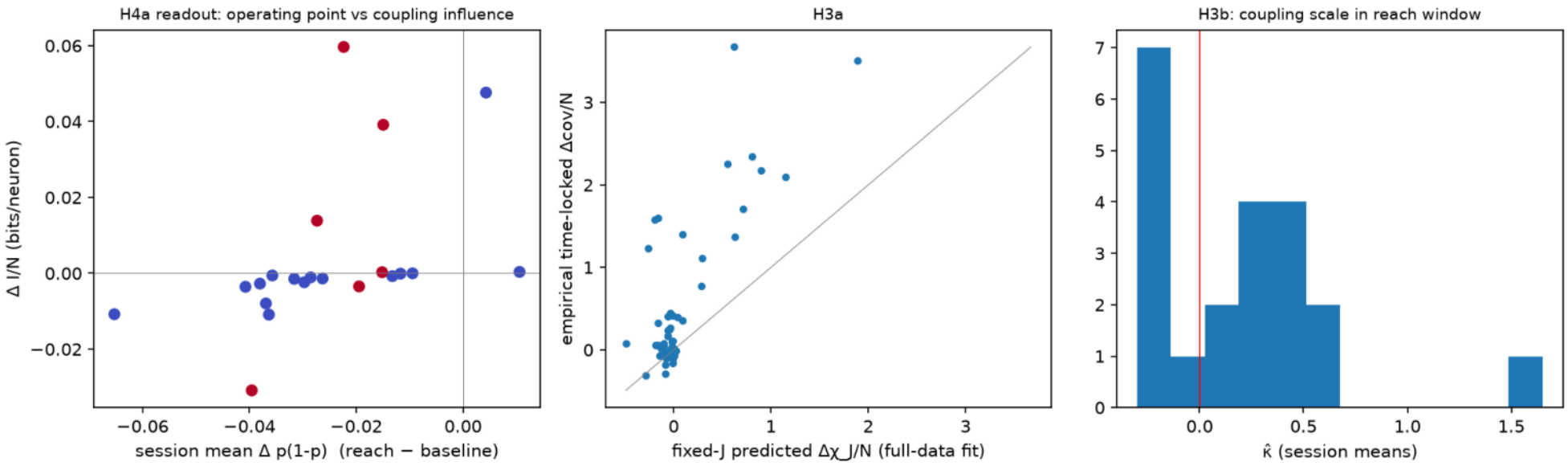
4_polarization_at_peak_acceleration / 07_calame_reconciliation [robust (E12 preregistered; not unique to the Ising model)]



EXPLORATORY. Pooled rate = mean over all cells of all 63 conditions (PSTH minus each cell's baseline, -1000..-410 ms from peak velocity); not velocity-binned. Field term and rate traces: E12 pipeline (regen). Shaded = [-100, +50] ms test window.

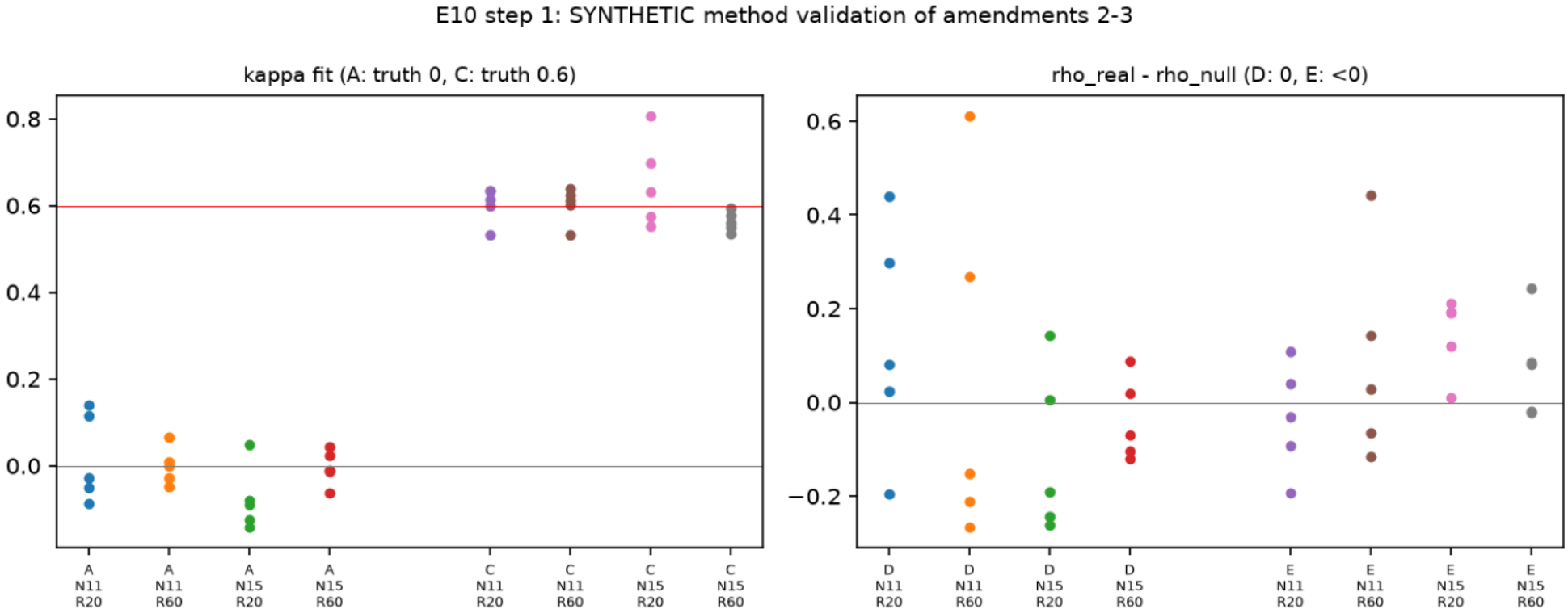
Shows: (a) Rate change (spikes/s) pooled over all 1,092 cell x condition PSTHs (Calame-style), vs ms from peak acceleration, session-bootstrap band. (b) Per session: change in the [-100, +50] ms window (baseline z) for the field term, the mean rate, and the field term with the rate regressed out. (c) Mean traces of the three signals. Pattern: Pooled rate steps down sharply at peak acceleration (Calame et al. reproduced) inside a broader decline that bottoms ~450 ms later. Per session the field-term dip is consistent; the rate change is not; regressing out the rate leaves the dip intact. Why: Reconciles P3 with Calame et al. 2023: the timing of a population event at peak acceleration is theirs; the per-session, pattern-level signal not explained by the rate is ours. Numbers: Field term dips 20/21 sessions (p = 2e-6); mean rate 13/21 (p = 0.055); field term with rate regressed out 20/21 (p = 2e-6). Pooled rate -4.2 Hz in the window, trough -10.3 Hz at +460 ms. Not velocity-binned. Use counts, not z depths (depth is basis-inflated, withdrawn). EXPLORATORY.

5_fixed_coupling_influence_E10 / 01_e10_h2_h4 [preregistered-null (REFUTED / NOT SUPPORTED overall)]



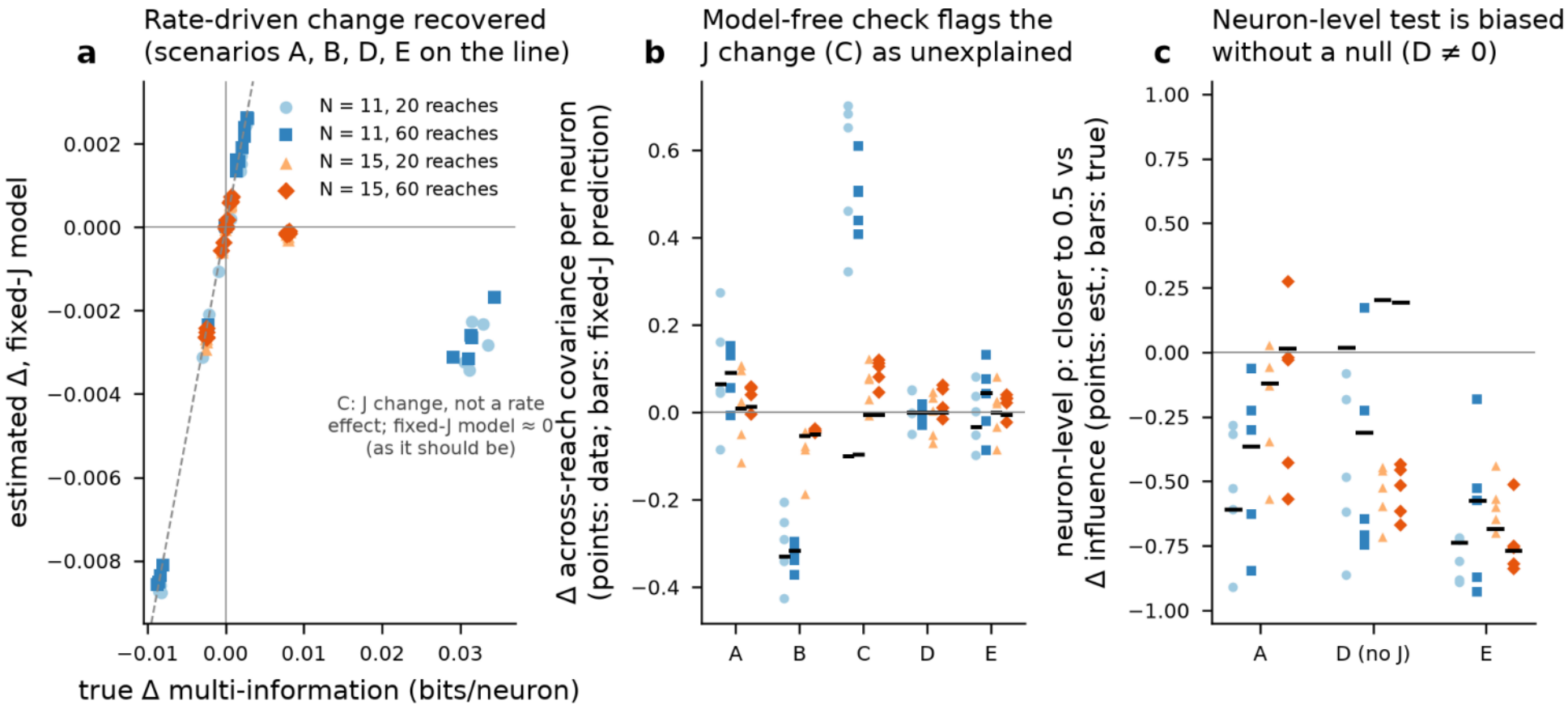
Shows: Left (H4a readout): session dI/N (bits/neuron) vs session mean change in p(1-p) (reach - baseline); red = 2022 cohort, blue = 2021. Middle (H3a): empirical time-locked dcov/N vs fixed-J predicted dchi_J/N (63 conditions). Right (H3b): histogram of session-mean coupling scale kappa-hat. Pattern: dI/N clusters near zero and is mostly slightly negative (2021); the 2022 sessions scatter positive. Predicted and empirical covariance changes correlate weakly; kappa-hat is mostly positive but wide. Why: The preregistered H2 test fails, H3a fails on sign agreement, so the fixed-coupling operating-point account is not supported. Numbers: H2 dI/N median -0.0011, 15/21 negative, p = 0.29; 2021 alone -0.0014, 13/15 negative, p = 0.015; 2022 +0.007 (6, p = 0.44). H3a rho = 0.34, sign agreement 13/21 (p = 0.38). H3b kappa median +0.19, 13/21, p = 0.055 (verdict: no change; but 2022 alone 6/6, p = 0.031). H4a rho = 0.70 (near-mechanical). Convergence: 2022 real fits not converged 54/99 tasks; 2021 19/180.

5_fixed_coupling_influence_E10 / 02_e10_step1_synthetic_amendments [preregistered-null (REFUTED / NOT SUPPORTED overall)]



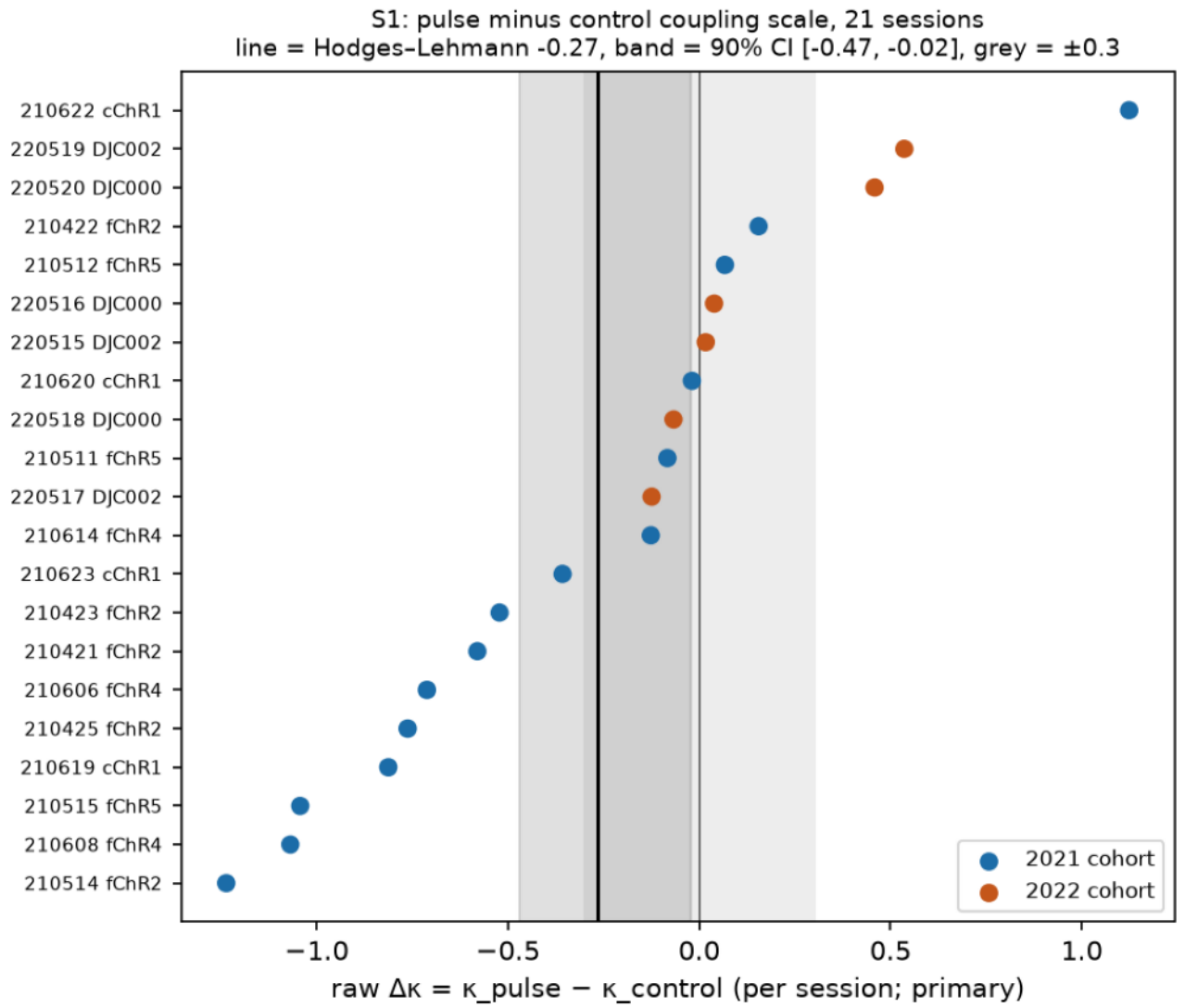
Shows: SYNTHETIC. Left: kappa fit in scenarios A (truth 0) and C (truth 0.6) by N and reaches. Right: rho_real - rho_null in scenarios D (truth 0) and E (< 0). Pattern: kappa recovers 0 and 0.6; rho_real - rho_null cannot detect scenario E. Why: Method validation; shows why the neuron-level test H4b was demoted to exploratory. Numbers: Synthetic only.

5_fixed_coupling_influence_E10 / 03_h0_synthetic_validation [preregistered-null (REFUTED / NOT SUPPORTED overall)]



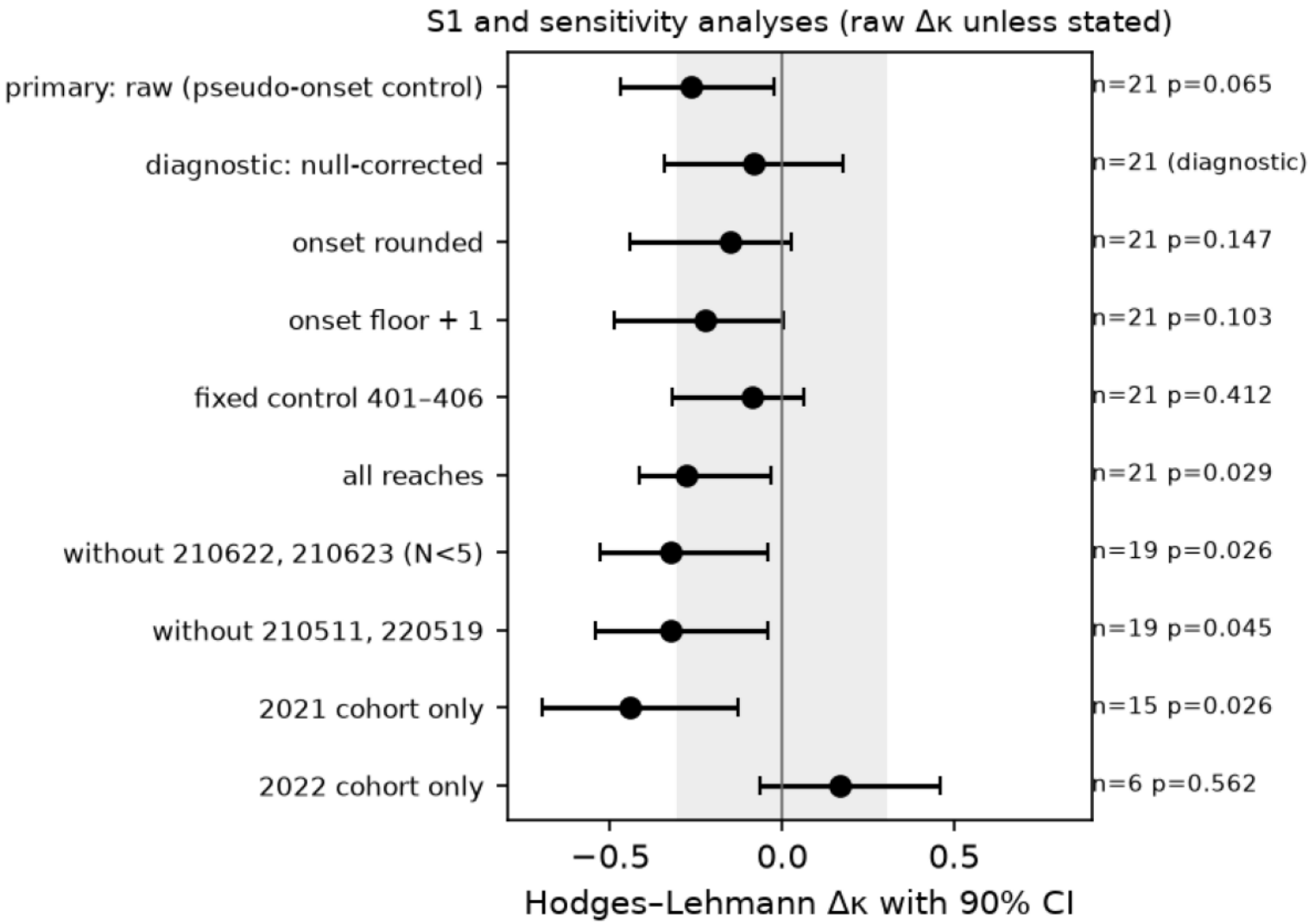
Shows: SYNTHETIC. (a) Estimated change from the fixed-J model vs true change in multi-information (bits/neuron), scenarios A-E. (b) Change in across-reach covariance per neuron (data points vs fixed-J prediction bars). (c) Neuron-level rho (closer to 0.5 vs change in influence). Pattern: Rate-driven changes are recovered on the identity line; a true J change (C) is flagged as unexplained; the neuron-level test is biased without a null. Why: Validation of the E10 measures before the real-data test. Numbers: Synthetic only. Labelled h0_synthetic_validation in 09_claude_science_handoff (panel a is redrawn with per-scenario counts as 1/05).

S1_stimulation_E11 / 01_E11_S1_dkappa_per_session [inconclusive (preregistered, refit under amendments A3-A5)]



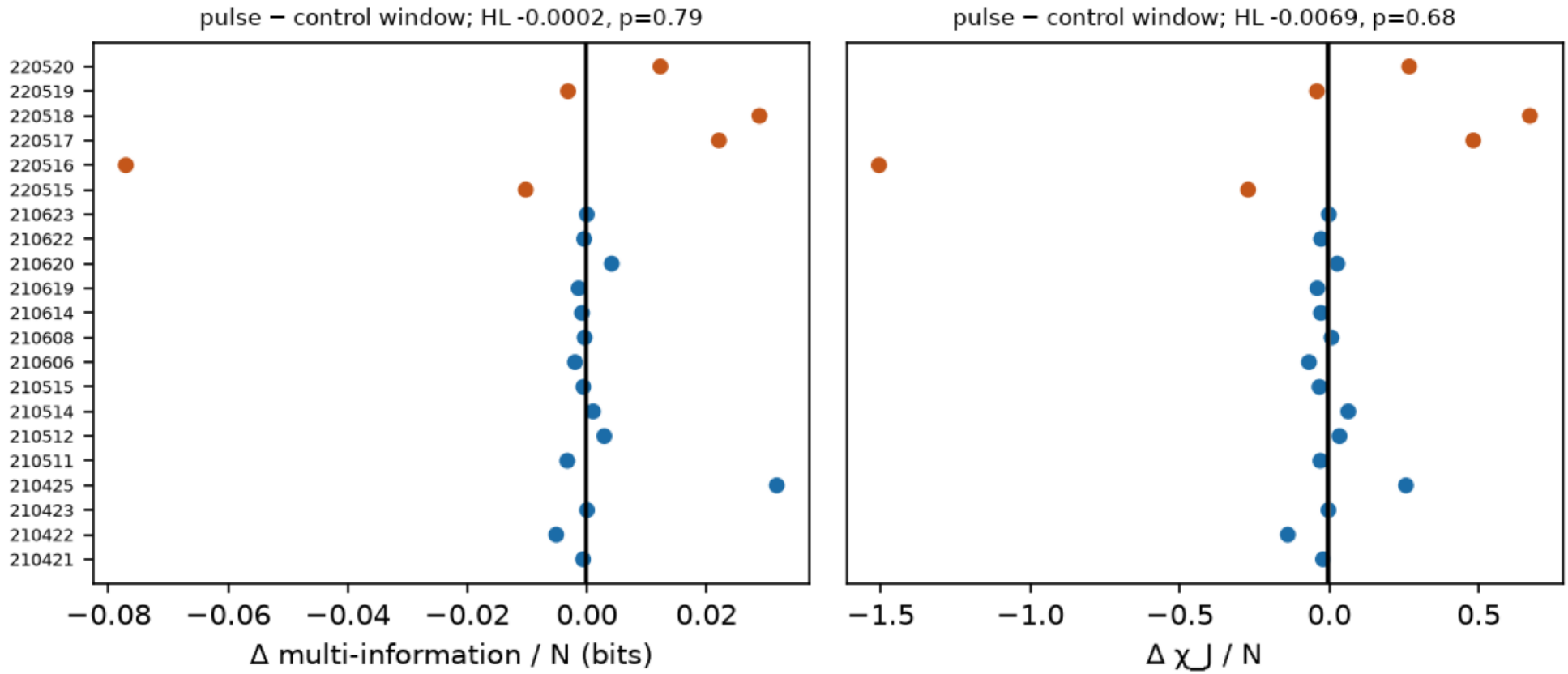
Shows: Raw dkappa = kappa_pulse - kappa_control per session (21 sessions, coloured by cohort); line = Hodges-Lehmann estimate, dark band = 90% CI, grey = +-0.3 equivalence band. Pattern: 14/21 sessions negative; CI excludes 0 but extends beyond -0.3. Why: Primary S1 result: outcome (c) inconclusive. Numbers: HL -0.266, 90% CI [-0.469, -0.022]; Wilcoxon p = 0.065, p_FDR = 0.13.

S1_stimulation_E11 / 02_E11_S1_sensitivity [inconclusive (preregistered, refit under amendments A3-A5)]



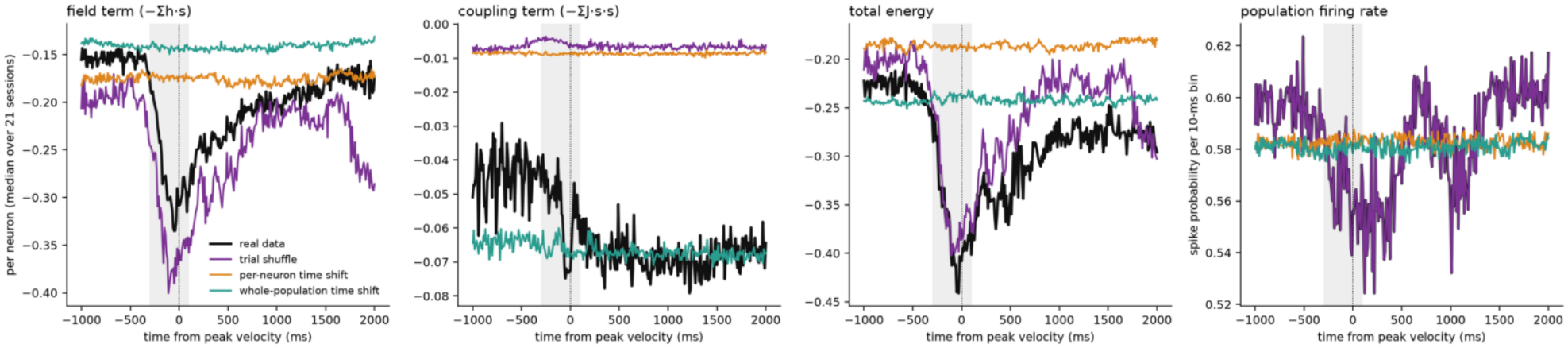
Shows: Hodges-Lehmann dkappa with 90% CI for the primary analysis and sensitivity variants (control window, onset rounding, exclusions, cohorts). Pattern: All variants lean negative except the 2022 cohort; the fixed control is near zero. Why: The lean depends on the control window and is carried by 2021. Numbers: 2021 -0.44 (n = 15, p = 0.026 nominal); 2022 +0.17 (n = 6, p = 0.56); fixed control -0.09, p = 0.41.

S1_stimulation_E11 / 03_E11_S2_multiinformation [inconclusive (preregistered, refit under amendments A3-A5)]



Shows: Per-session pulse - control change in multi-information / N (bits; left) and chi_J / N (right). Pattern: Centred on zero; outliers in 2022 (220516). Why: S2: no change in coupling influence in the pulse window. Numbers: dI/N HL -0.0002, p = 0.79; dchi_J/N HL -0.007, p = 0.68.

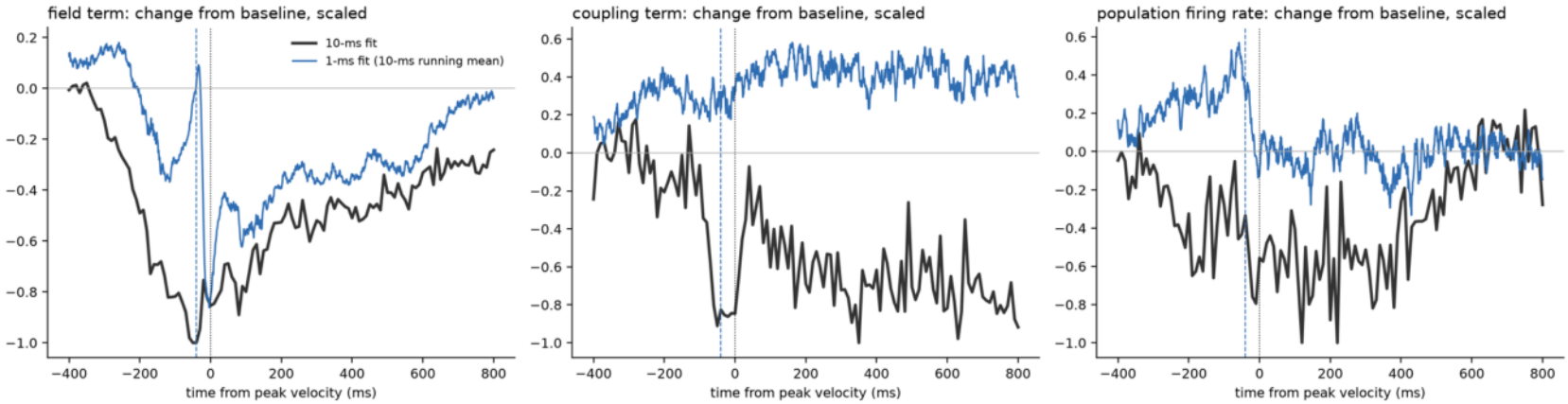
S2_shuffle_controls_energy_timeseries / 01_energy_timeseries_vs_shuffles [exploratory (shuffle controls requested by the PI; not preregistered)]



EXPLORATORY. M1 refit on each shuffle (20 per type per block); shuffle curves = mean over shuffles, median over sessions. In the FR panel the real curve lies under the trial-shuffle curve (identical by construction).

Shows: Field term, coupling term, total energy (per neuron) and population firing rate vs ms from peak velocity: real data and the time-varying-field model refitted on 20 shuffles of each type per block (trial shuffle = reach identity permuted per cell; per-neuron time shift; whole-population time shift). Median over 21 sessions. Pattern: Time shifts flatten the dip and the rate change; the trial shuffle keeps the dip (each cell's average time course is intact) but drives the coupling term to ~0; the population shift keeps the coupling level. Why: Answers the PI's shuffle question: the dip is a reach-locked property of each cell's time course; the couplings come from same-bin co-firing across cells within a reach, which the trial shuffle breaks. Numbers: Sessions beyond all 20 shuffles (dip = mean -300..+100 ms minus baseline -1000..-400 ms): field dip 18/21 per-neuron shift, 19/21 population shift, 6/21 trial shuffle (ns, p = 0.24); total dip 19, 19, 7/21 (ns); FR change 19/21 and 16/21 (time shifts); coupling level 21/21 trial, 19/21 per-neuron shift, 6/21 population shift (ns). Wilcoxon p < 1e-5 for every non-ns row.

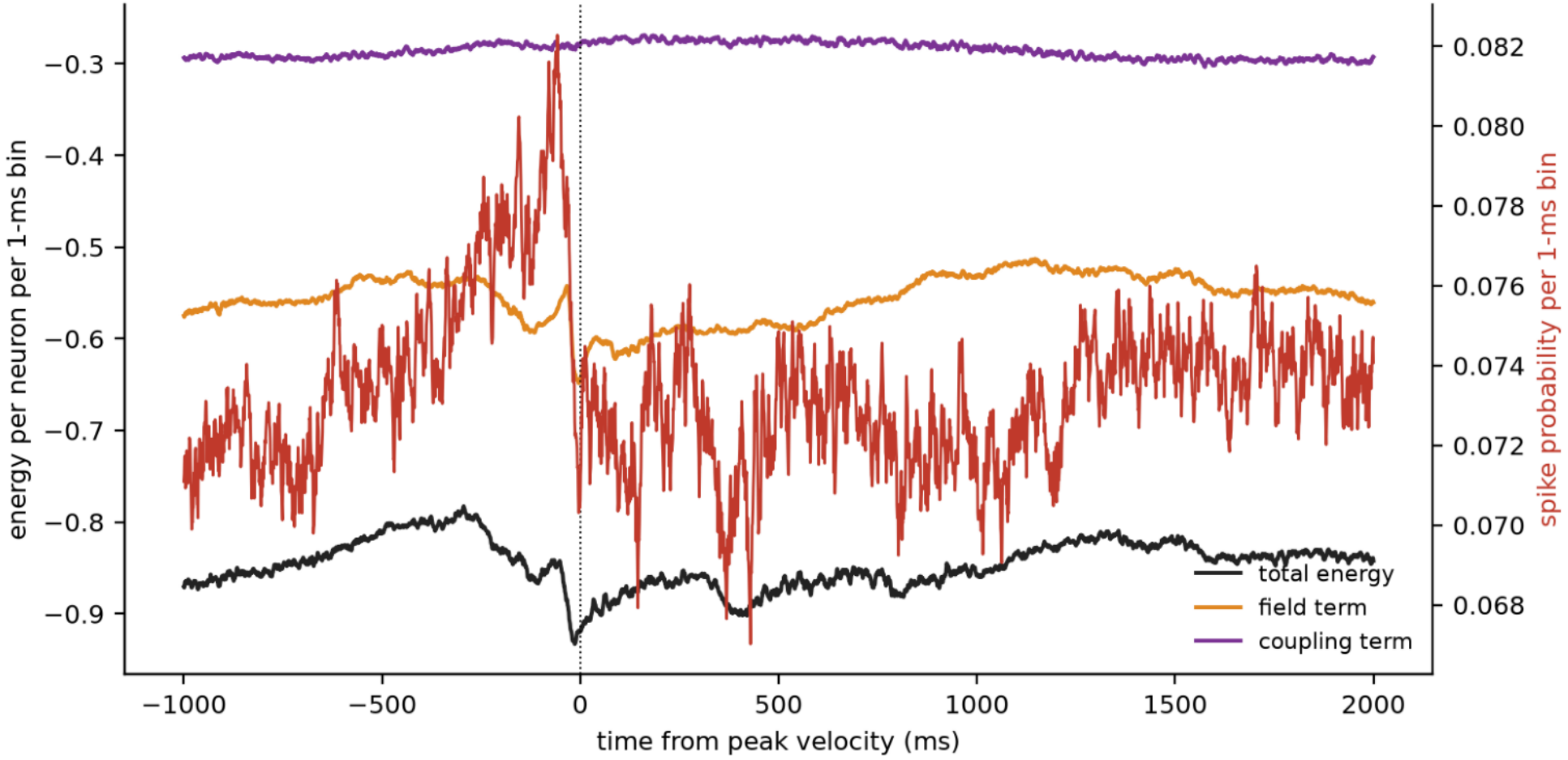
S3_bin_size_1ms / 01_compare_1ms_vs_10ms_scaled [exploratory]



EXPLORATORY. Same model fitted at 1-ms and 10-ms bins (21 sessions; 1-ms spikes re-binned reproduce the 10-ms data exactly). Curves = change from baseline (−1000..−400 ms), median over sessions, scaled to each curve's max |change|; energy scales are not comparable across bin sizes. Dashed = median peak acceleration.

Shows: Field term, coupling term and population firing rate, change from baseline scaled to each curve's maximum, 1-ms fit (10-ms running mean) vs 10-ms fit, vs ms from peak velocity; median over 21 sessions. Dashed = median peak acceleration. Pattern: The field-term dip is present at both bin sizes; at 1 ms its minimum is about 45 ms later. The coupling-term dip seen at 10 ms is absent at 1 ms. Why: The field-term result does not depend on the 10-ms binning; the coupling-term dip may. Numbers: Minimum of field term: -50 ms (10-ms bins) vs -4 ms (1-ms bins), CI wide. All gates passed (1-ms spikes re-binned reproduce the 10-ms data exactly); 2,184 fits converged. J at 1 ms vs 10 ms r = 0.61. Data flag: 210511 unit 11 looks duplicated (74% of active 1-ms bins have >= 2 spikes).

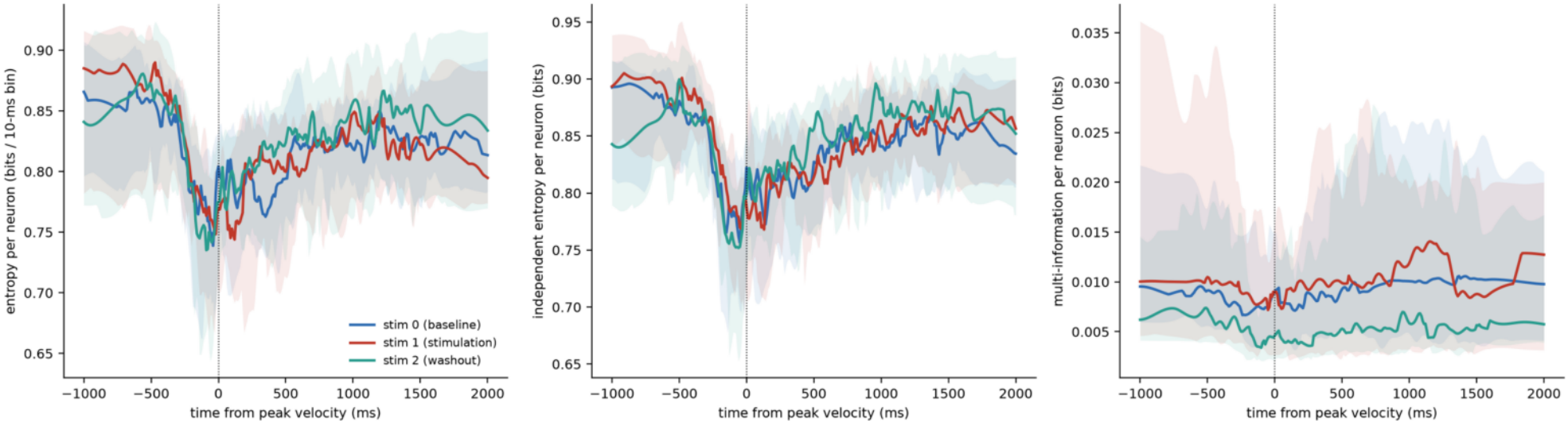
S3_bin_size_1ms / 02_energy_absolute_1ms [exploratory]



EXPLORATORY. 1-ms version of the absolute-energy panel (10-ms running mean for display); median over 21 sessions.

Shows: Absolute energy terms per neuron per 1-ms bin (total, field, coupling) and firing rate (right axis), median over 21 sessions, 10-ms running mean. Pattern: Same layout as 3/09 at 1 ms. Per-ms firing rises during the reach while P(>= 1 spike per 10 ms) falls, i.e. spikes cluster. Why: Context for 01. Numbers: Energy scales are not comparable across bin sizes.

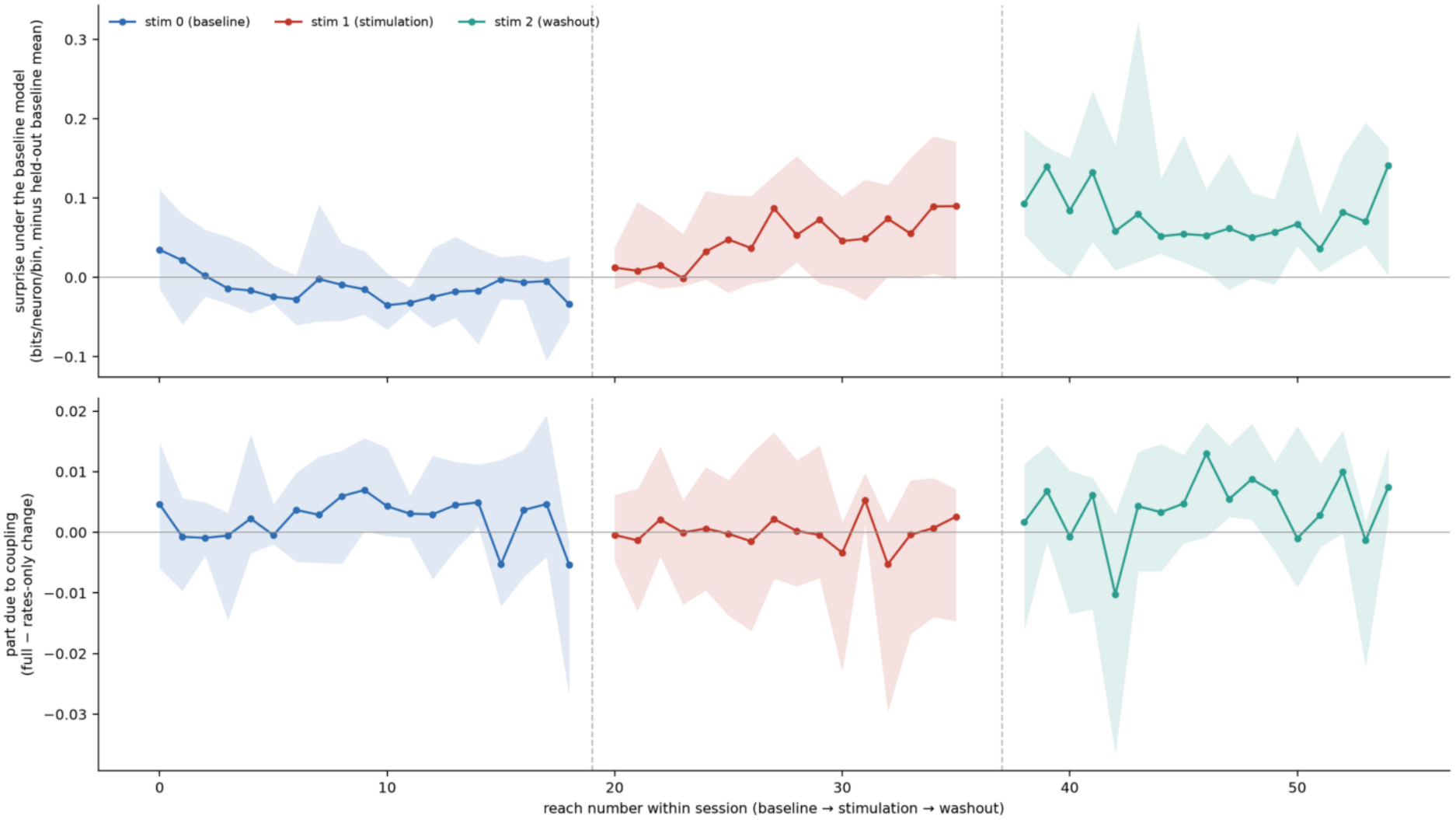
S4_entropy_and_learning_exploratory / 01_entropy_by_block [exploratory (confounded with time in session)]



EXPLORATORY. Each block fitted separately; entropy exact from the fitted model (sessions > 16 cells: one fixed 16-cell subset). Median over 21 sessions, IQR.

Shows: Entropy, independent entropy and multi-information per neuron (bits per 10-ms bin) vs ms from peak velocity, separate fits per block (stim 0 baseline, 1 stimulation, 2 washout); median and IQR over 21 sessions. Pattern: The three blocks overlap. Why: No block-level change in the model's entropy or in coupling-carried information. Numbers: Session-level Wilcoxon (window -1000..+2000 ms): all block contrasts p >= 0.20.

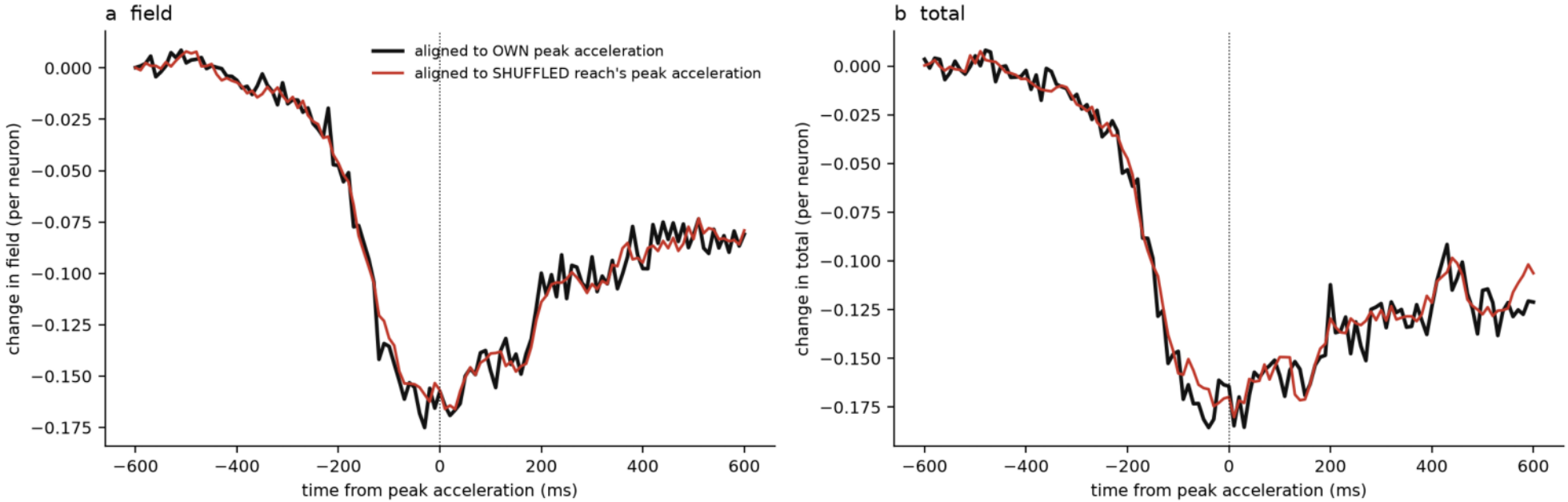
S4_entropy_and_learning_exploratory / 02_baseline_referenced_surprise [exploratory (confounded with time in session)]



EXPLORATORY. One model per session fitted on baseline reaches; later reaches scored against it (exact likelihood, ±200 ms around the crossing); baseline scored held-out. CAVEAT: confounded with time in session (recording drift / animal state).

Shows: Top: surprise of each reach under a model fitted on that session's baseline reaches, minus the held-out baseline mean (bits/neuron/bin), vs reach number across baseline, stimulation, washout. Bottom: part due to coupling (full-model change minus rates-only change). Median and IQR over sessions. Pattern: Surprise climbs through the stimulation block and stays high in washout; the coupling part stays near 0. Why: Suggests the population code moves away from baseline during stimulation without recovering, carried by rates. Could be learning, or simply time in session. Numbers: Stim vs baseline 17/21 above, p = 0.0004; within-stimulation trend 18/21 positive, p = 0.006; washout vs baseline 19/21, p < 1e-4, no recovery trend (p = 0.92); coupling part p = 0.52 (stim), 0.95 (washout). CAVEAT: drift.

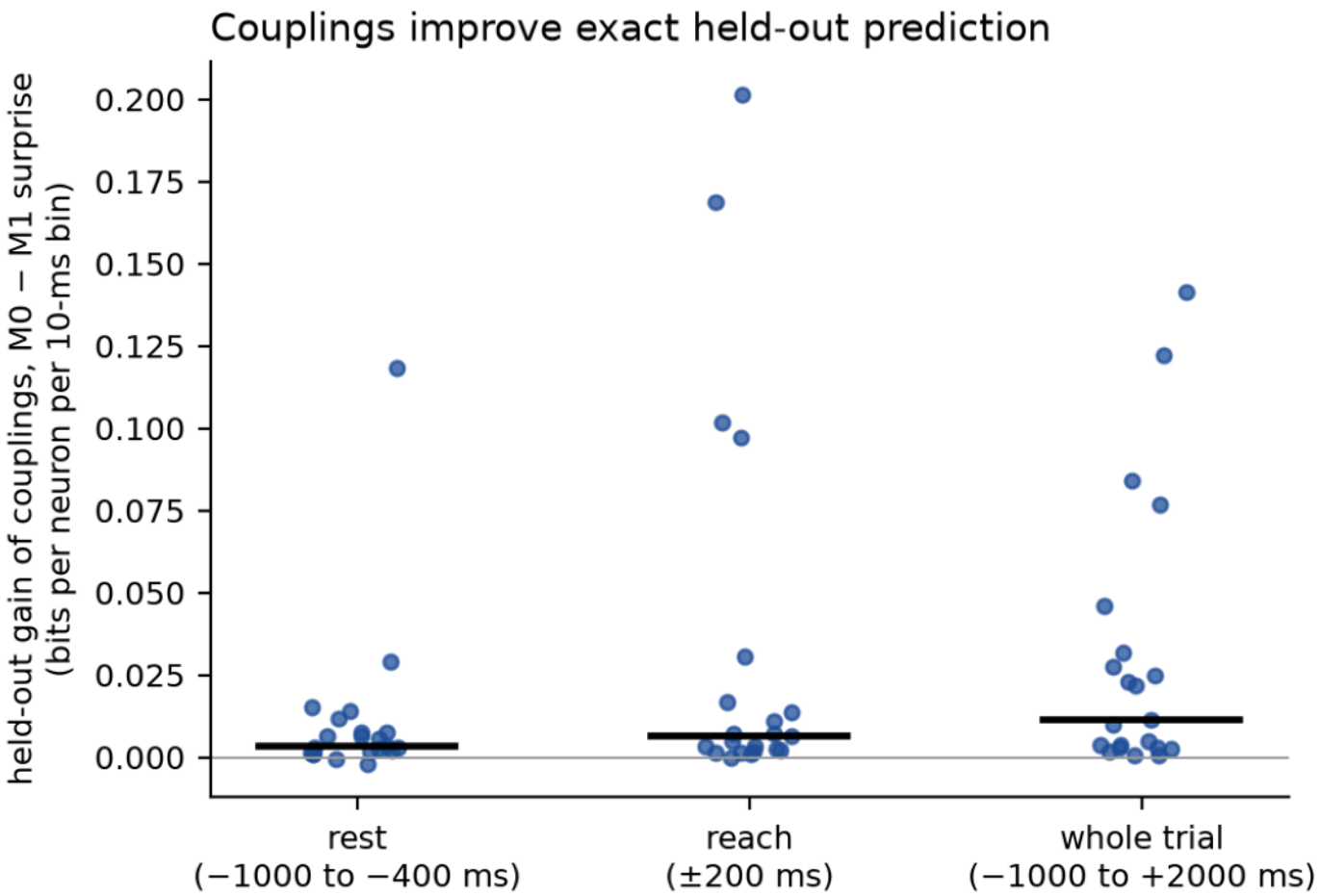
S5_per_reach_locking_exploratory / 01_realign_own_vs_shuffled [exploratory (low power)]



EXPLORATORY. Per-reach energy aligned to each reach's own (black) or a shuffled reach's (red; 1,000 shuffles) peak acceleration, averaged. Low power: peak-acceleration jitter ~±55 ms vs a ~300-ms-wide dip.

Shows: Field term and total energy per reach, aligned to each reach's own peak acceleration (black) vs a shuffled reach's peak acceleration (red, 1,000 shuffles), averaged; change from baseline, median over sessions. Pattern: The two curves are nearly identical. Why: Per-reach jitter of peak acceleration is small relative to the width of the dip, so this test has little power either way. Numbers: Depth own -0.160 vs shuffled -0.161 (field, 14/21 deeper, p = 0.37); total -0.173 vs -0.172 (p = 0.19). Per-reach rho(dip time, peak-acc time) field 0.09 vs null 0.00, 13/21, p = 0.04; distance 113 vs 119 ms, p = 0.12. Peak-acceleration jitter SD 54 ms vs a dip ~300 ms wide.

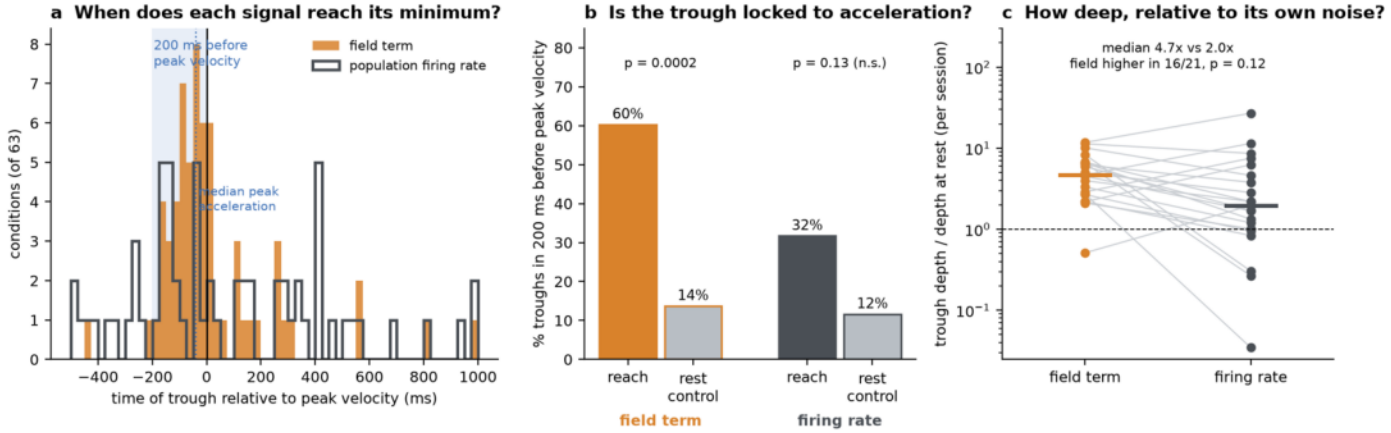
S6_model_comparison_M0_vs_M1 / 01_M1_vs_M0_exact [exploratory check (supports established A7)]



One dot per session (21). Times from the 1-cm crossing. 5-fold CV over reaches. Exact likelihood (sessions > 16 cells: fixed 16-cell subset).

Shows: Held-out gain of couplings (M0 minus M1 surprise, bits per neuron per 10-ms bin) per session, for rest (-1000..-400 ms), reach (+-200 ms) and whole trial (-1000..+2000 ms) around the 1-cm crossing. 5-fold CV over reaches, exact likelihood (fixed 16-cell subset for larger sessions). Pattern: Every session is above 0 over the whole trial; nearly all at rest and during the reach. Why: Exact-likelihood confirmation that couplings add predictive power beyond time-varying rates, including at rest (A7). Numbers: Whole trial 21/21 sessions, 63/63 conditions, median 0.0115 bits/neuron/bin (median relative gain 1.6% of M0 surprise), p = 9.5e-7; rest 19/21, 51/63, p = 1.8e-5; reach 20/21, 59/63, p = 1.9e-6.

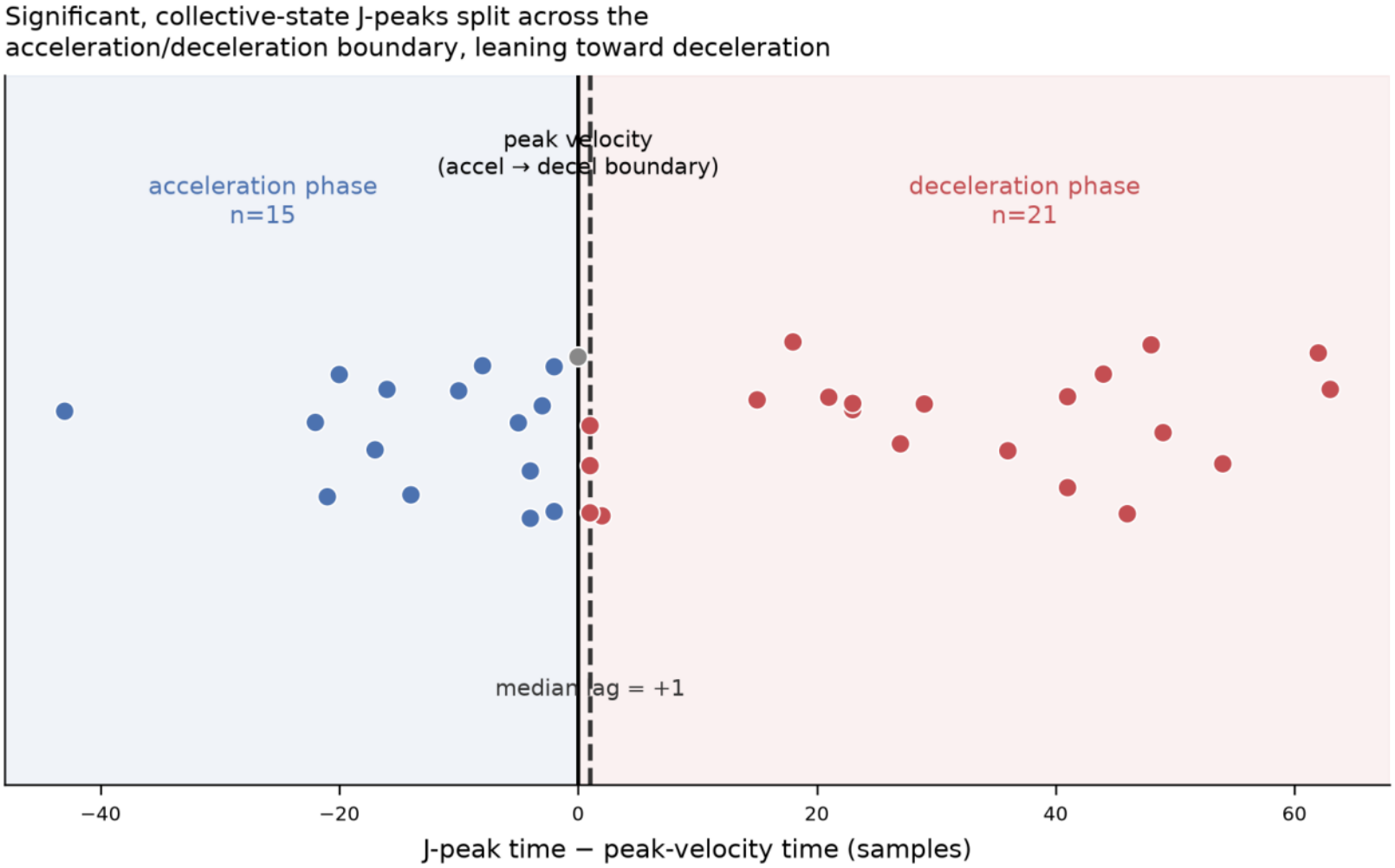
X_withdrawn_or_superseded / 01_WITHDRAWN_prelim_field_vs_rate_trough_E09 [withdrawn / superseded (do NOT cite as evidence)]



21 sessions x 3 blocks = 63 conditions, 10-ms bins. Field term = -sum_i h_i(t) s_i from the time-varying-field Ising model (fields h_i(t) follow the reach, J fixed; pseudo-likelihood fit, tvfield_model.py). Firing rate = population-average spike probability. Rest controls = the same search in windows shifted -1.5/-2/-2.5 s. Statistics are session-level (n = 21) two-sided Wilcoxon tests. EXPLORATORY: post hoc from the E09 table (its generating script and exclusion status were not preserved), not pre-registered, no FDR. Formal test: E12.

Shows: (a) Trough time histograms, (b) % troughs in 200 ms before peak velocity, (c) trough depth / rest depth per session. Pattern: Exploratory E09 version of the E12 test. Why: SUPERSEDED by E12 T1_T2_by_signal. Panel c depth ratio is WITHDRAWN (basis inflation, E12 R7). Numbers: -

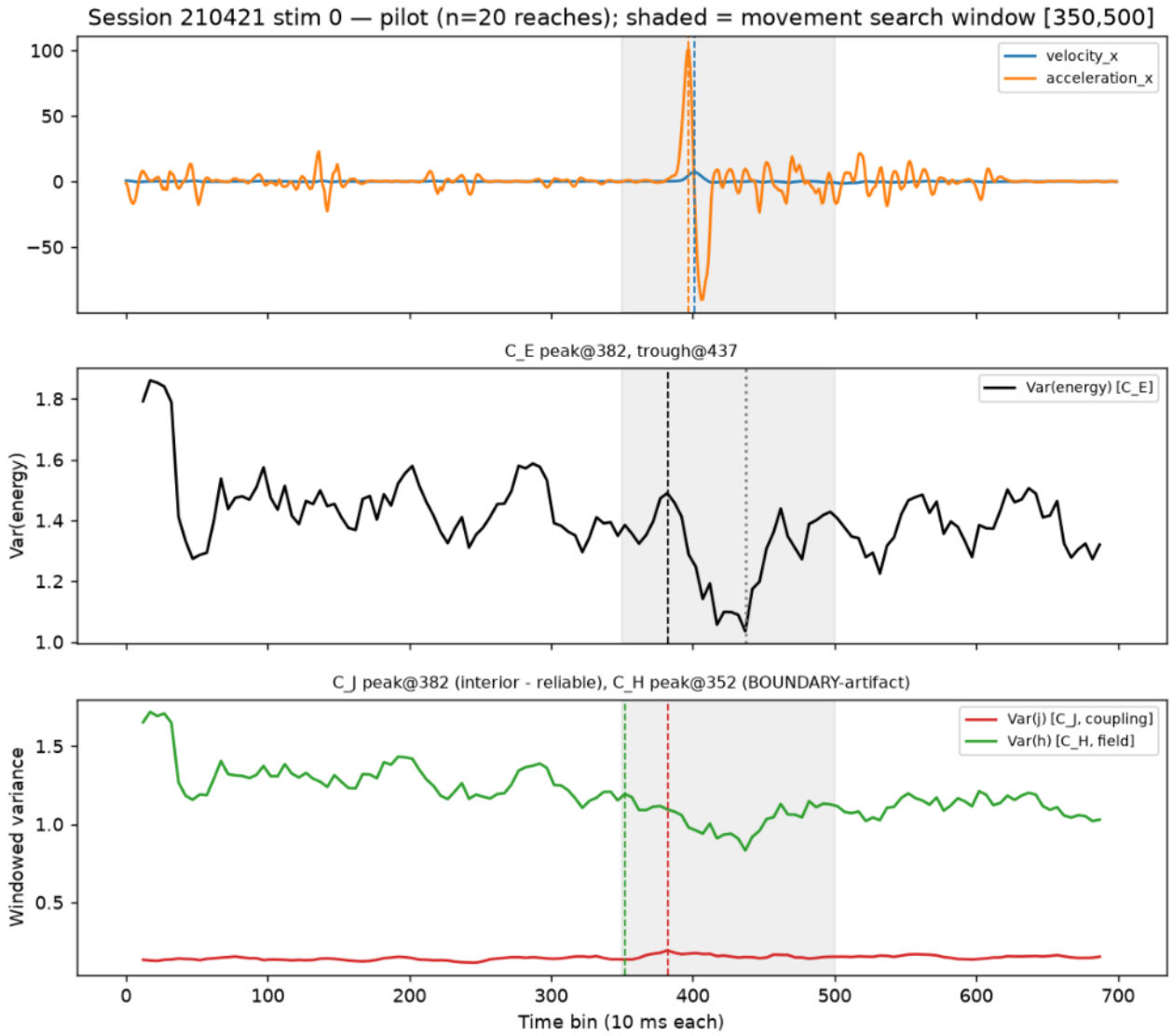
X_withdrawn_or_superseded / 02_WITHDRAWN_j_peak_timing_vs_velocity_peak [withdrawn / superseded (do NOT cite as evidence)]



n = 37 session x stimulus reaches with a detected J-peak AND is_collective = True (pk_j_summary, prominence-based detector)
Wilcoxon signed-rank on lag vs. 0: p = 0.035 | sign test on accel-vs-decel split: p = 0.405

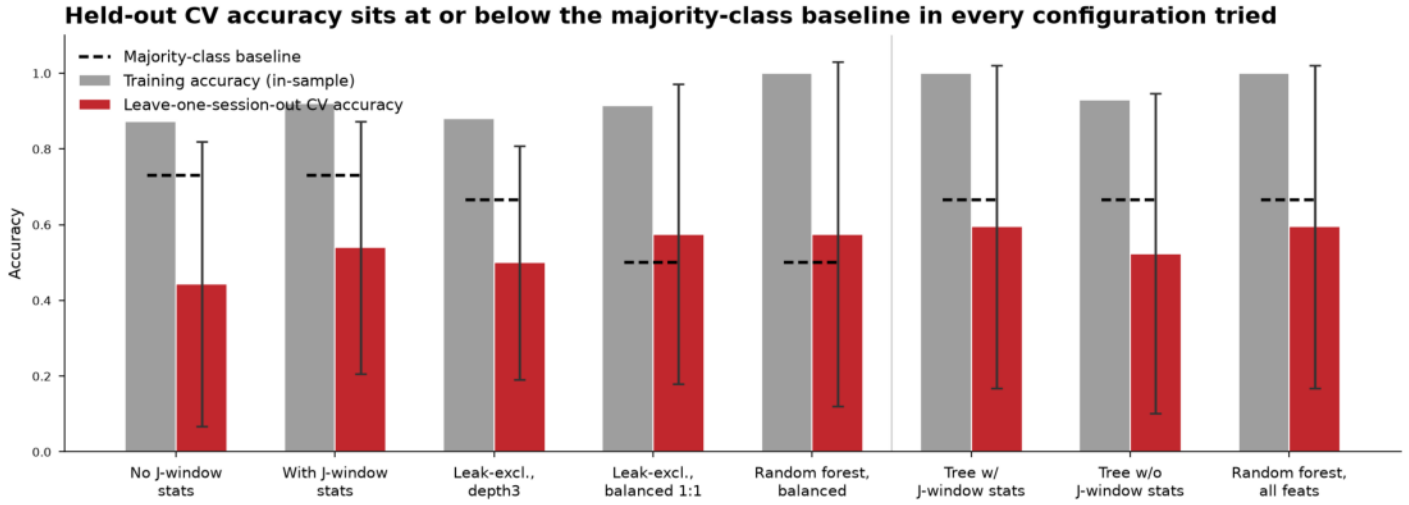
Shows: J-peak time minus peak-velocity time (samples) for 37 collective reaches. Pattern: Peaks split across the acc./dec. boundary. Why: WITHDRAWN: 'J peaks after peak acceleration' / 'J peaks in deceleration' (window-length artefact; log D). Numbers: -

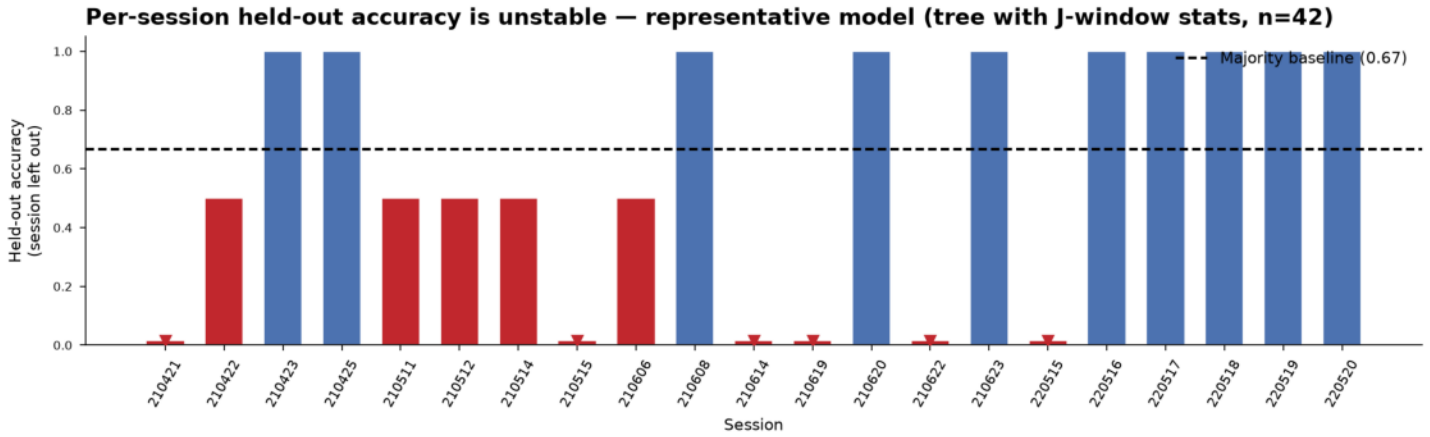
X_withdrawn_or_superseded / 03_WITHDRAWN_pilot_criticality_proxy [withdrawn / superseded (do NOT cite as evidence)]



Shows: Pilot session 210421: kinematics, Var(energy), Var(J) and Var(h) vs time bin. Pattern: Pilot variance peaks. Why: DID NOT REPLICATE: criticality proxy Var(J) peak (log D). Numbers:
-

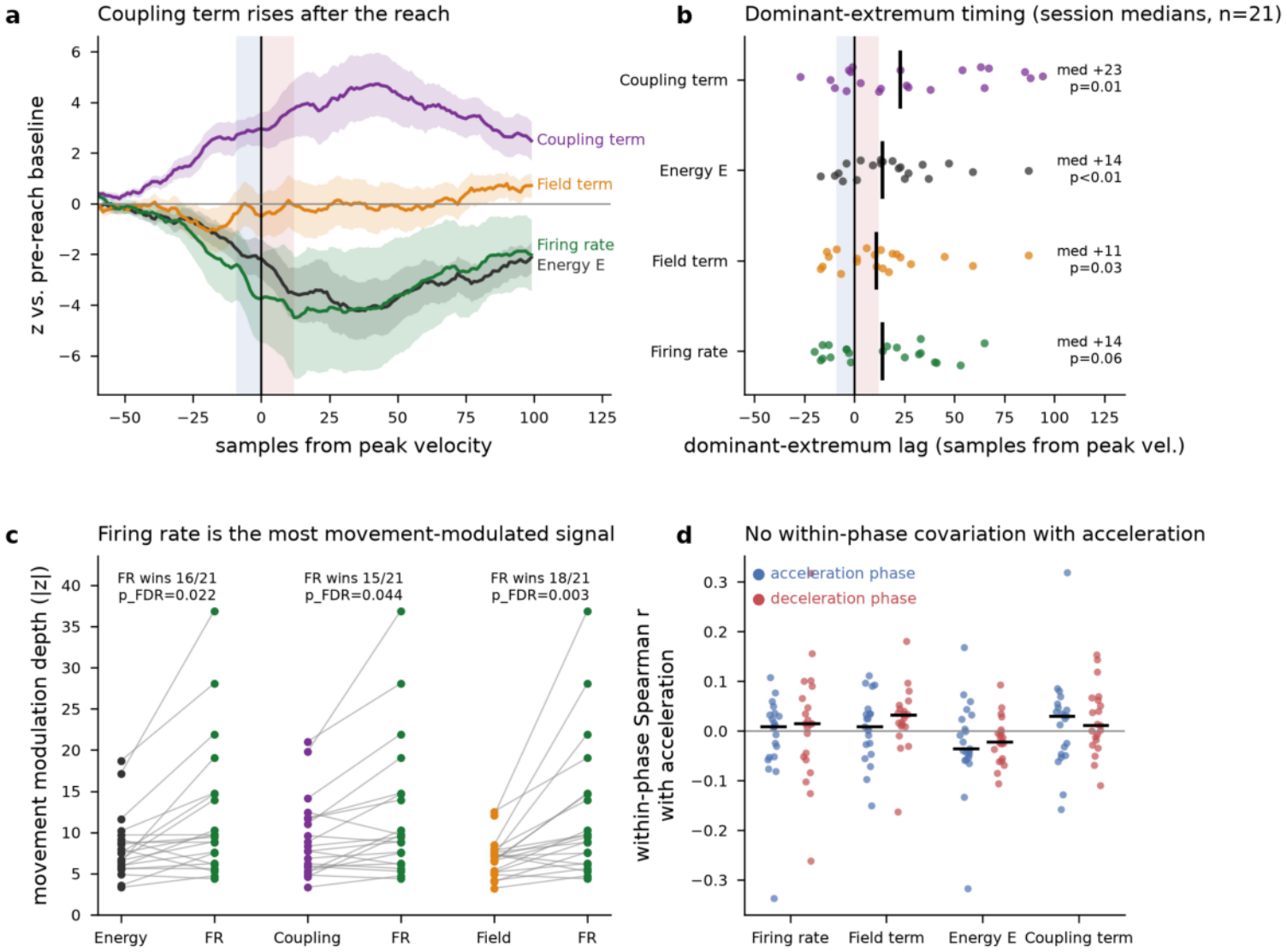
X_withdrawn_or_superseded / 04_WITHDRAWN_decision_tree_generalization [withdrawn / superseded (do NOT cite as evidence)]





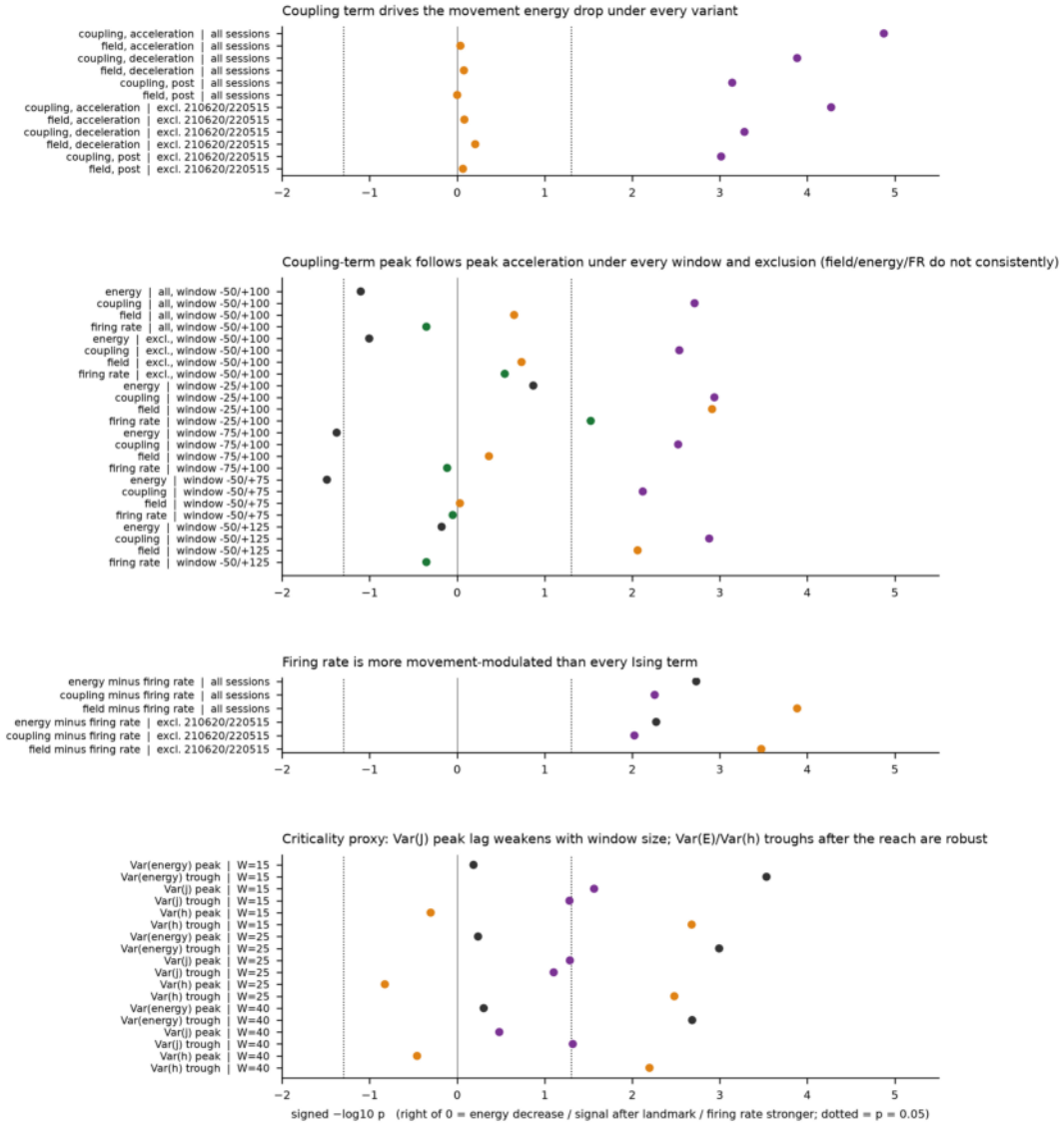
Shows: Training vs leave-one-session-out accuracy for decision-tree variants; per-session accuracy. Pattern: CV accuracy at or below majority baseline. Why: ABANDONED: decision tree predicting J peaks (log D). Numbers: -

X_withdrawn_or_superseded / 05_PARTLY_WITHDRAWN_phase_timing_summary [withdrawn / superseded (do NOT cite as evidence)]



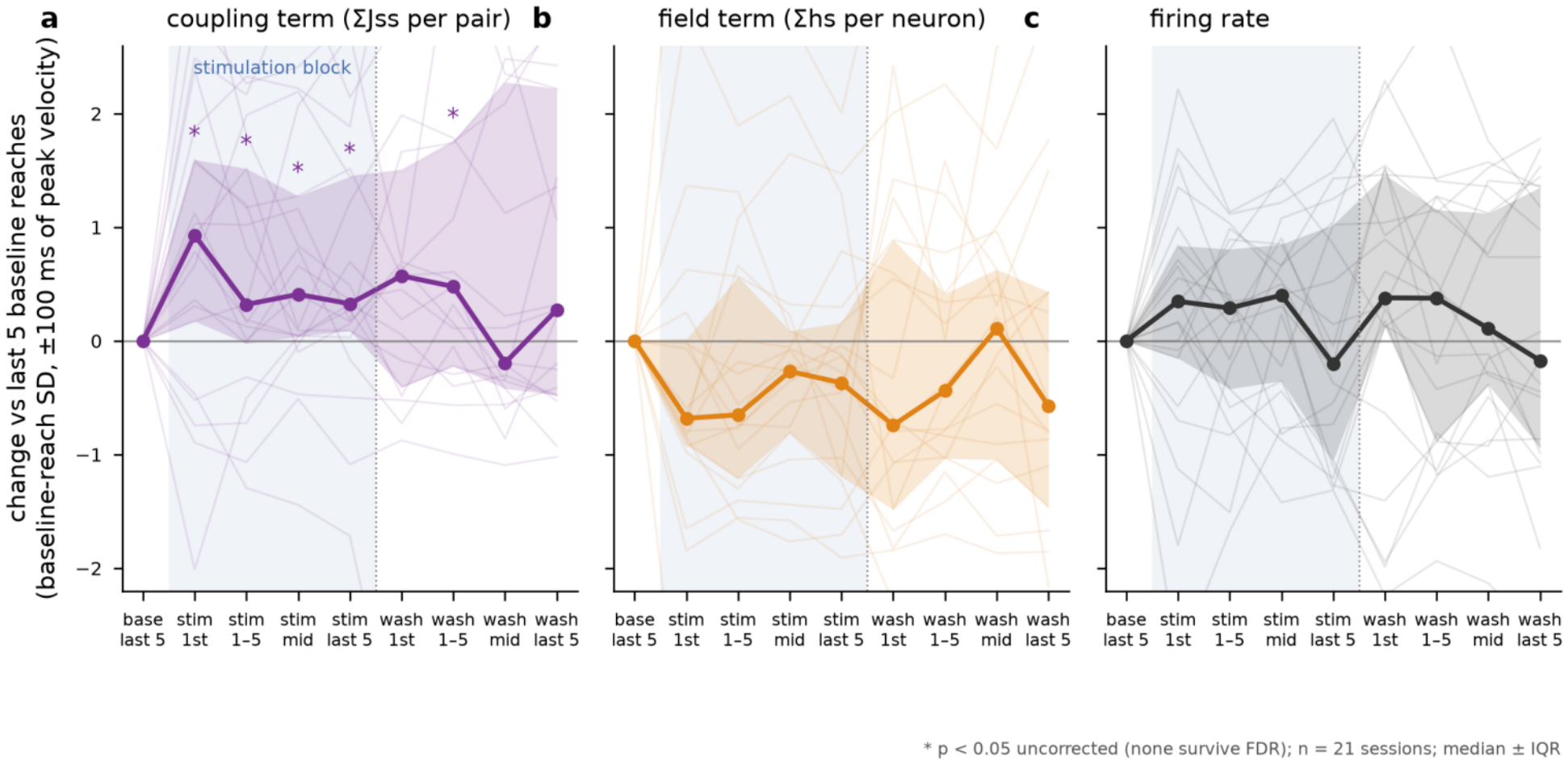
Shows: (a) z vs baseline of coupling, field, rate, energy; (b) dominant-extremum timing; (c) movement-modulation depth; (d) within-phase correlation with acceleration. Pattern: - Why: Panel b is DROPPED (superseded by 3/05 peak_timing_vs_control). Panels a, c, d still support B4/B6. Numbers: FR wins 15-18/21 (B6).

X_withdrawn_or_superseded / 06_SUPERSEDED_sensitivity_robustness [withdrawn / superseded (do NOT cite as evidence)]



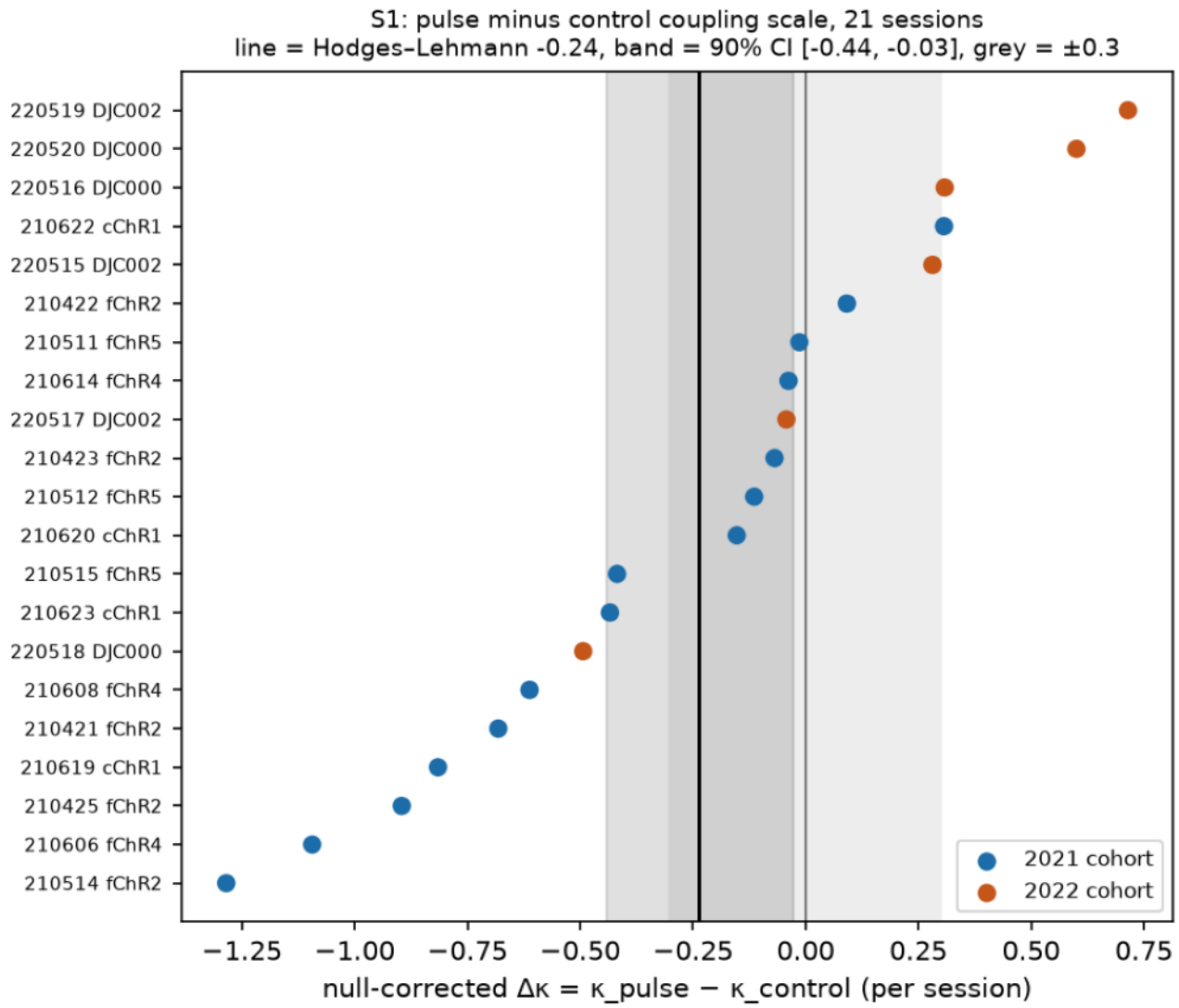
Shows: Signed -log10 p for many window/exclusion variants. Pattern: - Why: Peak-timing rows SUPERSEDED by peak_timing_vs_control; 'coupling term drives the energy drop' holds only for the stationary model. Numbers: -

X_withdrawn_or_superseded / 07_SUPERSEDED_stim_block_reach_trajectories [withdrawn / superseded (do NOT cite as evidence)]



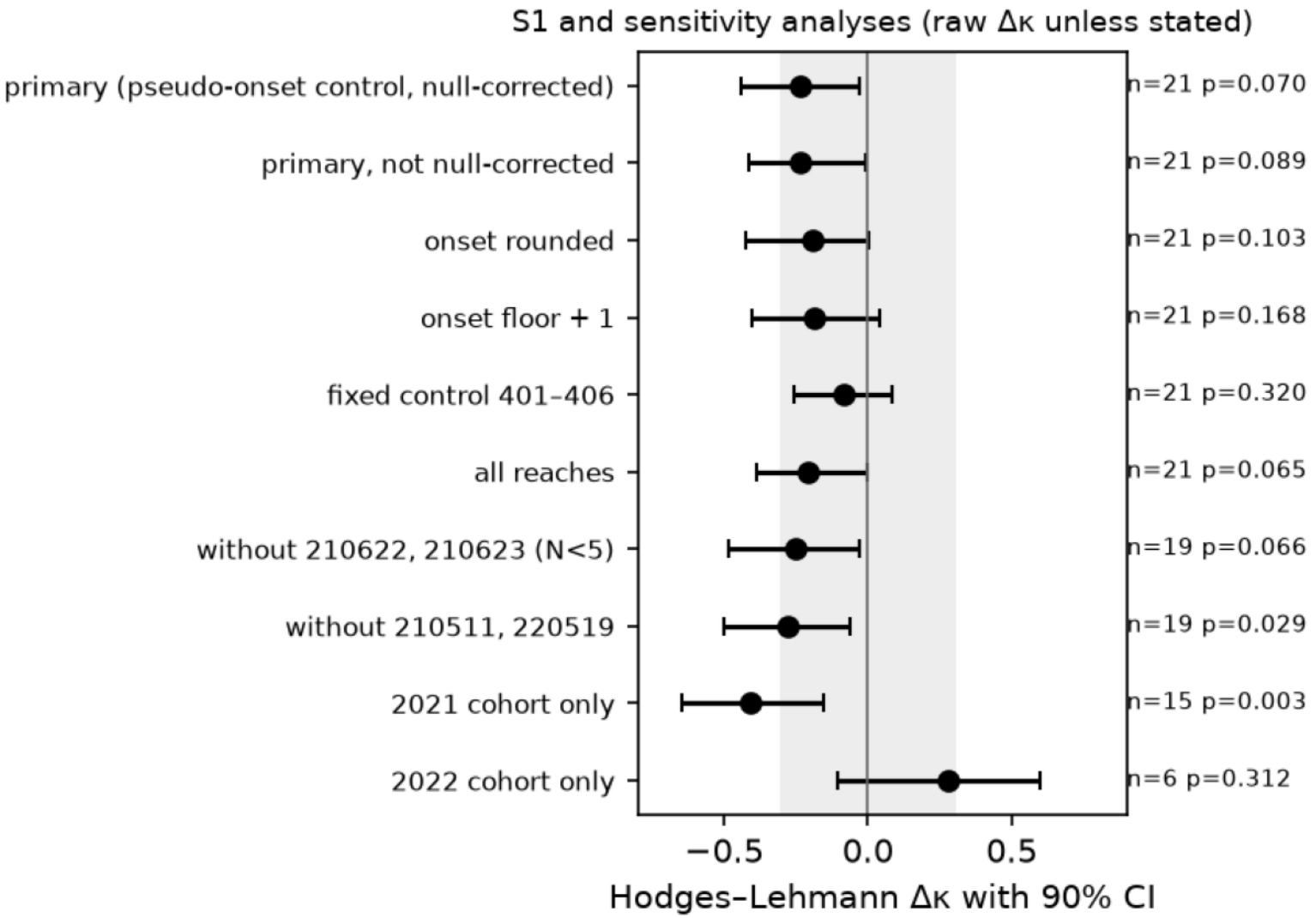
Shows: Change vs last 5 baseline reaches (baseline SD) of coupling term, field term and firing rate across stimulation and washout blocks. Pattern: Coupling term above baseline in the stim block (uncorrected). Why: PRELIMINARY per-block fits; none survive FDR; SUPERSEDED by E11 (S1). Numbers: -

X_withdrawn_or_superseded / 08_RETRACTED_E11_run1_fig1 [withdrawn / superseded (do NOT cite as evidence)]



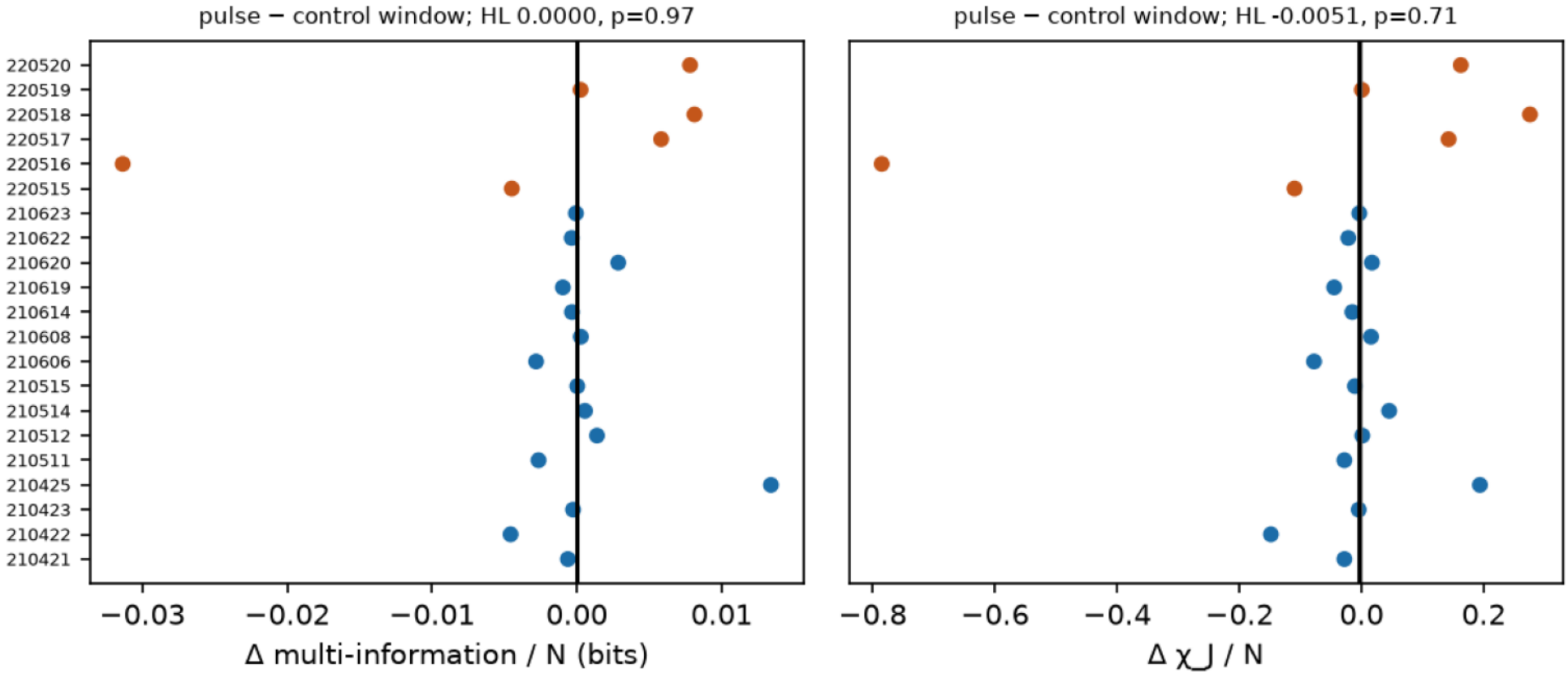
Shows: As S1/01 (run 1). Pattern: - Why: RETRACTED: run 1 fitted bins 0-699 instead of 300-559 and used a misaligned null. Numbers: -

X_withdrawn_or_superseded / 09_RETRACTED_E11_run1_fig2 [withdrawn / superseded (do NOT cite as evidence)]



Shows: As S1/02 (run 1). Pattern: - Why: RETRACTED (run 1). Numbers: -

X_withdrawn_or_superseded / 10_RETRACTED_E11_run1_fig3 [withdrawn / superseded (do NOT cite as evidence)]



Shows: As S1/03 (run 1). Pattern: - Why: RETRACTED (run 1). Numbers: -

Figure 1. The outward reach is brief

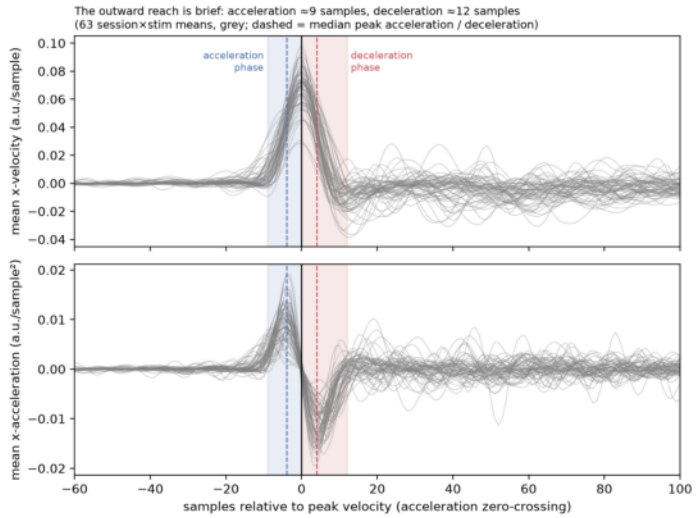


Figure 1. Mean x-velocity (top) and x-acceleration (bottom) of the hand for all 63 session × block conditions (grey), aligned to peak velocity (the acceleration zero-crossing, sample 0; 1 sample = 10 ms). Blue shading: acceleration phase; red shading: deceleration phase. Dashed lines: median peak acceleration and peak deceleration. Median acceleration phase lasts 90 ms and deceleration 120 ms. Position is min-max scaled within each reach, so units are arbitrary.

In plain terms. The actual reach is over in about a fifth of a second. That matters because any neural signal that lasts longer than this cannot be "locked" to acceleration or deceleration just by where its peak lands: those windows are too short. Every later timing test was built with this in mind.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 1: The outward reach is brief.

Figure 2. Population activity is more collective than independent neurons predict

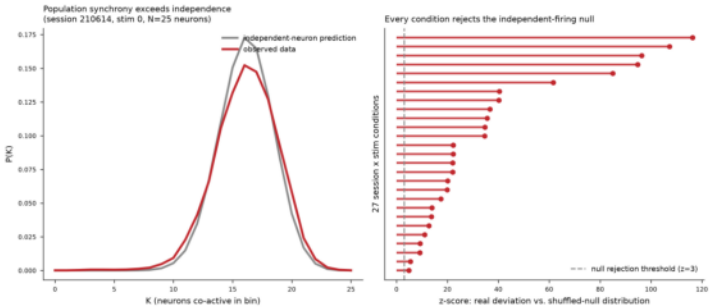


Figure 2. Left: probability that K of the N = 25 neurons are active in the same 10-ms bin, P(K), for session 210614 stim 0 (red), against the prediction for independent neurons with the same firing rates (grey). Right: deviation of the observed P(K) from a shuffled independent-firing null, as a z-score, for each of the 27 conditions with raw data (z = 4.9 to 116; dashed line z = 3). This is a model-free test. The shuffle also removes reach-locked rate modulation, so it establishes structure beyond stationary independence only.

In plain terms. If neurons fired independently, the count of neurons active at once would follow the grey curve. The real data have fatter tails: more bins where many or few neurons fire together. Every condition passes this test. It proves the population is not a set of independent coin flips, but not yet why.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 2: Population activity is more collective than independent.

Figure 3. Fitted couplings are not a fitting artefact

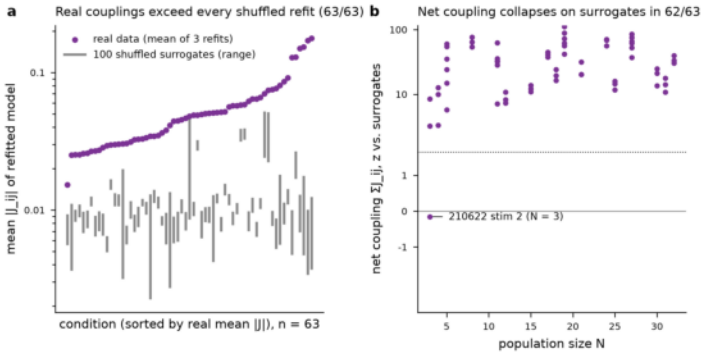


Figure 3. (a) Mean $|J_{ij}|$ of the refitted Ising model for the real data (purple; mean of 3 refits) and the range over 100 rate-preserving shuffled surrogates (grey bars) per condition, sorted by real value; the real value exceeds every surrogate in 63/63 conditions (median 4.6×). (b) Net coupling ΣJ_{ij} of the real fit as a z-score against the surrogates, against population size N. Dotted line z = 2. The one exception, 210622 stim 2, has only N = 3 neurons.

In plain terms. A fitting algorithm can report non-zero couplings even from pure noise. So we scrambled each neuron's spikes to destroy any co-firing while keeping its firing rate, then refit. The scrambled data give couplings several times smaller in every condition, so the real couplings reflect genuine co-firing.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 3: Fitted couplings are not a fitting artefact.

Figure 4. Validation of the time-varying-field model

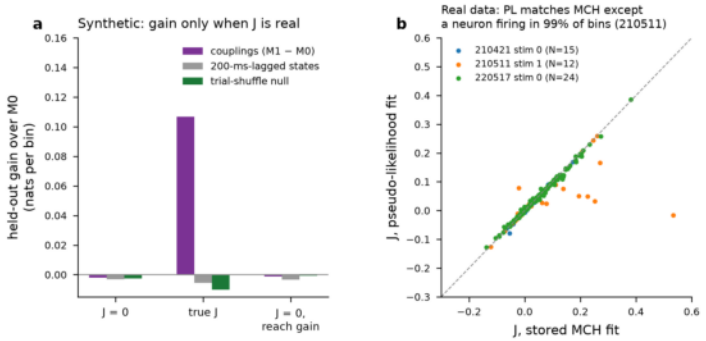


Figure 4. (a) Synthetic data with known parameters: held-out log-likelihood gain over the time-varying independent model (M0) from adding couplings (M1 − M0, purple), from other neurons' states lagged 200 ms (grey), and for the trial-shuffle null (green). The couplings help only when true J is present (+0.107 nats/bin; J recovered with r = 0.98), including when a per-reach gain fluctuation is simulated with J = 0. (b) Real data: couplings from the pseudo-likelihood fit against the stored MCH (coniii) fit. They agree (r = 0.995–0.997) except in 210511 stim 1 (r = 0.65), where one neuron fires in 99% of bins. There the pseudo-likelihood fit reproduces the data's pairwise correlations more closely (rms error 0.0006 vs 0.0011).

In plain terms. Before using a new tool on real data, we tested it on fake data where we know the answer. It finds couplings when they exist and reports none when they don't, even when the fake neurons share slow ups and downs. On real data it agrees with the lab's original fitting method.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 4: Validation of the time-varying-field model.

Figure 5. Couplings add prediction beyond shared rate changes, and the drop in energy moves to the fields

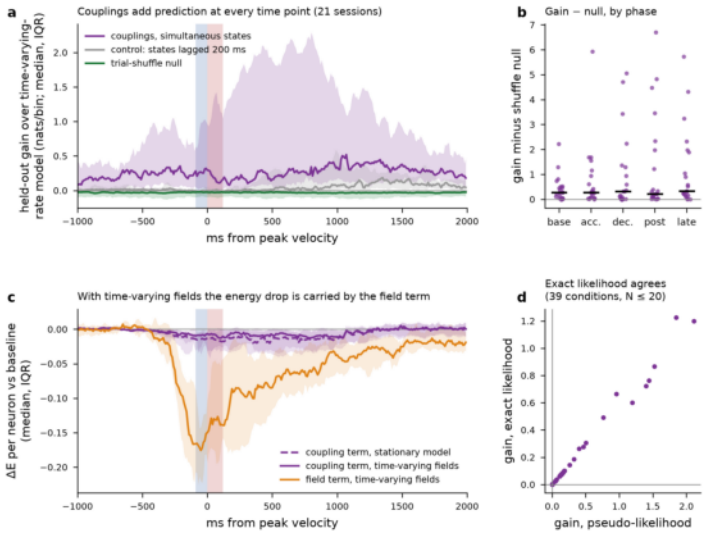


Figure 5. (a) Held-out gain over a time-varying-rate independent model (nats per 10-ms bin; session median and IQR, n = 21), for couplings on simultaneous states (purple), for a control using states lagged 200 ms (grey), and for the trial-shuffle null (green). The shading marks acceleration (blue) and deceleration (red). (b) Gain minus null, by phase, per session. The gain is above the null in 21/21 sessions and 63/63 conditions (median +0.26 nats/bin) and does not differ between baseline and the reach (change $p \geq 0.45$). The acceleration change relative to baseline is −15% (bootstrap 95% CI −28% to +93%), and couplings fitted separately for the reach epoch do not improve held-out prediction (9/21 sessions, $p = 0.71$). (c) Energy change per neuron against baseline: the coupling term in the stationary model (dashed), and the coupling term (purple) and field term (orange) in the time-varying-field model. With time-varying fields the reach-related energy drop is carried by the field term (acceleration 20/21 sessions, $p_{FDR} = 2\times10^{-5}$). (d) Pseudo-likelihood gain against exact-likelihood gain for the 39 conditions with $N \leq 20$ (38/39 positive; $\rho = 0.999$).

In plain terms. This is the main result. We gave the model the ability to follow each neuron's firing rate moment by moment during the reach. The couplings still helped predict reaches the model had never seen, in every session, so the neurons really are tied together beyond just speeding up and slowing down in sync. The size of that help is the same at rest and during the reach. Once rates are properly accounted for, the energy change during the reach comes from the volume knobs (fields), not the strings (couplings). We cannot rule out a moderate change in coupling during the reach, because the reach is short and each session has only about 60 reaches.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 5: Couplings add prediction beyond shared rate changes.

Figure 6. Where the stationary model's movement-time coupling signal comes from

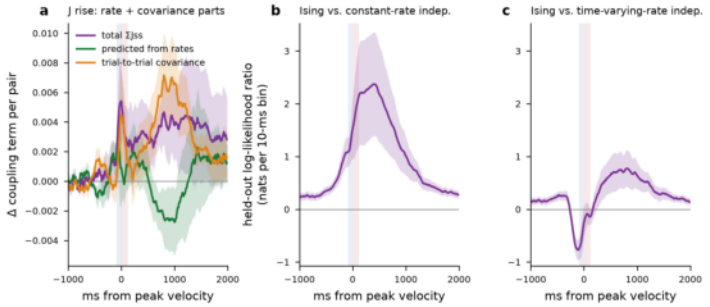


Figure 6. (a) Change in the coupling term per pair against baseline (total, purple), split into the part predicted from each neuron's rate alone (green) and the trial-to-trial covariance part (orange). A median of only 36% comes from covariance, and neither part is consistent across sessions ($p_{FDR} > 0.16$). (b) Held-out log-likelihood ratio of the stationary Ising model against a constant-rate independent model: its advantage rises during the reach (20/21 sessions). (c) The same model against a time-varying-rate independent model: the reach-time advantage disappears; it is worse during acceleration (2/21 sessions, $p_{FDR} = 0.004$) and rises again after the reach.

In plain terms. Our original model uses a fixed volume knob for each neuron. When all neurons change their rates together during the reach, that model has nowhere to put the change except the strings, so the strings look like they tighten. Once rates are allowed to move (panel c), the apparent tightening during the reach goes away. This is why the stationary-model result in Figure 7 has to be interpreted carefully.

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Figure 7. Energy decomposition in the stationary model

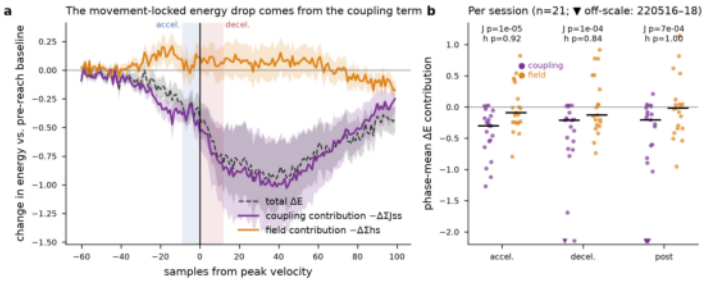


Figure 7. (a) Change in energy against the pre-reach baseline (dashed black), split into the coupling contribution $-\Delta\Sigma Jss$ (purple) and the field contribution $-\Delta\Sigma hs$ (orange), session mean ± SEM, n = 21; x-axis in 10-ms samples from peak velocity. (b) Phase-mean contributions per session. The coupling term falls below baseline in acceleration ($p = 10^{-5}$), deceleration ($p = 10^{-4}$) and post-movement ($p = 7\times10^{-4}$); field term $p \geq 0.84$. **Model dependence:** this holds for the stationary model only (per-block fits with fixed fields). With time-varying fields the drop moves to the field term (Fig. 5c). The panel title states the stationary-model result.

In plain terms. In the original model, the population's "energy" (lower means a more typical, more probable pattern) drops during the reach, and the drop comes from the coupling part. It is a real feature of that model. But Figures 5 and 6 show it happens because the fixed-rate model pushes the rate changes into the coupling part. So it is not evidence that neurons coordinate more during the reach.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 7: Energy decomposition in the stationary model.

Figure 8. J is a broad elevation around the reach, not a timed peak

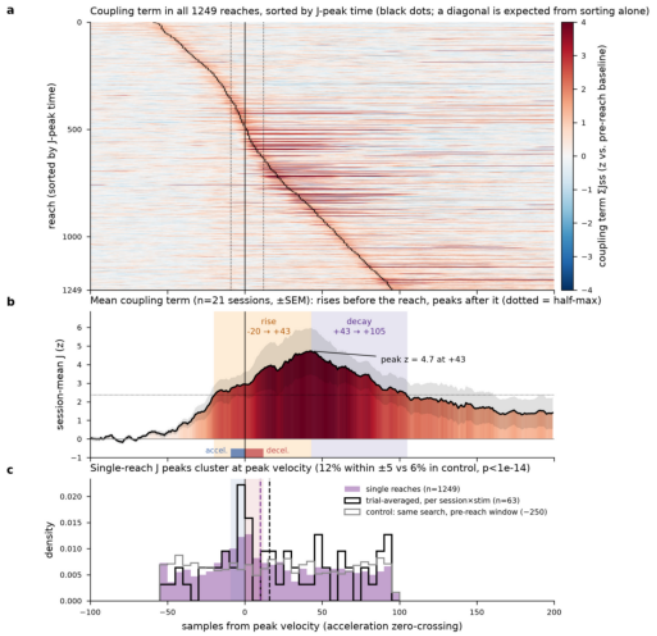


Figure 8. (a) Coupling term ΣJss (z against the pre-reach baseline) for all 1,249 reaches, sorted by the time of each reach's J peak (black dots). The diagonal is produced by the sorting and is not a result. (b) Session-mean J (n = 21, ± SEM): above z = 1 from −340 ms, half-maximum at −200 ms, peak z = 4.7 at +430 ms, back below half-maximum at +1,050 ms relative to peak velocity. (c) Distribution of single-reach J-peak times (purple), trial-averaged peaks per condition (black outline) and the same search in a pre-reach control window (grey). Single-reach peaks fall within ±50 ms of peak velocity in 12.4% of reaches against 6.4% in control windows ($p < 10^{-14}$), concentrated in 4 sessions (220516, 210425, 210608, 210512). Time axis in 10-ms samples.

In plain terms. The coupling term rises before the hand moves, stays up for about a second, and slowly comes down. There is no sharp spike at a particular instant of the movement. The one timing hint: single reaches have their J peak right at top speed about twice as often as chance, but mainly in four sessions, and we found no property that explains why.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 8: J is a broad elevation around the reach, not a timed peak.

Figure 9. Peak timing of J, h and firing rate matches chance

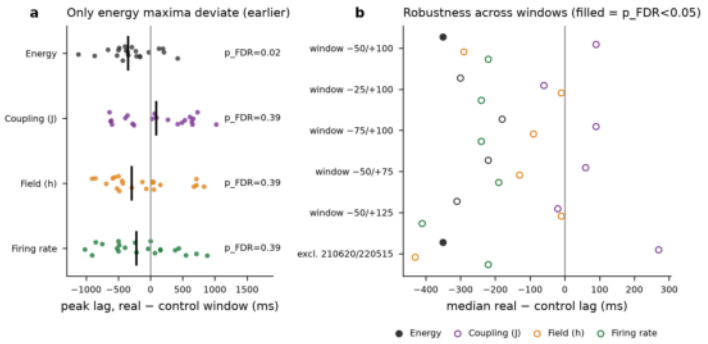


Figure 9. (a) Per-session lag of each signal's maximum after peak velocity minus the lag from the identical search in control windows at rest (shifted −1.5, −2 and −2.5 s). Black bars: medians. Only energy maxima deviate (earlier by about 350 ms, $p_{FDR} = 0.02$). (b) Median real − control lag across six search-window and exclusion variants; filled markers $p_{FDR} < 0.05$. The energy deviation holds in only 2 of 6 variants; J, h and firing rate never differ from control.

In plain terms. Earlier we reported that J peaks about 300 ms after acceleration. The search window was lopsided (it looked further forward than backward), so a peak would land "after" even in random noise. When the same search is run on quiet pre-reach data, the lags look the same, so that claim was withdrawn. Always compare against a fair control.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 9: Peak timing of J, h and firing rate matches chance.

Figure 10. Firing rate is the most movement-modulated signal

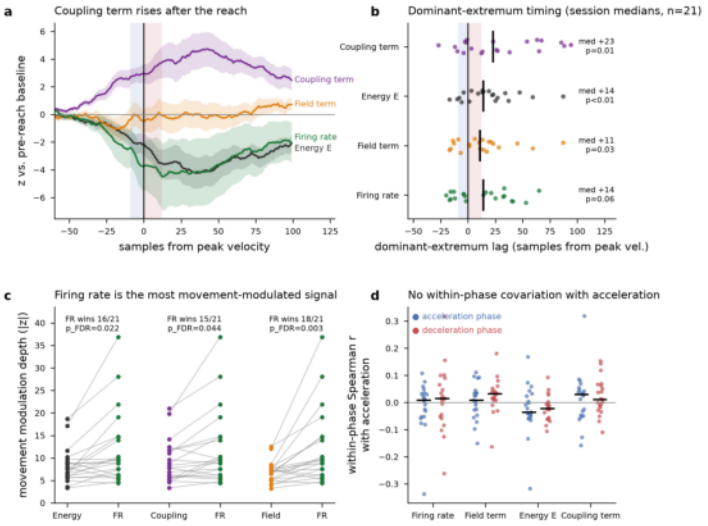


Figure 10. (a) Session-mean z-scored time courses of the coupling term, field term, firing rate and energy (x-axis in 10-ms samples from peak velocity). (b) Dominant-extremum lag per session. **Superseded by Fig. 9:** these p-values come from the asymmetric window and are withdrawn. (c) Movement modulation depth (peak |z|) of energy, coupling and field against firing rate in the same session; firing rate wins 16/21, 15/21 and 18/21 sessions (p_{FDR} = 0.022, 0.044, 0.003). (d) Within-phase Spearman correlation of each signal with acceleration: none differs from zero (all $p_{FDR} > 0.05$). The title of panel a reflects the earlier framing; see Fig. 8 for the current description (broad elevation spanning the reach).

In plain terms. Plain firing rate changes more strongly with the movement than any of the model's quantities. None of the signals rises and falls with acceleration within the reach, from moment to moment. The coupling term is interesting for what it measures (co-firing), not because it tracks the arm better.

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Figure 11. J changes without firing-rate changes are rare

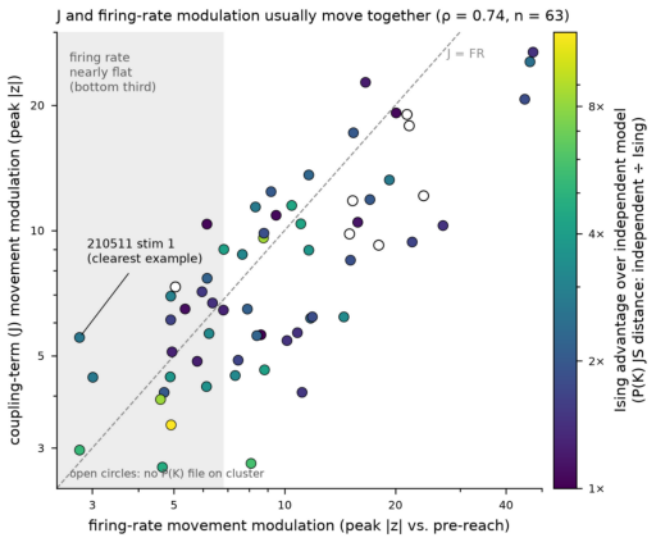


Figure 11. Coupling-term movement modulation against firing-rate movement modulation (peak |z| against the pre-reach baseline; log axes) for 63 conditions, coloured by the Ising model's P(K) advantage over the independent model (Jensen–Shannon distance ratio). Grey region: bottom third of firing-rate modulation (peak |z| ≤ 6.8). Dashed line: J = FR. Open circles: no P(K) file on the cluster. Spearman ρ = 0.74.

In plain terms. We looked for reaches where firing rates stay flat but coupling changes, which would be strong evidence of coordination on its own. They are rare: J and firing rate usually move together. One condition, 210511 stim 1, is a clear example (next figure).

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 11: J changes without firing-rate changes are rare.

Figure 12. The clearest dissociation: 210511 stim 1

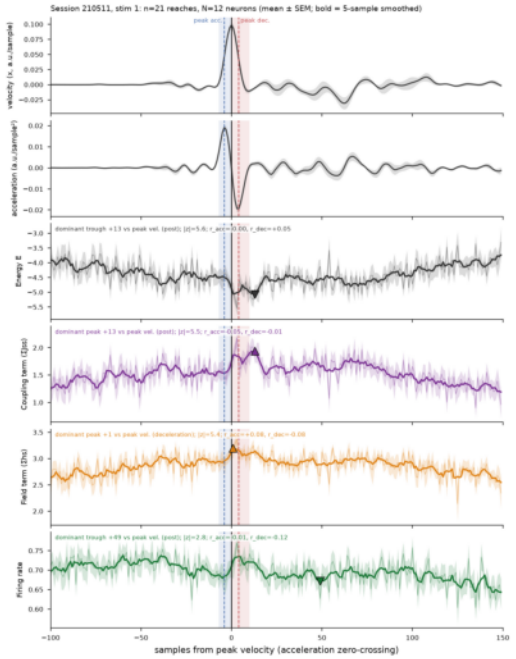


Figure 12. Session 210511, stim 1 (21 reaches, N = 12 neurons), mean ± SEM, bold lines 5-sample smoothed. From top: x-velocity, x-acceleration, energy, coupling term Σjss, field term Σhs, firing rate. Firing rate barely moves (peak |z| 2.8, about noise level), while the coupling term peaks 13 samples (130 ms) after peak velocity (|z| 5.5) and energy dips at the same time. One neuron in this condition fires in 99% of bins, which also affects the fit (Fig. 4b).

In plain terms. Here the neurons' overall activity stays roughly flat during the reach, yet the co-firing signal jumps. It is the kind of case one would want if coupling carried information on its own. But it is one condition out of 63, so it is an example, not a pattern.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 12: The clearest dissociation: 210511 stim 1.

Figure 13. Suggestive: a late post-reach coupling increase in the 2022 mice

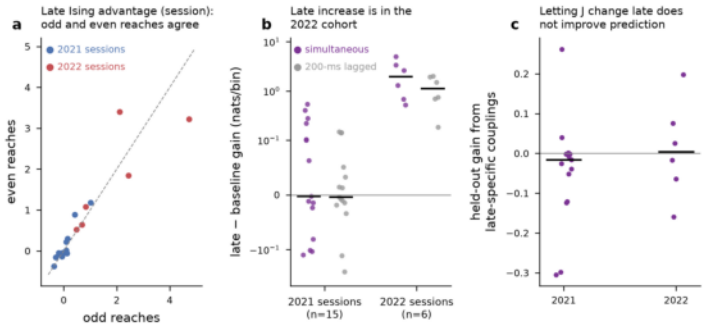


Figure 13. (a) Late-epoch (+0.5 to +1.5 s) Ising advantage over the time-varying independent model per session, from odd against even reaches (ρ = 0.89 across conditions). (b) Late minus baseline coupling gain for simultaneous states (purple) and 200-ms lagged states (grey), by cohort (symmetric-log axis). 2022: 6/6 sessions, median +1.97 nats/bin; 2021: 7/15, p = 0.45. The lagged control also rises in 2022 (6/6), so part of the effect is slow shared fluctuation. (c) Held-out gain from letting couplings change in the late epoch: no improvement in either cohort. The 2022 cohort is 2 mice (DJC000, DJC002).

In plain terms. About a second after the reach, a coupling increase shows up, but only in recordings from two mice made in 2022. It is consistent within those sessions, but some of it is slow drift, and letting the couplings change there does not improve predictions. Treat it as a lead to check in new recordings, not a finding.

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Figure 14. What distinguishes the sessions with the strongest effects

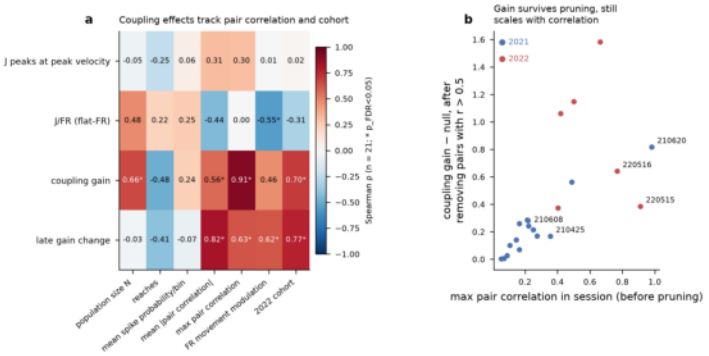


Figure 14. (a) Spearman ρ between each session-level effect (rows) and recording properties (columns), n = 21; * $p_{FDR} < 0.05$. Coupling gain tracks the maximum pair correlation (ρ = 0.91), N (0.66), mean |pair correlation| (0.56) and the 2022 cohort (0.70). (b) Coupling gain minus null after removing one unit of every pair with r > 0.5, against the session's maximum pair correlation before pruning. The gain stays above the null in 63/63 conditions after pruning.

In plain terms. Sessions with more strongly correlated neurons show more coupling, which makes sense. We removed near-duplicate neuron pairs, in case one cell had been recorded twice. The coupling survives that. The 2022 recordings stand out as more correlated overall; the lab records may show why (animals, site, sorting).

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 14: What distinguishes the sessions with the strongest effects.

Figure 15. Preliminary: changes across baseline, stimulation and washout

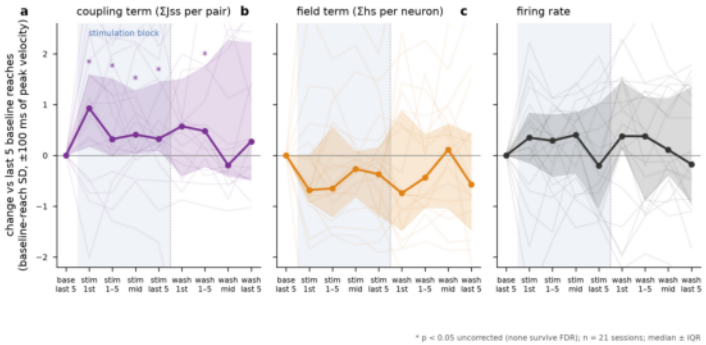


Figure 15. Change from the last 5 baseline reaches (in units of the SD across baseline reaches), in a ±100-ms window around each reach's peak velocity, for (a) the coupling term per pair, (b) the field term per neuron and (c) firing rate. Positions along x: last 5 baseline reaches, then the first reach, reaches 1–5, the middle 5 and the last 5 of the stimulation block (blue shading) and of the washout block. Thin lines: sessions; thick line: median; band: IQR; n = 21. * p < 0.05 uncorrected; none survives FDR. Coupling term: first stimulated reach +0.93 SD (16/21, p = 0.009), last 5 stimulated reaches +0.33 (17/21, p = 0.013), first 5 washout reaches +0.48 (14/21, p = 0.038). **Caveats:** each block had its own Ising fit, so the comparison mixes parameter changes with activity changes; the window is aligned to peak velocity as a proxy for the 1-cm stimulation trigger. A joint one-model-per-session test has been built and validated on synthetic data; it awaits the raw stimulation timing.

In plain terms. In the stimulation block, a brief light pulse drives the cerebellum's inputs as the hand passes 1 cm. The coupling term around that moment is higher from the very first stimulated reach and fades in washout, loosely like the behavioral adaptation in the dataset paper. But it is not statistically solid yet, and the comparison is not yet fair, because each block was fitted separately. The proper test, one model per session with the true pulse times, is next.

Shows: Page of the 2026-09-30 findings PDF: figure plus its caption. Pattern: - Why: SUPERSEDED by the 2026-10-13 RESULTS_LOG and the folders above (e.g. stationary-model energy decomposition is model-dependent; the field-dip depth and 'flat mean rate' claims are withdrawn). Numbers: Sept-30 Figure 15: Preliminary: changes across baseline, stimulation and washout.